Geology of Singapore

2nd Edition



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MESSAGE

In land-scarce Singapore, judicious land use is necessary to ensure that we continue to be a great place to live, work and play, despite the growing and often competing demands for physical developments. Such developments are driven by a diverse spectrum of requirements including those related to economic, environmental, social and sustainability considerations. The development of the Underground Ammunition Facility (UAF) by the Defence Science and Technology Agency (DSTA) for the Ministry of Defence is an excellent illustration of such judicious land use in Singapore. By exploiting the underground dimension for the UAF, we have created more space for our defence and released significant areas of surface space for other uses.

For our government and industry planners to continue to make prudent decisions related to land use, it is imperative that they have a reliable and accurate source of information on the geology of Singapore. Fortunately, technological advances over the years have provided us with the tools for more comprehensive geological investigations as well as wider and deeper coverage. I would like to commend DSTA and its partners from the Nanyang Technological University and the Building and Construction Authority for their concerted effort to update Singapore's geological data culminating in this report.

The development of the UAF has spurred great interest in our land use community to explore our underground space for other purposes such as deep cavern storage. I hope that they, as well as those pursuing research on the geology of Singapore, will find this report informative and useful. To ensure that our geological information remains accurate and updated, I encourage all our planners, users and researchers to contribute their geological data and expertise to our continuing effort to keep it current.

Quek Tong Boon
Chief Defence Scientist
Chief Research and Technology Officer
Ministry of Defence

FOREWORD

The first official publication on Singapore's geology was released in 1976. Known as the Geology of the Republic of Singapore, it has been a key reference resource for our professionals in the construction industry and government planners.

In recent years, more information on Singapore's geology has been uncovered from geology studies and underground construction projects undertaken by government agencies and local academic institutions.

Since the mid 1990s, the Defence Science and Technology Agency (DSTA) has been involved in underground rock cavern design and construction. Useful geological information was gathered from projects undertaken.

In 2003, DSTA and Nanyang Technological University jointly established the Underground Technology and Rock Engineering Programme to develop new technologies and sustain the long-term capability in underground cavern construction. This programme has also contributed significantly to our understanding of Singapore's geology.

The geological information compiled by DSTA and other organisations has made it possible to provide a major update. The publication of the new Geology of Singapore is our contribution to the building and construction industry and the geo-sciences community.

The updated geology report would be useful reference as Singapore gears up for more concerted development of underground space in the optimal use of land. The increased use of underground space will free up the limited surface land for more living, office and green space.

This publication would not be possible without the contributions and support from industry partners, government agencies and academic institutions. I wish to thank all who have contributed and I applaud their passion and dedication.

Soh Kong Pheng
Chief Executive
Defence Science and Technology Agency

FOREWORD

I am heartened by this update of Singapore's geological information. Our planners, designers and builders need reliable information on Singapore's geology and landscape to develop Singapore optimally. The Geology of Singapore will therefore be a very important resource to our building and construction industry. I am sure that users of this publication will also be delighted with the updated topography and other information included in the geological maps.

This publication is also timely, considering the recent changes in the building legislation for works related to deep excavations and more complex underground structures. Recognising the high safety impact associated with such works, the Building and Construction Authority (BCA) had recently amended the Building Control Act and Regulations to require geotechnical specialist inputs for their design and construction. The availability of accurate geological information from this publication is therefore well-timed. I am confident that this publication will be an indispensable reference for designers and practitioners in complying with the enhanced building control requirements and in contributing to a safe built environment.

On behalf of BCA, I congratulate the Defence Science and Technology Agency and all parties involved in this publication for their success in making this valuable and useful geological resource available to the building and construction industry.

Dr John Keung
Chief Executive Officer
Building and Construction Authority

PREFACE TO SECOND EDITION

The Geology of Singapore, based on the Geology of the Republic of Singapore published in 1976 by the then Public Works Department, presents an update of the Singapore geology based on new information and knowledge obtained over the last 30 years. It provides the various disciplines and communities involved in work associated with the ground, namely, civil engineering, tunnelling and rock engineering, land utilisation, urban planning, and agriculture, the much needed reference on the Singapore geology.

This second edition offers new information on the geology of the limestone and the Fort Canning Boulder Bed, the widespread occurrence of metamorphism, the geo-chronological history of the young Pleistocene sediments, and an appraisal of the engineering geological properties of our rocks. The construction of the Underground Ammunition Facility has also added new insights into the geology of the Bukit Timah Granite. The geology map has been updated extensively although the basic surface geology information in the 1976 edition has been retained. Some cross-sections to the surface maps have been redrawn to reflect the subsurface occurrences of the limestone and the Fort Canning Boulder Bed. Surface topography and contour lines, main road networks and aboveground structures, reclaimed land and boundaries of the shores have also been updated. In addition, we have included a compilation of the typical engineering properties of the main rock types found in the main geological formations of Singapore. This is a first attempt to fill the gaps between geology as a science and its engineering applications. In addition, the presentation of geological information and maps have been rearranged and made more readable.

Similar to the first edition, this publication comes in two parts. The first part comprises a book which presents the geology of Singapore. We have kept to the same chapter headings, but new content has been introduced to respective chapters. The second part is represented by six coloured geological maps at a scale of 1:25,000, one sheet of cross-section which reveals the subsurface distribution of geologic deposits along six lines of traverse across various areas in Singapore, one locality map and one geology map of Singapore at a scale of 1:75,000. The maps have adopted SVY21 - a coordinate system introduced by the Singapore Land Authority in 2004. The system is based on the WGS84 ellipsoid – the reference datum for Global Positioning System, and the Transverse Mercator Projection parameters.

This publication was prepared under a joint research programme between the Defence Science and Technology Agency (DSTA) and the Protective Technology Research Centre (PTRC) at the Nanyang Technological University (NTU), and made possible by the significant contributions of the following individuals : Zhou Yingxin (DSTA) for recognising the need and leading the efforts for updating the geology report; Zhao Jian (previously with NTU) and Nick Shirlaw (Golder Associates, previously with the Land Transport Authority) for much of the earlier effort in compiling new geological information; Lee Kim Woon (previously with the former Public Works Department) for a major contribution in updating the text and geological maps; Tor Yam Khoon and Li Juan (NTU) for digitising the geological maps; Ho Ai Phang and Pearly Chua (DSTA) for managing the design and production of the book; Ong Huei Luen (DSTA) for assisting in the technical editing and compilation of engineering geology data, and Chia Cheng Ghee (DSTA) for the retaking of many photographs.

Special thanks are due to the SAF Mapping Unit of the Ministry of Defence for providing the surface data, editing and printing the maps. Credit also goes to the Land Transport Authority, Building and Construction Authority and Jurong Town Corporation for contributing major new geological information, and the Society for Rock Mechanics & Engineering Geology (Singapore) for keeping alive the professional activities in engineering geology. We would like to thank Professor Pan Tso-Chien, Director of the NTU Protective Technology Research Centre, for his leadership at the PTRC and for the excellent collaboration in the execution of this project. Additionally, we wish to thank Richard Lim, former Chief Executive of DSTA, for his support and guidance in steering this geology report. Finally, we would like to express our appreciation to Professor Lui Pao Chuen, former Chief Defence Scientist of the Ministry of Defence, for his vision and encouragement which, through the years, have inspired us towards this effort and much more.

Lim Chee Hiong
Director, Protective Infrastructure & Estate
Defence Science and Technology Agency
March 2009

PREFACE TO FIRST EDITION

Reproduced from Geology of the Republic of Singapore

This book aims to provide a comprehensive account of the geology of the Republic of Singapore for the various disciplines that are involved with the ground, namely, civil engineering, land utilisation, urban planning, agriculture, etc. For the civil engineers, the nature, distribution and relationship of the various rocks and deposits described in the book will furnish geological information for preliminary planning and design of foundations, roads, water supply, drainage and sewerage works and others. Those concerned with land planning and uses will find the information helpful in delineating potential areas for granite quarrying, sand washing, agriculture, brick and pottery works, etc. Besides these practical objectives, the book aims to become the Republic of Singapore's contribution to the earth sciences in the South-east Asian Region as well as to acquaint the public, teachers and students with the geological constitution and evolution of Singapore over the past 500 million years.

The book comes in two parts. The first part features the text which relates in detail the geology of the Republic. The summary of the text is contained in the first chapter. Some relevant aspects of the country's geography and a review of previous geological investigations since 1924 are described in the following chapter. The third chapter presents, in chronological order, the definition, distribution, character, field relation, and age and correlation of each of the nine rock units, namely, Sajahat Formation, the Gombak Norite, the Palaeozoic Volcanics, the Bukit Timah Granite, Jurong Formation and its facies, the Old Alluvium, Huat Choe Formation, Tekong Formation and Kallang Formation and its members. Many of these units are introduced for the first time. Petrographic descriptions form an integral part of the presentation in this chapter. The next chapter is concerned with fold and fault structures that affected the older rock formations. The geologic evolution is discussed in the final chapter. The text's appendices include a list of references, analyses of rocks and water from hot-springs and a list of fossils and their locations.

The second part of the book is represented by eight coloured geological maps at a scale of 1:25,000; one sheet of cross-sections which reveal the subsurface distribution of geologic deposits along six lines of traverse across various areas of the Republic of Singapore and a locality map at a scale of 1:75,000. The geological maps can be assembled into a convenient wall map.

This book is prepared with the assistance of the United Nations and the Government of New Zealand. Special gratitude is due to the following persons: Dr Peter Morris who worked from March 1972 to March 1973; Dr Hans Bader from March 1974 to February 1975 and from October 1975 to November 1975, both United Nations Office of Technical Co-Operation Experts and Mr Graham Mansergh, Colombo Plan Expert from the Geological Survey of New Zealand from August 1974 to November 1975. Gratitude is also due to Dr W.A. Watters, Dr I.G. Speden and Dr B.W. Collins who are staff members of the Geological Survey of New Zealand.

The assistance of other Government Departments and Statutory Bodies, namely, Mapping Unit of the Ministry of Defence, Marine Police Department, Jurong Town Corporation, Public Utilities Board, Housing and Development Board and Port Authority of Singapore is also acknowledged.

Thanks are also due to the Department of Geology of the University of Malaya, for making relevant theses available.

The following staff of the Geological Unit of the Public Works Department have participated in no small measure in the preparation of this book: Lee Kim Woon (September, 1972 – Present), Loy Wei Choo (August, 1975 – Present), Pun Vun Tat (November, 1973 – May, 1975). The untiring support given by the other staff members is greatly appreciated.

Dr Tan Swan Beng
Senior Executive Engineer
Former Public Works Department
November 1976

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SUMMARY

A new geological map of Singapore at a scale of 1:25,000 is presented. The new map will reflect the latest changes to the coastlines of the main island of Singapore and her surrounding islands, the extensive subsurface occurrence of limestone in southwest main island Singapore, and the relatively restrictive, essentially subsurface occurrence of colluvial material in downtown Singapore.

The Sajahat Formation is a metamorphosed sedimentary deposit of quartzite, quartz sandstone, and argillite intruded by acid and basic dykes. It is given a Lower Palaeozoic age, but it could be Upper Palaeozoic or even Triassic in age if the pink porphyritic granite from Pulau Sekudu were a Cretaceous granite.

The Gombak Norite is geographically separated from the Sajahat and the Palaeozoic Volcanics. It is given a Lower Palaeozoic age as it probably represents a part of the ophiolite suite of the Lower Palaeozoic geosyncline, but it would however be younger if the Gombak Norite is interpreted as a 'basic differentiate' of the Triassic granite magma.

The Palaeozoic Volcanics comprises metamorphosed volcanic agglomerate with andesitic fragments, andesitic ash and tuff. It is mapped as overlying the Sajahat Formation and thought to be Upper Palaeozoic on correlation with similar rocks in Malaya. It is however possible that the Sajahat Formation and Palaeozoic Volcanics are of the same age.

The Bukit Timah Granite shows considerable hybridisation and much evidence of assimilation within the formation. Inclusions of Gombak Norite in various stages of assililation, large rafted inclusions of basic rocks, and smaller inclusions of rocks of sedimentary origin can be found within the granite. The Bukit Timah Granite is assigned an early to mid-Triassic age by isotope dating. Two phases of dyke intrusion are recognised within the granite: a first phase of acid dyke emplacement took place during the final stages of granite emplacement, and a second basic phase took place shortly after the first phase but before the deposition of the overlying Triassic sedimentary rocks.

The late Triassic to possibly early Jurassic Jurong Formation overlies the granite although the contact is never seen. Seven sedimentary facies, including one that is characterised by limestone, are recognised within the formation. Volcanics such as tuff, spilitic lava, and dyke rocks are recognised within the formation and are contemporaneous with the formation. The Murai Schist, previously thought to represent the oldest rock in the formation and to have a different origin from the rest of the Jurong Formation, is a product of dynamic metamorphism within the formation.

The Fort Canning Boulder Bed is a colluvial deposit of slipped material of the Jurong Formation found in the central business district of downtown Singapore. It was probably laid down in late Cretaceous, but it could also have been deposited anytime between late Triassic (when the Jurong Formation was deposited and deformed) and late Tertiary (when block faulting occurred and the Old Alluvium first deposited).

There was further geological activity in early Tertiary, late Tertiary and early Pleistocene. Block faulting and renewed movement along pre-existing faults resulted. In late Tertiary, a downwarp occurred and the resulting trough was back-filled with a coarse sand-gravel unit referred to as the Old Alluvium. (The base of the Old Alluvium is below any reasonable base level for erosion associated with Pleistocene low sea levels, and its top may be associated with a minimum high sea level stand of 70 m*.)

The Huat Choe Formation, a lacustrine deposit, was probably laid down as kaolin-rich clay in a fault-controlled depression in mid-Pleistocene times.

**Unless otherwise stated, heights given are relative to present-day sea level. Positive values refer to heights above present-day sea level and negative values to those below present-day sea level.*

Alluvial, littoral and inshore marine sediments have been laid down from late Pleistocene through to present day. These sediments have been assigned to the Tekong Formation and the Kallang Formation. The Tekong Formation is a mid-Holocene coastal terrace deposit associated with a sea level of 6 m. The Kallang Formation is an extensive late Pleistocene to Holocene deposit of marine, littoral and alluvial origin, and it is found in offshore areas, coastal areas and onshore valleys. Five members have been mapped within the Kallang Formation. The members have been differentiated based on their present-day depositional environment and sediment content.

The structures in the Palaeozoic sedimentary rocks indicate a longer and more complex tectonic history than that affecting the other rocks, but there is insufficient data to establish the full history. The intrusion and uplift of the granite presumably started in the Upper Palaeozoic and became strongly evident in Triassic time. The Triassic sediments were laid down in a mobile, northwesterly trending trough bounded on either side by rising granitic hills. The lithology of the coarser clasts in the sediments suggests that the bulk of the granite was still buried at that time. The rate of uplift of the Main Range Granite in the area southwest of the trough appears to have been more rapid. This lopsided uplift caused the trough and the sediments within it to tilt towards the northeast and pushed the sediments against the uplifted Bukit Timah Granite, folding them in the process. The dynamic metamorphism that gave rise to the Murai Schist accompanied this sliding. There is no evidence that sedimentation persisted after early Jurassic time, and it is assumed from evidence in Malaya that the Malay Peninsula, including Singapore, was raised above sea level from that time.

Alternatively, the Triassic sediments in the Jurong Formation could have been uplifted, tilted, dislocated, moved, and piled on top of one another in late Cretaceous when the Woyla continental fragment collided with the Sibumasu/East Malaya block. This event has probably given rise to the Fort Canning Boulder Bed. There was also widespread shallow thrust faulting, shallow folding and metamorphism. Such faulting could have given rise to the major faults in the sediments, and metamorphism has given rise to the occurrences of low-grade metamorphic rocks such as slates, phyllites, and schistosed sandstones at many locations away from the Murai Schist.

Tectonic activity recommenced in the late Tertiary with block faulting and warping. The Old Alluvium and Huat Choe Formation were deposited in the depressions that had formed. After the cessation of warping, river valleys were cut in the Old Alluvium and older rocks. The valleys and coastal areas were subsequently backfilled with sediments of the Kallang Formation during late Pleistocene and Holocene. During mid-Holocene, coastal areas were covered with sediments of the Tekong Formation. Sediments of the Kallang Formation, except the Lower Marine Member, continue to be deposited in present day.

1 INTRODUCTION

INTRODUCTION

LOCATION

Singapore, consisting of Singapore Island and several smaller islands and lying at the southern end of the Malay Peninsula, is a roughly diamond-shaped area positioned between latitude 1°09'N and 1°28'N and longitude 103° 38'E and 104° 06'E. Along its northern border, Singapore is surrounded and separated from West Malaysia by the Selat Johor, a stretch of water body varying from 0.7 to 2.5 km in width. Some 12 to 15 km to the south of Singapore is the Riau Archipelago of Indonesia.

Apart from Singapore Island, offshore islands lie in four groups to its northeast, south, southwest and west. The northeastern group includes Pulau Ubin, Pulau Tekong, Pulau Sajahat and smaller associated islands. The southern group includes Sentosa, Pulau Brani, Pulau Tekukor, Pulau Subar Darat, Pulau Subar Laut, St. John's Island, Lazarus Island and smaller islands. The western group once consisted of many islands surrounding Pulau Ayer Chawan. Some of them include Pulau Merlimau, Pulau Seraya, Pulau Ayer Merbau, and Pulau Sakra (Map Sheet 4). These islands and Pulau Ayer Chawan have been joined by reclamation and they are now collectively called 'Jurong Island'. The southwestern group consists of islands between Pulau Bukum and Pulau Satumu. Some of these islands include Pulau Hantu, Pulau Semakau, Pulau Jong, Pulau Sebarok, Pulau Salu, Pulau Sudong, Pulau Pawai, Pulau Senang and Pulau Biola.

PHYSIOGRAPHY

Singapore has moderately low relief. Singapore Island and the offshore islands cover an area of about 700 sq km. The climate is hot and humid with an annual rainfall ranging from 1,600 mm in the southwest to 2,500 mm in the central regions. Under these conditions, the rocks are deeply weathered and the drainage has developed to a stage where the rivers are of low gradient with a mature profile. The drainage patterns are either consequent or structurally controlled.

Seven discrete physiographic areas can be recognised on Singapore Island. Fig. 1.1 shows the distribution of these areas.

Area 1, covering grounds to the north and west of Nanyang Technological University (NTU) and including the pronounced Pasir Laba Ridge, was an area of moderate relief with hills rising to a maximum of 85 m. The hills were aligned to the geological structure and were steep but covered with soil. Drainage was influenced by its geological structure.

Area 2, covering grounds to the east and south of Area 1, is of low relief and had rolling hills rising to about 28 m. The area was crossed by a northwesterly trending line of hills which reached 70 m high. Drainage is partly consequent and partly controlled by structure.

Area 3 is the low-lying Sungei Jurong River Valley. It has a flat relief and a consequent drainage pattern modified by man.



Fig. 1.1 Map of Singapore Island showing the area extent of the seven physiographic units. Pulau Brani and Sentosa are in Area 4. Pulau Ubin is not classified.

Area 4 is the largest physiographic area defined for Singapore. It is an area of high relief. Bukit Timah Hill, the highest hill in Singapore rising to 166 m, lies within this area. Slopes are generally steep. Drainage is controlled by the faults and joints in the granite and by the folds, faults and cross-joints in the Triassic sedimentary rocks to the south. A belt of relatively low relief cuts across the southern end of this area in a northwest direction along the line of Singapore River and Sungei Ulu Pandan.

Area 5 lies in the central north area and has similar characteristics to those of Area 2. Most of the area was below the 20 m contour, but hills once rose to 32 m at Punggol and 33 m at Seletar North Hill in North Seletar. Drainage is consequent to the northeast.

Area 6 is the low-lying Kallang River Basin and it shows similar characteristics to Area 3.

Area 7 lies to the east and southwest of Paya Lebar Air Base. It could be described as a deeply dissected plateau. The relief was high with steep slopes with a well-developed dendritic drainage pattern consequent on the surface. The highest point of 46 m is found just east of the Paya Lebar Airbase.

Wong (1969) made a more detailed analysis of the physiography of Singapore. He recognised nine discrete morphological units (including river valley floors) based on height, slope, relief, and dominance of hills or ridges. More than 50 areas were mapped. The reader is referred to his work for a fuller description of the areas.

SOURCES OF GEOLOGICAL INFORMATION

The most comprehensive papers on the general geology of the Republic of Singapore prior to 1976 are those of J.B. Scrivenor (1924), F.E.S. Alexander (1950), Public Works Department (PWD), Singapore (1976), and Pitts (1984). Other contributions have been made by Leow (1962), Burton (1964), and Hutchison (1964), and theses have been prepared by Ignatius Wong (1960), Chin Fatt (1965), Tai Say Ann (1972), Seet Chin Peng (1974), and Lim Meng Sze Wu (1974) on areas within the Republic. These latter works are of a detailed nature, and they concentrate on specific aspects of the geology. J.B. Scrivenor (1924) also prepared bibliographies of the geology pertaining to Singapore. D.J. Gobbett (1968) prepared bibliographies pertaining to West Malaysia and Singapore.

Scrivenor (1924) mapped three rock units in Singapore: Granite, Shale and Sandstone, and High Level Alluvium plus Recent Alluvium (Fig. 1.2). According to him, the

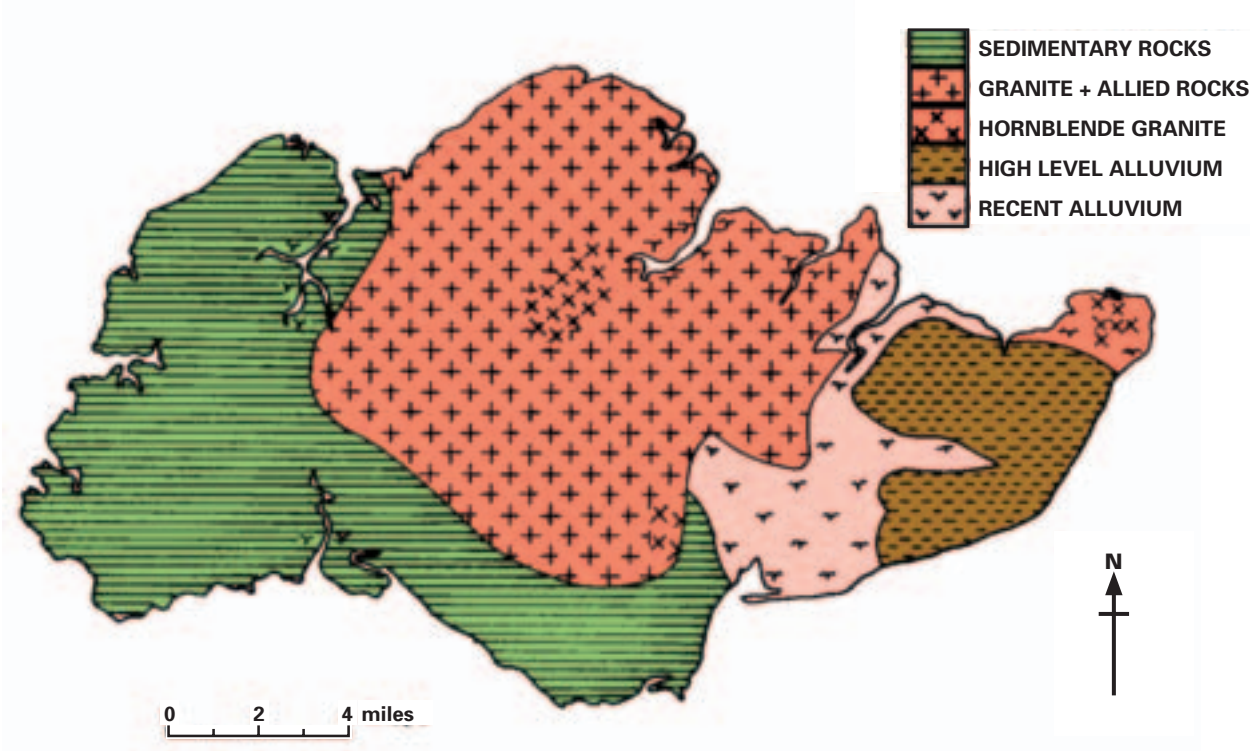


Fig. 1.2 Simplified geological map of Singapore after Scrivenor (1924)

the granite appears as a central granite mass east of a northwest diagonal crossing Singapore Island and appears again on Pulau Ubin and at Changi. The shale and sandstone unit lies west of the northwest diagonal. The legend on the map in Scrivenor (1924) describes the occurrence of limestone in the shale and sandstone unit. However, the text in Scrivenor (1924) made no mention of the limestone. The High Level Alluvium is found to the east of the central granite mass and separates it from the granite at Changi. Other igneous rocks including quartz-norite, quartz-norite-gabbro, fine-grained rocks, and enstatite-spessartite were reported from Pulau Ubin.

Leaf impressions from plant remains, molluscs and fossil woods were found in the sedimentary rocks at different localities in Singapore by Scrivenor. The fossil remains were described in Newton (1923). They were poorly preserved, and Newton suggested (with some reservation) a Rhaetic age for them. Scrivenor believed that the high inclination of the sedimentary beds near the contact with the granite in many places indicated that the sedimentary rocks were older than the granite. He suggested that the sedimentary rocks were shallow-water deposits and that some were probably estuarine.

Alexander (1950) recognised the same three units as Scrivenor, namely the Older Sedimentary Rocks, the Igneous Complex, and the Older Alluvium plus a Recent Alluvium (Fig. 1.3). She divided the Triassic sedimentary rocks into three series: an Old Schist which is highly contorted, an Argillaceous Series which is mainly black or grey shale unconformably overlying the Old Schist, and an Arenaceous Series which is mainly sandstone and conglomerate with some shale and associated silty beds. Thin beds of coal are occasionally found in the Arenaceous Series. Her text on the Older Sedimentary Rocks or the map that accompanies her report made no mention of limestone. Alexander divided the igneous rocks into two groups: the Central Singapore Igenous Rocks and the Changi-Pulau Ubin Igneous Rocks. The Central Singapore Igneous Rocks consists dominantly of granite with less abundant norite and diorite whereas the Changi-Pulau Ubin Igneous Rocks are mainly hornblende-soda granite and fine-grained soda granophyre. She postulated two magmas (an earlier gabbroic and a later granitic one) for the Central Singapore Igneous Rocks and described a zone of hybrid rocks between the two intrusions of these two magmas. The High Level Alluvium described by Scrivenor (1924) was renamed the Older Alluvium by Alexander (1950). The name change resulted from observations that the

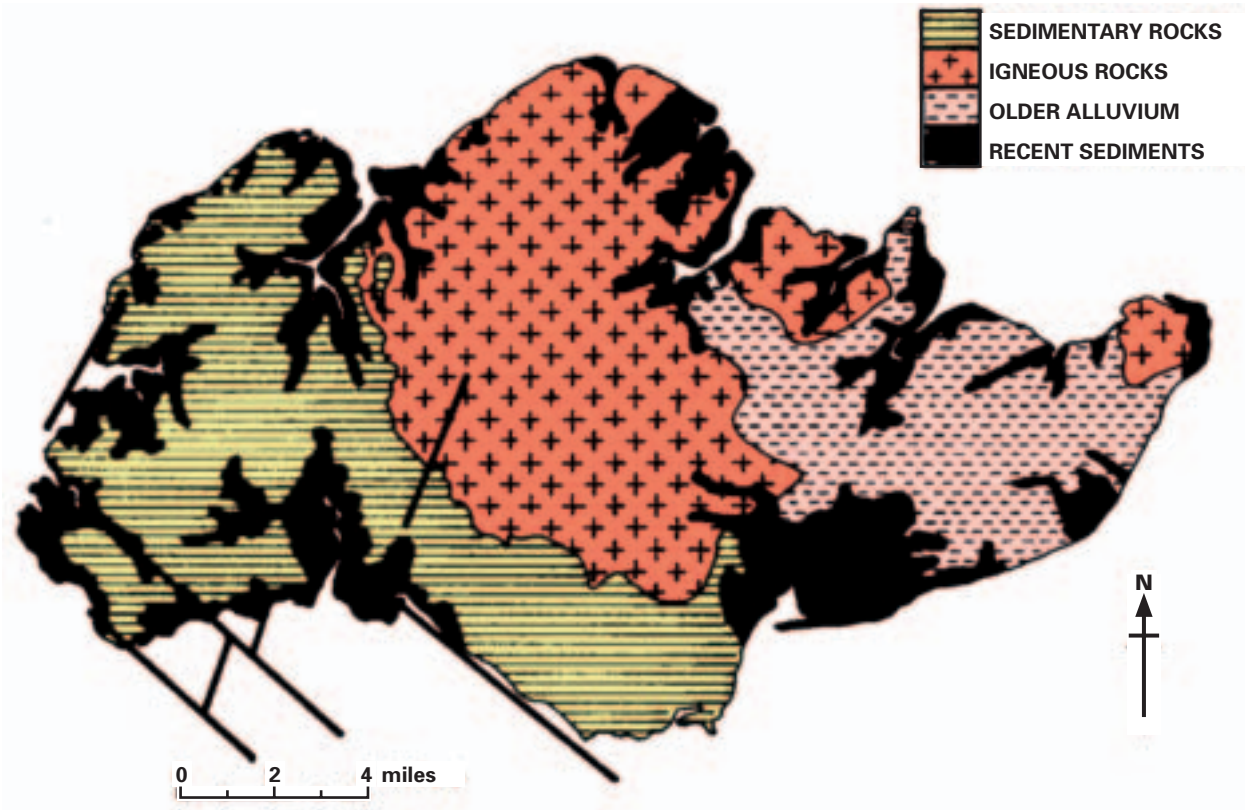


Fig. 1.3 Simplified geological map of Singapore after Alexander (1950)

High Level Alluvium has its base lying below present sea level and that it was older than the recent alluvial deposits. She also recorded her observations on raised beaches and discussed briefly the effects of sea level fluctuations on these deposits.

PWD (1976) does not report on the extensive subsurface occurrence of carbonate rocks in the Jurong Formation and the occurrence of the Fort Canning Boulder Bed in downtown Singapore.

Wong (1960) mapped the southwestern region of Singapore. He reported three distinct conglomerate-sandstone-shale sets of beds with each set beginning with a basal conglomerate. Leow (1962), following his study on the heavy mineral in the sedimentary rocks, could not accept the division of the Argillaceous and Arenaceous Series postulated by Alexander (1950). He believed that there were several periods of deposition during the Triassic and each period gave rise to a distinct set of beds beginning with a basal conglomerate. Later, Chin (1965) and Lim (1974) divided the Triassic sedimentary sequence into the Pasir Panjang and Jurong Formations. A detailed stratigraphic column for the Jurong and Pasir Panjang Formations is illustrated in Lim's thesis. Lithologically, the Pasir Panjang and Jurong Formations are similar to the Arenaceous and Argillaceous Series postulated by Alexander in 1950.

Well-defined, northwesterly trending fold axes in the Triassic sedimentary rocks were recognised by Wong (1960). Leow (1962) concluded that the Triassic rocks contain a series of folds forming part of a major fold system which plunges toward the northwest. He postulated a regional movement followed by tilting as the reason for such a fold system. Parasitic folds due to incompetency between the strata and many isoclinal folds, postulated as resulting from igneous intrusion in the area, were observed by Chin (1965) and Lim (1974). Both collected species of Myophoria, Pecten, and Pteria which suggested a late Triassic age for the sedimentary rocks.

Hutchison (1964) believed that the marginal occurrence of a gabbroic body to the granodiorite batholith in Singapore was related to the Thai-Malay Orogeny and that the gabbroic rocks are pre-orogenic. Based on field relations as well as the chemical and petrological characteristics of the gabbro and granodiorite in this area, Hutchison concluded that the granodiorite intruded the gabbro in the form of irregular dykes. The intrusion has, by thermal metamorphism and hybridisation, produced a variety of hornblende-rich rocks along the contact. Granodiorite dykes intruding the gabbro are crowded with gabbro xenoliths that show different stages of assimilation.

Seet (1974) carried out detailed mineralogical, petrographical, and chemical studies on the igneous rocks of Pulau Ubin and deduced a complex geological history for the igneous rocks on the island. He thought that the granite and granodiorite were emplaced at a high level as indicated by the low triclinicities of the alkali feldspars. He observed high aluminium oxide (Al_2O_3) but low total alkali percentage in the granite compared with the microgranodiorite. Based on chemical variation of the rocks, he concluded that the granite and granodiorite were not comagmatic and that they were probably derived by anatexis of sialic crustal material and not by fractional crystallisation or differentiation. No correlation of this granite on Pulau Ubin with those at Changi or other parts of Singapore was attempted by Seet for his thesis.

Burton (1964), in his report on the Older Alluvium of Johor and Singapore, agreed with the observations of Scrivenor (1924) and particularly the observations of Alexander (1950). He noted the distinct deficiency of manganese in soil developed on the Old Alluvium, the characteristic development of ferricrete within the staining zone, and the presence of tabular layers of pebbles at depths varying from several centimetres to three metres in the weathered members of the alluvium. From the distribution, extension, and other evidences observed, Burton concluded that the Older Alluvium was related to a 75 m sea-level stand during the pre-glacial of the First Interglacial (Gunz-Mindel or Aftonian) at the end of early mid-Pleistocene but added that the formation may have further dated back to late Pliocene.

Tai (1972) reported a marine fossil found in the Old Alluvium at a locality near the eastern end of the island.

Three unpublished geological reports were prepared by the Geological Unit of the PWD, Republic of Singapore (1973 A, 1973 B, 1974). These reports presented detailed geological maps of the Jurong-Tuas, Jurong-Pandan, and the Pasir Panjang-Tanglin areas. An appendix of field photographs and geological maps at scales of 1:25,000 and 1:2,500 were included in each of these reports.

The Jurong-Tuas area was found to consist of materials of sedimentary origin. Three units were recognised.

The first unit was described as an older and thicker sequence of black shale, grey mudstone, and grey tuffaceous agglomerate and the second unit, a younger and thinner sequence of sandstone, conglomerate, and siltstone. The younger sequence was thought to lie disconformably on the older sequence. The third unit was described as unconsolidated Holocene sediment.

The Jurong-Pandan and Pasir Panjang-Tanglin areas, which lie east and adjacent to the Jurong-Tuas area, were reported to be underlain by sedimentary rock with some granite lying in the northeastern and northern sectors of the areas. Four sedimentary units were recognised and mapped in these two areas. The first and oldest unit is made up of purple to red shale or mudstone, siltstone, and fine- to coarse-grained sandstone. Overlying the oldest unit is a second, younger unit containing black shale, mudstone, conglomerate, and tuff. Overlying the second unit is the third unit made up of conglomerate, sandstone, siltstone with minor red mudstone, and tuff. The fourth and youngest unit is unconsolidated Holocene sediment. The lithological changes from the Jurong-Tuas to the Jurong-Pandan and Pasir Panjang-Tanglin areas were interpreted as facies changes.

The three reports showed that most folds found in the three areas are open and asymmetrical and have their fold axes trending northwest. The dips of the beds in the Jurong-Pandan and Pasir Panjang-Tanglin areas were found to be steep. Major and minor faults in the Jurong-Tuas area strike mainly to the northeast and northwest. In the Jurong-Pandan area, northwesterly trending faults were more common than the northeasterly trending faults. Small faults were reported from the Pasir Panjang-Tanglin area.

The papers prior to 1976 form the major sources of information for the Report on Geology of the Republic of Singapore by the Singapore PWD (1976). The report is the basis and major source of information for this publication.

Since 1976, papers containing substantial references and discussions on the geology of Singapore have been prepared by Pitts (1983, 1984, 1992); Gupta (1986); Gupta, et al. (1987); Fontaine and Lee (1993); Han, et al. (1994); Redding and Christensen (1999); Taylor (2001); Bird, et al. (2003); Chiam, et al. (2003); Jeyatharan, et al. (2003); Ong, et al. (2003); and Shirlaw, et al. (2003). Vilpponen (1988), Chang (1995), and Guo (1998) have prepared unpublished theses.

The geology and engineering geology of Singapore is reviewed in Pitts (1984). Vilpponen (1988) describes the sedimentology and stratigraphy of the Jurong Formation. Fontaine and Lee (1993) define the limestone at Pandan Road. Zhou, et al. (1993 A, B) discusses the geological and geotechnical properties of the Bukit Timah Granite for a site investigated for cavern construction at Lorong Asrama, north of Mandai Road. The variety and composition of carbonate rocks in the Jurong Formation is presented in Guo (1998). Based on results of seismic investigation works, Redding and Christensen (1999) discussed the depositional setting of the Jurong Formation and the tectonic setting that had affected it. Heavy straining and micro-fracturing of minerals in the Bukit Timah Granite at the Underground Ammunition Storage Facility was highlighted in a 1999 unpublished report by the Lands & Estates Organisation, Ministry of Defence. The engineering geology and the rock mass properties of the granite at the Mandai Quarry site is discussed in Zhou (2001). A hydrogeological study of the Sembawang hot spring (the larger of two hot springs in Singapore) is reported in Zhao, et. al (2002). Ong, et al. (2003) describes the topography of the granite floor on which the Old Alluvium rests in eastern Singapore Island. The Fort Canning Boulder Bed is described in Han, et al. (1994), and Shirlaw, et al. (2003), and Singapore's Quaternary deposits are described in Chang (1995); Gupta (1986); Gupta, et al. (1987); Pitts (1983, 1992); Taylor, et al. (2001); Chiam, et al. (2003); and Bird, et al. (2003).

The papers published for the workshop 'Updating the Engineering Geology of Singapore' have also provided valuable information. The workshop, an event in conjunction with Underground Singapore 2003, was organised by the Tunnelling and Underground Construction Society (Singapore), the Centre for Soft Ground Engineering in the National University of Singapore, and the Geotechnical Research Centre in the Nanyang Technological University, Singapore.

2 STRATIGRAPHY

STRATIGRAPHY

OUTLINE OF STRATIGRAPHY

Ten separate formations are recognised in Singapore. Three of the ten, the Sajahat Formation (S) (a partially metamorphosed sedimentary unit in eastern Singapore), the Palaeozoic Volcanics (PV) (a partially metamorphosed unit of andesitic ash and tuff also in eastern Singapore), and the Gombak Norite (GN) (a body of noritic and gabbroic rock in central Singapore), date from Lower Palaeozoic times. The relationship between them however cannot be determined based on field evidence. The Bukit Timah Granite (BT) was emplaced next. In the process, it has intruded and altered the Gombak Norite. A number of hybrid rocks are recognised within the Bukit Timah Granite. Granite emplacement is believed to have taken place in early to mid-Triassic time. The dykes associated with this granite can be divided into two groups: an older acid group assumed to be a cooling phase and a slightly younger basic group seen cutting the dykes of the older acid group. Both groups of dykes cut the Sajahat Formation, Gombak Norite, and Bukit Timah Granite.

The Jurong Formation of late Triassic to early Jurassic age has been deposited on the granite. The formation is divided into seven facies: the Queenstown, Jong, Ayer Chawan, Pandan, Rimau, St. John, and Tengah Facies. The division is based on lithology, and rocks mapped in any one facies do not necessarily form a continuous body. Deposition was terrestrial, transitional, and shallow marine. Schist related to faulting and volcanic rocks interbedded with the facies are included in the formation.

The Fort Canning Boulder Bed (FC) is a small colluvial deposit of materials derived from the Jurong Formation. It is found in the central business district of downtown Singapore. It lies on the Jurong Formation and is probably of late Cretaceous age.

The Old Alluvium (OA) is a dominantly terrestrial deposit of late Tertiary to early mid-Pleistocene age. It lies unconformably on older rock units or in fault contact with them. No subdivision has been made within the Old Alluvium.

The Huat Choe Formation (HC) is a small lacustrine deposit probably of early to mid-Pleistocene age. It lies unconformably on the Jurong Formation.

The Tekong Formation (T) is a mid-Holocene marine and littoral deposit whose upper depositional surface is found lying at elevations of between 3.6 and 5.5 m above present sea level.

The Kallang Formation includes both marine and terrestrial sediment laid down from late Pleistocene to present day. Five members are recognised within the Kallang Formation. The members are the Marine Member (Km), the Littoral Member (Kl), the Transitional Member (Kt) , the Alluvial Member (Ka), and the Reef Member (Kr). Further work may allow the establishment of some of these members as formations.

SAJAHAT FORMATION

Definition and Distribution

The Sajahat Formation is defined as those variably metamorphosed sedimentary rocks comprising quartzite, sandstone, and argillite found on Pulau Sajahat and Sajahat Kechil. Similar rocks are found along the north coast of Pulau Tekong and as far to the east as Tanjong Renggam. Quartz-rich sandstones are exposed at Tanjong Batu Koyok on the southwestern tip of Pulau Tekong (Fig. 2.1) and on the western side of Tanjong Punggol on Singapore Island. These rocks have also been mapped as part of the Sajahat Formation, but they generally show less deformation and a lower degree of lithification than the rocks from the type locality.



Fig. 2.1 Outcrop of very thickly bedded sandstones at Tanjong Batu Koyok, Pulau Tekong

A recrystallised quartzite similar to the quartzite of Pulau Sajahat was found in the Public Utilities Board Test Hole No. 1 at Bedok (GR 598465) beneath the Old Alluvium at a depth of -149 m. A similar rock is exposed in a single outcrop at location GR 466437 near Republic Polytechnic in Tanglin. These rocks are also considered as part of the Sajahat Formation.

Content

The rocks are well lithified and vary from quartzite through quartz sandstone to argillite. Five thin sections have been prepared from rocks assigned to the Sajahat Formation. Two of the five sections are from quartzites from the type locality. The third section is from quartz sandstone from the type locality (Fig. 2.2). The fourth section is from a quartzite recovered from the Public Utilities Board Test Hole No. 1 at Bedok, and the fifth is from a spotted argillite from Tanjong Renggam.

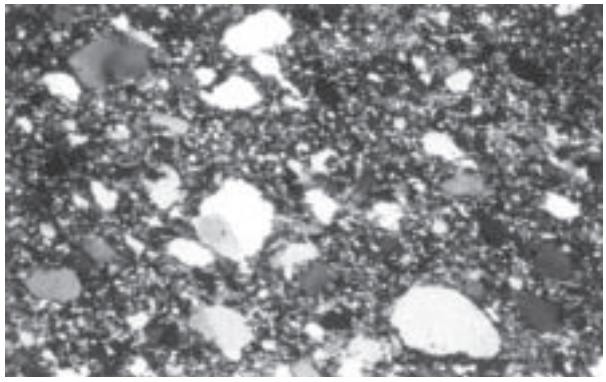


Fig. 2.2 Poorly sorted quartz sandstone from Pulau Sajahat 45X Crossed Nicols

Quartz makes up more than an estimated 97% in the quartzites that have been sectioned. Biotite and an unidentified amphibole (possibly tremolite) make up most of the remainder. Feldspar (usually represented by a rather clouded alteration product) is also present. In purer quartzites, the grains often show good sutured contacts between them. In quartz sandstone, the grains are well rounded but poorly sorted and show undulose extinction in each section. Grains up to 0.75 mm in diameter were observed.

The texture of the rocks suggests they are hornfelsic, and this texture would have been acquired by their proximity to granite and dykes cutting the formation.

The sample from the Public Utilities Board Test Hole No. 1 at Bedok is more highly altered than samples from Pulau Sajahat. It shows signs of shear and granulation (Fig. 2.3) as well as the effects of thermal metamorphism. From its position close to the granite, such increase in metamorphic rank is expected.

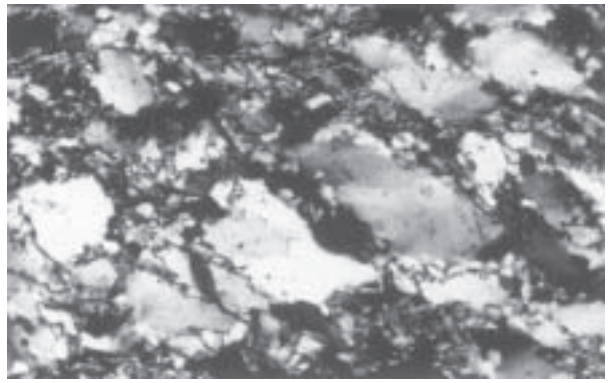


Fig. 2.3 Recrystallised quartzite of the Sajahat Formation from the Public Utilities Board Test Hole No. 1 at Bedok 120X Crossed Nicols

An argillite found at Tanjong Renggam is sheared and shows prominent spotting (Fig. 2.4). In thin section, the rock shows a distinct foliation. Its porphyroblasts are strongly altered to secondary minerals. They suggest that the porphyroblasts could be cordierite (Fig. 2.5).

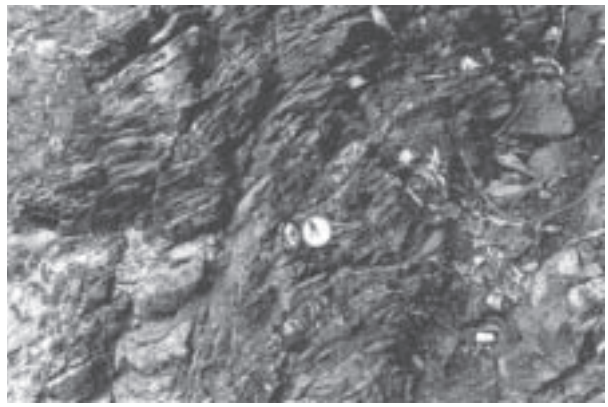


Fig. 2.4 Well-bedded and sheared argillite of the Sajahat Formation at Tanjong Renggam, Pulau Tekong



Fig. 2.5 Spotted argillite at Tanjong Renggam, Pulau Tekong 45X Plain Polarised Light.

The rocks are well bedded and have bedding thicknesses that range from 2 cm for the finer sediments to 4 to 5 m for the quartzite. The beds are folded and sheared with boudinage developing in them.

A number of dykes were mapped intruding the formation on Pulau Sajahat, Pulau Sajahat Kechil, and along the northern coast of Pulau Tekong. No dykes were seen in contact with the sediment on the south side or at Tanjong Punggol. In these latter areas, the bedding is more uniform and the rock less well lithified. In contrast, the rock at Tanglin is well lithified, but no dykes were seen.

Material recorded as tuff was seen in highly weathered exposures on the north coast of Pulau Tekong Kechil and north of Kampong Salabin on Pulau Tekong. No tuff was seen in fresh exposures.

Field Relations

The Sajahat Formation is mapped as being in contact with the Palaeozoic Volcanics at eastern Pulau Tekong. The presence of the Palaeozoic Volcanics is postulated on the identification of highly weathered material at the eastern tip of the island. Contact between the two formations is defined by a change in topographic expression on Pulau Tekong. The area of Palaeozoic Volcanics showed a rounded hill form with the peaks usually below 21 m whereas the area of Sajahat Formation had steeper hill forms and sharp ridges often rising to 30 m or more.

It may be argued that the two formations interdigitate and that the tuff, tentatively recognised in the Sajahat Formation, is actually an extension of the Palaeozoic Volcanics.

Age and Correlation

There is no direct evidence of the age of the Sajahat Formation. It is correlated here with sedimentary and volcanic rocks described by Grubb (1968) from the Pengerang area at the southern tip of Malaya. Although they were not identified as such, the rocks of the Sajahat Formation were mapped as undifferentiated middle and late Triassic rocks by Gobbett (1972). Chung (1973) excluded Singapore in the 7th Edition of the Geological Map of West Malaysia but mapped the surrounding areas described by Grubb (op cit) as Permian.

Grubb (1968) found no direct evidence for the age of the older rocks in the Pengerang area. He mapped a belt of metasediment on the eastern side of the peninsula as older than the volcanics. The volcanics were mapped as correlatives of the 'Pahang Volcanic Series' of Carboniferous to Triassic age. Hutchison (1973 B) later restricted the age of the Pahang Volcanic Series to Carboniferous to Permian.

In this report, the Sajahat Formation is placed as older than the Jurong Formation for these reasons: its higher degree of deformation, the presence of dykes which show petrographic similarities to the dykes intruding the Bukit Timah Granite, and the evidence of contact metamorphism. Contact metamorphism has probably resulted from the emplacement of the Bukit Timah Granite about 220 million years ago.

The deformation of the rocks in the Sajahat Formation indicates that it must have been deposited at least before the tectonism preceding the emplacement of the granite, and thus it must predate an Upper Palaeozoic tectonic event.

Hutchison (1973 A), in a reconstruction of the arc system of Malaya, plots a geosynclinal trough as lying east of Bukit Timah in Lower Palaeozoic time and states that the subduction zone on the western side had migrated east to Kalimantan and the Anambas Islands by late Carboniferous time. If the deposition of the Sajahat Formation was in this trough, its deposition must be restricted to Lower Palaeozoic. However, deposition in Upper Palaeozoic may have occurred in the later miogeosynclinal trough that lays between the Upper Palaeozoic Volcanics to the east and the granite intrusion postulated as lying to the west (Hutchison, op cit).

Alternatively, if the Sajahat Formation and the Palaeozoic Volcanics interdigitate, the age of the Sajahat Formation is then the time equivalent (Permian-Carboniferous) of the Pahang Volcanic Series.

The age of the Sajahat Formation is thus considered as probably Lower Palaeozoic, but it can be Carboniferous to Permian.

GOMBAK NORITE

Definition and Distribution

The term norite is used here in a general sense to include the entire suite of noritic and gabbroic rocks.

The Gombak Norite (GN) is an association of norite and gabbroic rock found on the western side of the Bukit Timah Granite on Singapore Island. The unit is named after Bukit Gombak where noritic and gabbroic rocks are well exposed in a number of quarries. Gabbro and norite have also been reported from Pulau Ubin. Alexander (1950) reported the presence of a solitary boulder of norite on the north coast of the island at a locality about 1.5 km east of Tanjong Tajam. Norite boulders in the area could not be located. However, the presence of deep red quartz-free clay soil in the general vicinity suggests a basic parental material for the soil. It is suggested that rafted bodies of basic material within the granite in this area have weathered to give this soil. Hutchison (1973 C) also mentioned gabbro and norite as occurring near the south coast of Pulau Ubin, but this occurrence was not relocated.

The area of the gabbroic rocks on Singapore Island is small in comparison to that of the granite. It occurs adjacent to the granite and forms an alignment of hills of which Bukit Panjang and Bukit Gombak are the most prominent. The hills form a ridge which measures some 2.5 km in length and attains a maximum width of 1 km. The hills once supported nine quarries. They were namely Bluestone, Lian Hup, Swee Construction, Yun Onn, Gim Huat, Chua Chai Seng, Peng Seng, Chia Oh Kang, and Poh Hua (Map Sheets 1 and 4). The quarries have been reclaimed for building and recreational uses.

Chemical analyses of a norite from the Gim Huat Quarry and a hornblende-gabbro from the Yun Onn Quarry are given in Appendix 1.

Content

The noritic body is rather variable in composition. Its overall composition can be described as ranging from norite through noritic gabbro to gabbro, the latter being least abundant. Noritic gabbro was found in all quarries. Norite occurs less frequently and has been sampled from the south and southeastern faces of the Yun Onn and Peng Seng quarries respectively. Gabbro is largely confined to the Bluestone Quarry (Hutchison, 1964).

The noritic and gabbroic rocks are coarse-grained and plagioclase-rich with varying amounts of clino- and orthopyroxene minerals appearing as interstitial grains giving an intergranular texture to the rock. Ophitic texture, when present, is feebly developed. The plagioclase is usually euhedral labradorite with diffuse twinning following the albite law. The feldspar is normally fresh, and any significant alteration is believed to be a localised feature as only patches of the host rock in the Yun Onn Quarry show an alteration of the plagioclase to clinozoisite, epidote, and calcite. Sericitisation of the plagioclase appears to be appreciable near basic and acid dykes only.

The pyroxene minerals are represented by hypersthene and augite. Unlike the irregular crystal outlines of the augite, the hypersthene often occurs in euhedral elongated prisms and can also be characteristically finely fibrous. In thin section, the hypersthene shows a dark grey birefringent colour. Augite is the typical clinopyroxene. It frequently shows finely spaced lamellar structure that has been caused by the exsolution of less calcic pyroxene. This feature is best observed in norite and noritic gabbro from the Peng Seng, Poh Hua, and Chia Oh Kang quarries. Unlike the plagioclase, the pyroxene minerals are always altered. Serpentinisation has affected many of the hypersthene minerals present in the norite and noritic gabbro in quarries situated on the southern slopes of Bukit Gombak. Sericitisation of the pyroxene minerals is seldom observed, but this alteration is seen together with saussuritisation of the accompanying plagioclase in the Yun Onn Quarry. The most noticeable alteration of pyroxene is to amphibole in which it forms conspicuous rims on the pyroxene (Fig. 2.6). The amphibole may be coarse or fibrous and is usually pale brown to green. The coarse variety is generally hornblende, and this

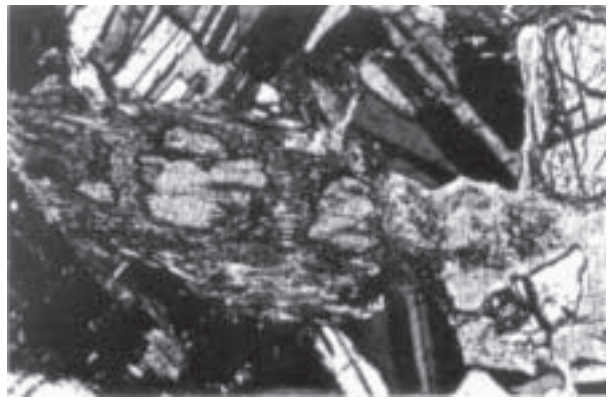


Fig. 2.6 Alteration of pyroxene to fibrous amphibole in norite gabbro from the Swee Construction Quarry, Bukit Panjang 120X Crossed Nicols

texture is characteristic of the noritic gabbro and norite in the Bluestone, Swee Construction, Yun Onn, and Peng Seng quarries. Fibrous actinolite is more often associated with hypersthene and forms rims and veins on and through the hypersthene. Although pyroxene replacement by amphibole is best observed in gabbroic rocks adjacent to the granite, such alteration (although less spectacular) is present in all the quarries. This characteristic enrichment of amphibole reflects a slight retrogressive metamorphism of the gabbro caused by the intrusion of the granite (Hutchison, 1974).

Quartz has been identified in some samples of noritic gabbro and norite obtained from the Bluestone and Peng Seng quarries. A few flakes of reddish-brown biotite, probably secondary in origin, accompany the accessory quartz. Serpentinised crystals of olivine associated with many opaque grains have been identified in noritic gabbro samples from the Yun Onn Quarry.

Field Relations

The Gombak Norite lies on the western edge of the Bukit Timah Granite. Further north in Johor, similar gabbroic bodies lie as satellite bodies along the western margin or just within the northern correlatives of the Bukit Timah Granite. The Gombak Norite in Singapore has been penetrated by the Bukit Timah Granite, and the resulting dykes and zones of mixed rock associated with blocks of gabbro and norite at various stages of digestion can be seen in the granite (Fig. 2.7). The greatest mixing is seen in the quarries at the southern end of Bukit Gombak. Acid dykes have been observed in all quarries. A younger set of basic dykes, postdating the Bukit Timah Granite, also penetrates the Gombak Norite.



Fig. 2.7 Granite intruding norite and containing xenoliths of norite at the Peng Seng Quarry, Bukit Gombak

A zone of altered gabbro and norite characterised by hornblende gabbro has been delineated by Hutchison (1964). Such a rock is exposed behind some houses on the eastern side of Bukit Gombak (GR 404504). The rock looks deceptively like a coarse-grained, ferromagnesian-rich granodiorite, but it consists dominantly of labradorite and coarse, pale green hornblende. A few of the hornblende crystals pass through to an almost colourless fine-grained fibrous amphibole. Minor chlorite and sericite are also present (Fig. 2.8).



Fig. 2.8 Large plates of hornblende with augite cores in hornblende gabbro from Bukit Gombak (GR 404504) 45X Crossed Nicols

Age and Correlation

The Gombak Norite can be correlated with bodies of similar rock found along the western edge of the granite in the Malay Peninsula. Burton (1973 A) mapped several such bodies in Johor and named them the Linden Hill Gabbro. Hutchison (1973 C) listed other occurrences on the Tembeling River and Rompin River in Pahang, on Sungei Segamat and Sungei Simat in Negri Sembilan, and near Segamat in Johor.

The age relationship of the norite and the granite on Singapore Island is discussed in Hutchison (1964). The evidence available suggests that the norite body predates the granite batholith. This is suggested by the presence of many microgranite dykes cutting the norite. These dykes often contain numerous basic xenoliths. Elsewhere, partially assimilated rocks of basic origin can be found in the granite quarries. These represent true hybrid rocks and not intermediate varieties that have crystallised in-situ (Hutchison, 1964). Hutchison also pointed to the enrichment of hornblende in the norite which he stated was caused by its reaction with a silica rich fluid which had invaded the basic body before the intrusion of the granite magma. This event has caused the replacement of the pyroxene in the norite by hornblende without altering the texture.

Isotopic ages of the Bukit Timah Granite range from 210 to 230 million years. The upper limit of the range provides a minimum age for the emplacement of the Gombak Norite. Hutchison (1973 C) suggested that the norite may have represented part of the ophiolite suite of the Lower Palaeozoic trench situated to the west of a geosyclinal trough in which the Sajahat Formation was deposited.

An alternative hypothesis is that the Gombak Norite represents a part of the basic differentiate of the magma which gave rise to the Bukit Timah Granite, and thus the age of the norite is associated with a Permian to an early Triassic event.

PALAEOZOIC VOLCANICS

Definition and Distribution

Only three exposures of the Palaeozoic Volcanics were seen. Two exposures are on reefs to the northeast of Pulau Tekong and one very weathered exposure is at the eastern end of Pulau Tekong. The exposures were small, and the rocks showed differences in general texture. Hence, no formal name or type area is proposed for these rocks. The formation has also been mapped as forming the reefs in Selat Johor and the eastern area of Pulau Tekong about 2 km inward from the shore.

Content

On the unnamed reef in Selat Johor (GR 745588), the Palaeozoic Volcanics occurs as a massive volcanic agglomerate crossed by well-developed sets of vertical joints striking at southeast and north. The rock fragments embedded in the agglomerate are usually small, being less than 5 cm in diameter. On the second reef of Malang Si Ajar (GR 766576), the volcanics are represented by a tuff. Deep weathering at the third exposure at GR 762568 on Pulau Tekong has destroyed the parent lithological texture.

Only one thin section has been made from the volcanic agglomerate. The section shows a lithic tuff with andesitic fragments set in a groundmass of irregular feldspar laths, glass, and opaque ore. Diaspore is also present in high concentrations, particularly in the lithic fragments.

Field Relations

The Palaeozoic Volcanics is not seen in contact with any other rock type in Singapore unless tuffs that occur north of Kampong Salabin on Pulau Tekong, along the northwest coast of Pulau Tekong, and along the west coast of Pulau Tekong Kechil are considered as part of the formation. The tuffs are considered as part of the Sajahat Formation. The field relation between the Palaeozoic Volcanic and the Sajahat Formation and the criteria used to define the boundary for the two formations is discussed under the Sajahat Formation.

Age and Correlation

There is no direct evidence for the age of the Palaeozoic Volcanics in the area mapped. Grubb (1968) mapped andesite and tuff on the eastern side of Selat Johor through to the eastern coast of Malaya where it is intruded by granite. He recognised two types of tuff which he termed 'ashy tuff' and 'agglomeratic tuff' and a metamorphosed zone extending up the eastern side of Selat Johor in which pyrophyllite and diaspore occur as alteration products in the tuff. The Palaeozoic Volcanics is mapped here as an extension of Grubb's 'Volcanic rocks'. Grubb considered the volcanic rocks of andesite and tuff as correlatives of the Pahang Volcanic Series. The series was given a Permian-Carboniferous age in Hutchison (1973 B). From Hutchison's (1973 A) palaeo-tectonic synthesis, the Palaeozoic Volcanics would appear to represent the volcanic arc rocks to the southwest of the subduction zone in the Natuna islands.

If it is assumed that the granite intruding the andesite is the same as that at Bukit Lanchu in Johor and at Bukit Timah in Singapore, the age of the andesite and tuff can be restricted to the Permian-Carboniferous. The granite was dated as having been emplaced approximately 220 million years ago (Hutchison, 1973 C). Grubb (op cit) considered the tuff to be younger than the andesite but made no suggestion of any time interval between the two. Hence, the Permian-Carboniferous age cannot be refined further for volcanic agglomerate and tuff found in eastern Singapore.

BUKIT TIMAH GRANITE

The term granite is used here in a general sense for the entire suite of acid rocks including granite, adamellite, granodiorite, and the acid and intermediate hybrids (mainly of granodioritic and dioritic composition) which resulted from the assimilation of basic rock within the granite.

The Bukit Timah Granite for the purposes of discussion here is divided informally into two groups. The first group, the Central Singapore Granite, lies to the west of a line running from Siglap to Punggol; the second group, the Pulau Ubin Granite, lies to the east of the line. The Pulau Ubin Granite appears to be richer in ferromagnesian minerals, and it might have reached a more advanced stage of hybridisation. To date, there is insufficient data to separate the two groups formally, and thus they are not distinguished on the map sheets.

Chemical analyses of rocks of the Bukit Timah Granite and of inclusions and dykes within the granite are given in Appendix 1.

CENTRAL SINGAPORE GRANITE

Distribution

The Central Singapore Granite (BT, BTb, BTh, BThy, BTgdt) occupies an area extending some 8 km in a northerly direction and 7 km in a westerly direction in the centre of Singapore Island where it forms hills and valleys of both high and low relief. Over most of the area, the hills are less than 60 m high. The granite near its western contact with other formations forms steeper and more prominent hills that reach a maximum height of 166 m at Bukit Timah Hill. Less extensive areas of the Central Singapore Granite occur in the vicinity of Braddell Heights Estate, Woodleigh Park, and around Seletar Airport beside Lower Seletar Reservoir in South Seletar.

The Central Singapore Granite is well represented by the rocks exposed in nine quarries which were once located on the slopes of prominent hills to the east of the Gombak Norite. The Seng Kee, Housing Development Board (HDB), Singapore Granite (Mandai), and Public Works Department (PWD) (Mandai) quarries were established on hills in Mandai. Quarries established on hills in Bukit Timah were the PWD Rural Depot, Singapore Granite (Bukit Timah), Hindhede, and Sin Seng. Swee Construction was established in Bukit Batok. Samples taken show that granite (in its restricted sense) is the most common rock found in all quarries excepting Seng Kee, HDB, and Swee Construction (Bukit Batok) where granodiorite is predominant. Porphyritic microgranite was sampled from the Singapore Granite (Mandai), PWD Rural Depot, and Sin Seng quarries. Adamellite was located only in the PWD (Mandai) Quarry. All nine quarries are now defunct, and they have long since been reclaimed for building and recreational uses.

Away from the quarries, outcrops of the Central Singapore Granite are rare. The presence of granite is being indicated by weathered residual boulders particularly on the undulating terrain around north of Mandai Road and by the cores of granitic rocks obtained from boreholes. Boulders of granodiorite have been discovered along Marsiling Road in Woodlands (GR 428591) and in excavation near the coast in Kranji (GR 396571) between Kranji Reservoir and Woodlands Road. Boulders of hornblende-biotite granite have been found in a small abandoned quarry off Mandai Road (GR 461568) and near Tagore Industrial Estate (GR 481531). The small quarry off Mandai Road is assumed to be the quarry described by Scrivenor (1924) as yielding granite with abundant dark mica, hornblende (which sometimes showed a core of pyroxene), feldspar, and a fair amount of quartz. Pink-spotted, fine-grained adamellite rock chips have been recovered from a shallow drillhole at a depth of 6 m at location GR 491540 near Bullion Park in Upper Thomson - Yio Chu Kang and about 1 km west of the contact with the Old Alluvium.

Hybrid rocks associated with the Central Singapore Granite are well exposed in the vicinity of the gabbro-granite contact. Near this contact, numerous xenoliths varying in their stage of hybridisation can be found in microgranite dykes. In the Swee Construction Quarry at Bukit Batok, hybrid inclusions of quartz diorite occur in granodiorite. These inclusions are spotted with dark minerals and are finer grained than the coarse-grained granodioritic host rock.

Alexander (1950) first recorded hybrid rocks of dioritic composition on Bukit Panjang, Bukit Gombak, and on the site of the municipal flats at Monk's Hill (GR 495450). Subsequently, removed for reclamation, the rocks could not be relocated at Monk's Hill. This site is located near Newton Circus. The other occurrences on Bukit Panjang and Bukit Gombak were confirmed.

The thickness of residual soils derived from weathered Bukit Timah Granite ranges from a few metres to 70 m and the average thickness is 30 m (Zhou, et al. 1993 A). They also observed that the granite rock head generally follows the ground surface elevation and that the completely weathered granite rapidly becomes slightly weathered granite with the heavily to moderately weathered intermediate zones often missing or thin. Such weathering profiles are especially common in low-lying areas (Zhou, et al. 1993 B). Zhou (2001) also documented for the first time in Singapore three deep weathered trenches reaching to depths of 40, 44 and 80 m in the Bukit Timah Granite at the Mandai Quarry site (Fig. 2.9). They share similar orientations of the major sub-vertical joints present at the site. The trenches are sited below ponded valleys and ravines. Strips of densely spaced joints with relatively softened infill material are also conspicuous features on the quarried faces at the Mandai Quarry site (Fig. 2.10). They are generally sub-vertical and sparsely distributed at intermittent distances of less than 5 m.



Fig. 2.9 Deep weathered trench (now covered with lean concrete) at the Mandai Quarry site

Content

Granite, adamellite, and granodiorite together with various hybrid rocks are found in the area. Their boundaries have not been mapped because it is difficult to do so due to the limited extent of the outcrops and to gradational variations in the distribution of quartz, alkali-feldspar, sodic plagioclase, and the ferromagnesian minerals in these rock types.

Zhou (2001) reported that granite, adamellite, and granodiorite are present at the Mandai Quarry site.



Fig. 2.10 A strip of densely spaced joints at the Mandai Quarry site

Granite

The granite is generally light grey and medium-grained with grain sizes measuring from 3 to 5 mm. The main minerals can be distinguished easily by the naked eye. Quartz, which often accounts for 30% of the minerals present, has a glassy grey appearance and a rough surface. It occurs interstitially to the feldspar crystals and has interlocking boundaries with them. Feldspar is the most abundant mineral in the sample, and the mineral often constitutes 60% to 65% of the rock. It is commonly cream in appearance, with the more weathered feldspars being pale to brownish yellow. The pink variety of orthoclase is present in granite at the PWD (Mandai) Quarry and in granite boulders at the Tagore Industrial Estate. Biotite and hornblende, which make up the remaining constituents, are easily recognised by their dark brown colour and their cleavage.

Microscopically, the granite consists predominantly of quartz and varying amounts of alkali-feldspar and acid plagioclase with subordinate amounts of biotite and hornblende.

Quartz occurs mainly as anhedral grains interstitial to the feldspar and the ferromagnesian minerals. It occurs less frequently as excellent lobate intergrowths

in alkali feldspar. Such texture is only well developed in granite and porphyritic microgranite samples from the Hindhede and Sin Seng quarries.

The alkali feldspar is frequently a microperthitic orthoclase. It occurs as anhedral or rarely as subhedral grains. The exsolution texture of the mineral is best seen in the less altered grains of the granite from the Hindhede and Sin Seng quarries. The remaining orthoclase is extensively clouded with very fine-grained opaque dust, but in rocks obtained outside the Hindhede and Sin Seng quarries, the orthoclase is comparatively lightly affected.

The plagioclase is somewhat variable in composition ranging from oligoclase to andesine. Oligoclase is the more common variety. Andesine is present in the porphyritic microgranite from the Singapore Granite (Mandai) Quarry. In the Seng Kee Quarry, the plagioclase is albitic in composition (Hutchison, 1964). Crude oscillatory zoning of the mineral has been observed in the porphyritic microgranite of the Singapore Granite (Mandai) Quarry and in the pink hornblende-biotite granite from Tagore Industrial Estate. One of the plagioclase phenocrysts present in the porphyritic microgranite from the former locality contains small crystals of hornblende. A ring of sericite flakes has formed as a replacement after a zone within the feldspar phenocryst (Fig. 2.11). The porphyritic microgranite was possibly derived from basic rock during intrusion of the Bukit Timah Granite. Deuteric alteration of the plagioclase to calcite is infrequent, but sericitisation is evident in many of the rocks. Such alteration of the plagioclase is most widespread in the granite from the Hindhede and Sin Seng quarries. In the rocks from these quarries, the alteration has produced a very dense concentration of sericite in the cores of the plagioclase.

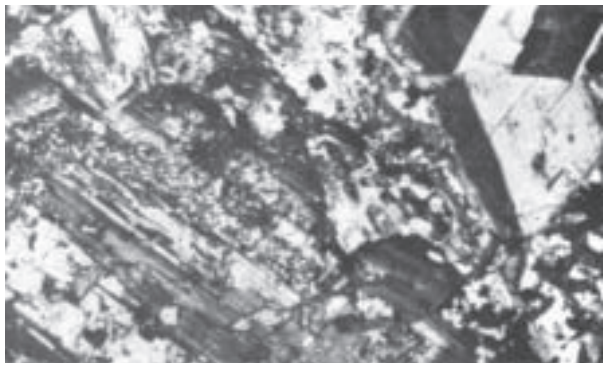


Fig. 2.11 Zoning in andesine. Sericite after a zone within the plagioclase in porphyritic microgranite from the Singapore Granite (Mandai) Quarry 45X Crossed Nicols

The most common ferromagnesian mineral present is reddish-brown biotite with hornblende occurring in subordinate amounts. Much of the biotite is in part chloritised. The most severely altered ferromagnesian minerals are to be found in the granite from the Hindhede Quarry and in the granite and porphyritic microgranite from the Sin Seng Quarry. In these rocks, the biotite has been replaced around the rims and along the cleavage by a mass of unidentified brown cloudy material, sericite, and chlorite. In addition, muscovite may occur as pseudomorphs after biotite. The occurrence of secondary muscovite as pseudomorphs of biotite and the extensive alteration of microperthitic orthoclase, oligoclase, and biotite in the rocks from the Hindhede and Sin Seng Quarries may have been caused by an episode of mineralisation. Mineralisation was also responsible for the emplacement of veins of quartz, molybdenite and pyrite, and probable bornite in these two quarries.

The usual accessory minerals in the granite are apatite and zircon. A few grains of epidote are found associated with calcite in oligoclase in the porphyritic microgranite of the Singapore Granite (Mandai) Quarry. Epidote in biotite has also been identified in the granite at Tagore Industrial Estate.

Adamellite

In the field, adamellite is indistinguishable from granite and thus cannot be mapped separately. It has been recognised in thin-section, and thus the rock warrants a separate description. It is distinguished from granite if between one-third and two-thirds of its feldspar is alkali.

The adamellite from the PWD (Mandai) Quarry is characterised by blobs of quartz in crudely developed myrmekitic intergrowths. The rock chips of adamellite from a drillhole near Bullion Park consist predominantly of quartz and sericitised oligoclase with lesser amounts of clouded alkali feldspar and green chloritised biotite. The latter is associated with some black opaque grains.

Granodiorite and Diorite

There are gradational changes from granodiorite to diorite. Rock with an alkali feldspar content of less than one-third of its total feldspar content is a granodiorite.

It is believed that the granodiorite (BTgdt) is a product of hybridisation of granite and basic rock. This belief

is supported by field evidence in many quarries where basic rock can be seen in different stages of assimilation within the granite (Fig. 2.12).



Fig. 2.12 Rounded, well-assimilated, basic inclusions in adamellite at the PWD (Mandai) Quarry

In the granodiorite from the Seng Kee and HDB quarries, much of the biotite is in part chloritised. Similar rock from Marsiling Road consists of anhedral quartz, clouded microperthitic orthoclase, sericitised plagioclase of oligoclase to andesine composition (some of which exhibit zoning), and subordinate amounts of well-developed, brown biotite flakes which are often chloritised in part.

Granodiorite from the location at GR 396571 in Kranji contains quartz and plagioclase of oligoclase to andesine composition. The crystals are often crudely anhedral in shape, and the feldspar has been slightly sericitised. Orthoclase is present in subordinate amounts and appears interstitially. Green hornblende dominates over partially chloritised brown biotite.

In hand specimen, the granodiorite along Marsiling Road and Kranji appears darker than the specimens from the Seng Kee and HDB quarries, and the rock is seen in thin-section to contain more ferromagnesian minerals.

The rock near Newton Circus has been described in Alexander (1950) as a diorite consisting of xenomorphic quartz grains containing inclusions of oligoclase, biotite and amphibole, and minute needles which were thought to be apatite. She stated that the larger quartz grains did not carry large inclusions and that these grains were usually rimmed with indistinct bands of small feldspar, biotite, and amphibole crystals.

The granodiorite rock in the Swee Construction quarry at Bukit Batok consists of sericitised andesine, quartz, subordinate orthoclase, red-brown biotite occurring usually in clusters, and locally fibrous, pale green amphibole. Hutchison (1964) noted that this granodiorite was foliated. He delineated a zone of foliation adjacent to and east of the altered zone of gabbro and norite.

The granodiorite from the Mandai Quarry site shows micro-fracturing (Fig. 2.13 and Fig. 2.14) and straining of its minerals (Fig. 2.15). These features indicate that the Bukit Timah Granite had been subjected to stress in the geologic past (Lands & Estates Organisation, Ministry of Defence, 1999).

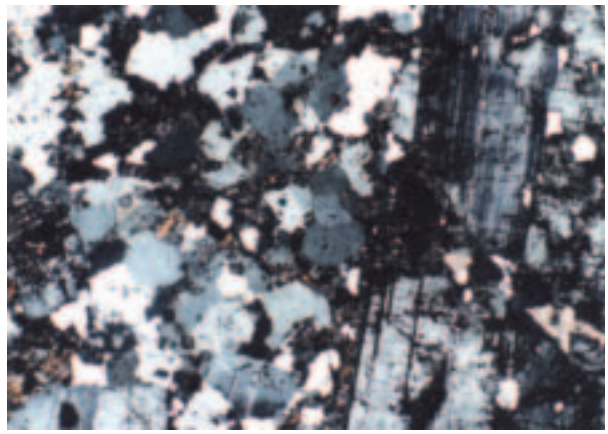


Fig. 2.13 Micro-fractured plagioclase phenocryst in granodiorite from the Mandai Quarry site. Large elongated plagioclase was cut and displaced by two micro-fractures. On the left of the plagioclase are interlocking quartz and feldspars. Small yellow minerals are biotite. 20X Crossed Nicols

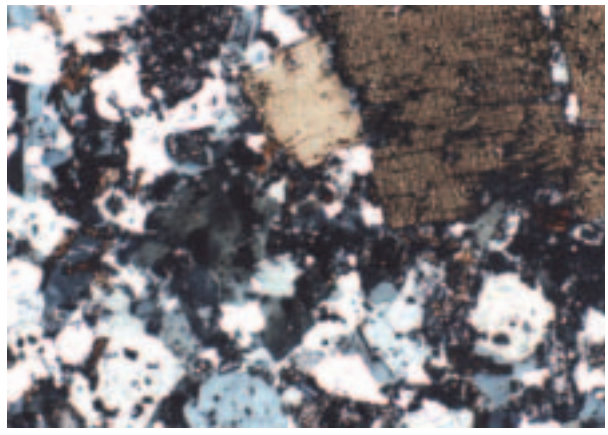


Fig. 2.14 Micro-fractured biotite phenocryst in granodiorite from the Mandai Quarry site. Large speckled brown biotite at top left had an early fracture which has been in-filled by quartz. The groundmass consists of interlocking feldspar and quartz. Small brown grains are biotite. 20X Crossed Nicols

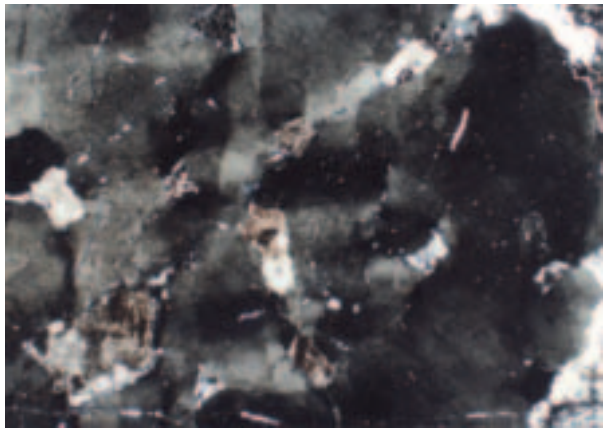


Fig. 2.15 Strongly strained quartz phenocryst in granodiorite from the Mandai Quarry site. The quartz is strongly strained as indicated by its undulose extinction. It is also fractured and these fractures have been in-filled with feldspar and biotite. 20X Crossed Nicols

Field Relations

The contact at its western edge between the Central Singapore Granite and the Gombak Norite is not exposed. However, there is sufficient evidence within quarries in the noritic body to establish the fact that the granite is younger. This has been discussed under Gombak Norite.

The contact between the granite and the Triassic sediments is also not exposed. Alexander (1950) described a core boulder of granite, located about 60 cm from the granite boundary, which she interpreted as residual material from an intrusive dyke. It is suggested here that the relict boulders observed by Alexander was not in-situ in a dyke but were deposited in weathered granitic material at the base of the Triassic sedimentary pile.

INCLUSIONS IN THE CENTRAL SINGAPORE GRANITE

Definition and Distribution

Inclusions are basic xenoliths with distinguishable boundaries that can be seen floating in the granitic rock.

Numerous ovoid-shaped inclusions which appear darker than the granitic host rock have been observed in the PWD (Mandai) Quarry (Fig. 2.12) and Sin Seng Quarry and in residual boulders remote from the granite-gabbro contact at Kranji (GR 396571) and Woodlands (GR 428591).

Composition

Inclusions of hybrid rock from the Swee Construction Quarry at Bukit Batok contain euhedral plagioclase (heavily sericitised in places) in a groundmass of anhedral quartz and numerous reddish brown biotite flakes. The biotite, tending to be considerably coarser grained than the feldspar and quartz, is widely altered to chlorite and associated opaque grains.

The faint ovoid inclusions found in the PWD (Mandai) and Sin Seng quarries have been referred to as quartz monzonite by van Bemmelen (1940) and Hutchison (1964).

DYKE ROCKS ASSOCIATED WITH THE CENTRAL SINGAPORE GRANITE

Classification and Distribution

Dykes associated with the Central Singapore Granite are divided into two groups: one with acid (Dd) and the other with basic affinities (Db).

There are four varieties of acid dykes: microgranite, granite porphyry, granophyre, and trachyte. The microgranite variety forms the most conspicuous dykes and is common in the norite at the Chia Oh Kang, Peng Seng, and Lian Hup quarries. Far less conspicuous than the microgranite are the granophyre dykes that cut the norite body in the Peng Seng, Chia Oh Kang, and Yun Onn quarries that and the granodiorite in the Swee Construction Quarry at Bukit Batok. The granophyre is only slightly lighter in colour than the dolerite and it may therefore be mistaken for dolerite in the field. Granite porphyry dykes (Dp) have been found only in

the Hindhede and Singapore Granite (Bukit Timah) quarries where they cut the Central Singapore Granite. Trachyte dykes (Dt) have been recognised cutting granodiorite in the Swee Construction Quarry in Bukit Batok. Like the granophyre, the trachyte also looks deceptively like dolerite in hand specimen.

Basic dykes, mainly dolerite (Dd), have been observed cutting the Gombak Norite in the Bluestone, Poh Hua, Chia Oh Kang, Chua Chai Seng, and Lian Hup quarries and cutting the Central Singapore Granite in the Singapore Granite (Mandai), Swee Construction (Bukit Batok), and PWD (Mandai) quarries. One dyke in the PWD (Mandai) Quarry has been identified as a spessartite.

Further north, a set of deeply weathered dykes is exposed in an excavation at Admiralty Road West (GR 450611) between Sembawang and Woodlands. The dykes are weathered to deep-red clay in contrast to the pale brownish yellow, coarse sandy clay soil derived from their granite host (Fig. 2.16). Two other deeply weathered dykes, one in Whampoa Valley at location GR 495472 in Toa Payoh and the other just north of Upper Kallang Valley at location GR 503513 in Bishan, have been mapped along the eastern edge of the granite.



Fig. 2.16 Weathered basic dyke in granite soil at Admiralty Road, (GR 450611)

A spessartitic intrusive (Dsp) in quartz-mica diorite was recorded in Seletar by Scrivenor (1931). He also recorded boulders of dolerite and quartz porphyry associated with andesitic ash near the junction of Orchard Road and Orchard Link in the city area (GR 490440). None of these rocks could be relocated.

Composition

Dykes of Acid Affinity

Microgranite is distinguished by its fine-grained light-coloured appearance and granite porphyry by its quartz phenocryts whose crystals can measure up to 2 mm in size.

The main minerals in the microgranite dykes are quartz, orthoclase, plagioclase of albite to oligoclase composition, and biotite. No hornblende has been identified. The orthoclase is heavily clouded, and the acid plagioclase is often crowded with individual sericite flakes. Many of the biotite minerals are chloritised, and epidote or sphene occurs occasionally.

The granite porphyry dykes, found intruding the granite at the Hindhede and Singapore Granite (Bukit Timah) quarries, have a distinctive porphyritic appearance in the field. Phenocrysts of quartz and acid plagioclase up to 2 mm long are set in a fine-grained groundmass consisting of clouded orthoclase and feldspar, quartz, and subordinate muscovite (Fig. 2.17). At its contact, the dyke rock shows excellent micrographic intergrowths between feldspar and quartz. The dyke in the Singapore Granite (Bukit Timah) Quarry is more severely altered than the ones in the Hindhede Quarry. It contains more phenocrysts of sericitised acid plagioclase and numerous tiny flakes of muscovite in a finer-grained groundmass.

The granophyre is often feebly porphyritic. Under the microscope, it is seen to contain euhedral acid plagioclase that is often heavily sericitised and replaced by calcite. Its groundmass invariably consists of radiating structures of slender feldspar and quartz



Fig. 2.17 Phenocrysts of rounded quartz and sericitised acid plagioclase in granite porphyry from the Hindhede Quarry 45X Crossed Nicols

(Fig. 2.18). Small, irregularly shaped, chloritised green hornblende is present elsewhere in the groundmass.

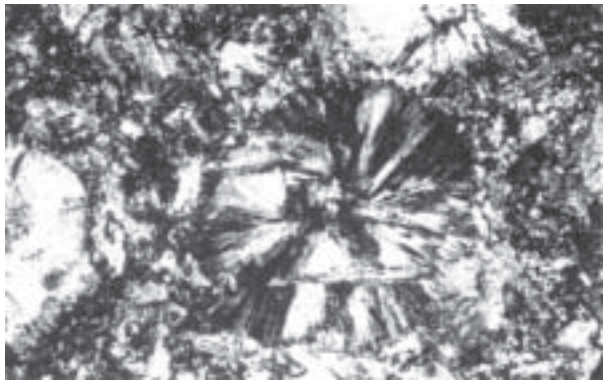


Fig. 2.18 Radiating structure of slender feldspar and quartz in granophyre from the Yun Onn Quarry, Bukit Gombak 120X Crossed Nicols

The trachyte in the Swee Construction Quarry at Bukit Batok looks deceptively like a dolerite in the field. Microscopically, this rock is composed of narrow, elongate prisms of clouded acid plagioclase and a few patches of quartz. A few of the tabular feldspar phenocrysts are heavily dusted with fine-grained alteration products, and there is widespread fine-grained chlorite with minor epidote along with a few sericite patches in the rock (Fig. 2.19).

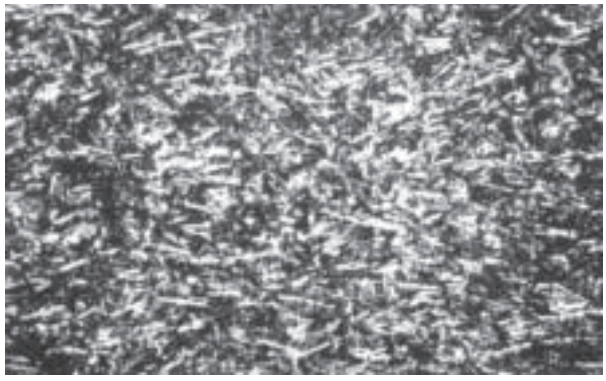


Fig. 2.19 Subparallel alignment of acid plagioclase laths in trachyte from the Swee Construction Quarry, Bukit Batok 45X Crossed Nicols

Dykes of Basic Affinity

The dolerite is dark green and possesses a fine-grained groundmass. It is generally porphyritic with phenocrysts of plagioclase and augite. The plagioclase ranges in composition from labradorite to andesine, and it is frequently altered to sericite. The groundmass is made up of altered small plagioclase laths, numerous small crystals of colourless augite, and abundant calcite (Fig. 2.20). Ilmenite, sphene, magnetite, epiodote, and sericite are common secondary minerals.

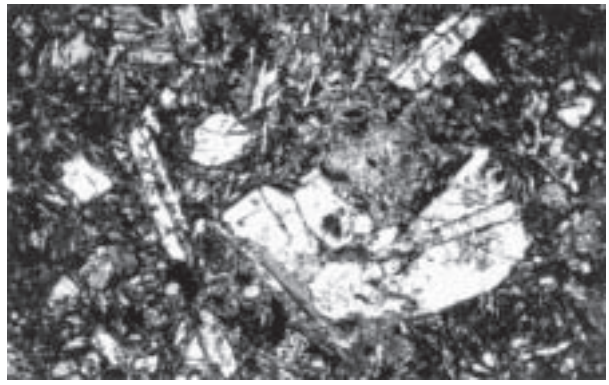


Fig. 2.20 Altered calcic plagioclase, smaller augite crystals, and occasional quartz in altered dolerite from the Swee Construction Quarry, Bukit Batok. The habit of augite is in marked contrast to that of the younger dolerite cutting the Jurong Formation at Pulau Senang. 45X Plain Polarised Light

Phenocrysts of augite and plagioclase occur in dykes found in the Singapore Granite (Mandai) and Bluestone quarries. Dyke rocks in the former quarry have also olivine in their groundmass.

In the more severely altered dolerite present in the Poh Hua, Lian Hup, Swee Construction, and Chua Chai Seng quarries, most of the augite has been replaced by pale green actinolitic amphibole.

The spessartite from the PWD (Mandai) Quarry possesses a panidiomorphic texture typical of lamprophyres. It contains numerous well-formed small crystals of augite and small flakes of red-brown biotite, all set in a base of larger irregular crystals of sericitised andesine. There are numerous irregular patches of calcite and numerous tiny needles of apatite (Fig. 2.21).

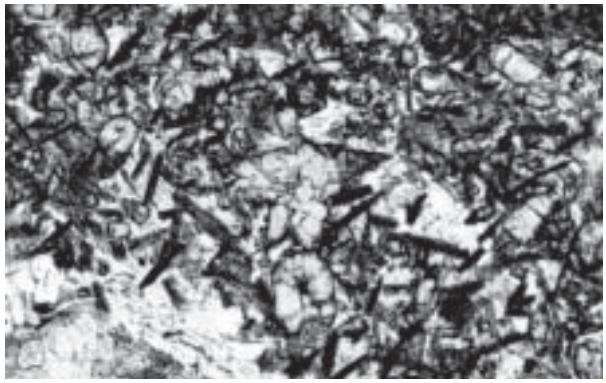


Fig. 2.21 Crystals of augite, slim laths of biotite, and patches of calcite in spessartite from the PWD (Mandai) Quarry 45X Plain Polarised Light

Field Relations

Relations between the microgranite, granophyre, and dolerite dykes were seen in the Chia Oh Kang and Peng Seng quarries. A twenty-five-metre-wide microgranite dyke intrusive into norite could be followed from the Chia Oh Kang Quarry through to Peng Seng Quarry. In the Peng Seng Quarry, a granophyre cuts this microgranite as a sinuous, two-metre-wide dyke. In the Chia Oh Kang Quarry, a dolerite dyke cuts the microgranite (Fig. 2.22).



Fig. 2.22 Dolerite cutting a twenty-five-metre-wide microgranite dyke in norite at the Chia Oh Kang Quarry, Bukit Gombak. The yellowish-brown colouration is a stain on the acid dyke and the norite

In the Chua Chai Seng Quarry, a set of dolerite dykes appears to be mapped as terminating against a granite dyke. This feature is apparent. The granite dyke has been undercut from the north and the dolerite is exposed at a lower level than the granite on the south side. A fault along the northern side of the granite, shearing the granite, has removed the dolerite from the north. A conspicious microgranite dyke cutting into norite can be observed at Little Guilin, a lake which was once a group of quarry pits (Fig. 2.23)

No observations were made within the Central Singapore Granite on the relative ages of the trachyte and granite porphyry.



Fig. 2.23 Microgranite cutting norite at Little Guilin, Bukit Gombak

PULAU UBIN GRANITE

Distribution

The Pulau Ubin Granite (BT, BTb, BTh, BThy, BTgdt), the second group of the Bukit Timah Granite, occurs to the east of the line running from Siglap to Punggol.

In Changi on Singapore Island, the Pulau Ubin Granite was found outcropping along the beach from Fairy Point to Changi Jetty (Fig. 2.24). Public Utilities Board Test hole No. 4, a deep borehole through the Old Alluvium in Pasir Ris near Loyang Avenue (GR 633515), penetrated the Bukit Timah Granite at -54 m. Adamellite was also found in the test hole at -76 m. Other outcrops occur on Pulau Sekudu and Pulau Ubin.



Fig. 2.24 Outcrop of granite near Fairy Point at Changi (GR 646539)

Hybrid granodiorite was also found in the Lian Moh, Lip Seng, and Gim Huat quarries. These quarries were operating on Pulau Ubin. Granodiorite has been sampled from a low hill (GR 628557) to the northwest of Sungei Jelutong and from an outcrop (GR 638557) at the headwater of Sungei Pulau Ubin.

Content

Granite

The Pulau Ubin Granite shows some distinctive differences when compared with the Central Singapore Granite. Hand specimens of rocks from Pulau Ubin are generally richer in green hornblende and brown biotite that characteristically occur in clusters. Under the microscope, hornblende and biotite are the dominant ferromagnesian minerals, and they occur as rather small irregular crystals marked with opaque spots of probable magnetite granules. Microperthitic texture in the orthoclase and heavy sericitisation of the plagioclase is prominent. Chloritisation of the biotite is comparatively rare in the Pulau Ubin Granite. The occurrence of allanite, pyroxene grains, and abundant magnetite granules as accessory minerals in the Pulau Ubin Granite is yet another distinctive feature.

The granite on Pulau Sekudu is distinguished from the others by its strongly porphyritic texture with pink euhedral orthoclase crystals, up to 4 cm in length, set in a coarse-grained groundmass. As in the other granites, this rock also contains clusters of ferromagnesian minerals.

Adamellite

The adamellite consists predominantly of quartz and clouded orthoclase in graphic intergrowth. Oligoclase occurs as euhedral phenocrysts (up to 4 mm in length) and also as subhedral to anhedral forms in the groundmass. The feldspar has been only slightly sericitised.

The sole mafic mineral present is some irregular shaped, green hornblende. Much of the hornblende has been chloritised, and epidote is often associated with the altered hornblende.

Hybrid rocks

A granodioritic rock from the Lian Moh Quarry appears very similar in hand specimen to coarse-grained granite by containing numerous clusters of ferromagnesian minerals. In thin section, the rock shows a consistent mineralogy and texture. The light-coloured minerals are andesine (strongly zoned in places), interstitial quartz, and orthoclase (in occasionally large irregular plates) enclosing plagioclase grains. Some of the orthoclase crystals have irregular margins and may have been corroded. Other minerals are large flakes of brown biotite and numerous crystals of hornblende, and these two minerals are often associated in clusters. Many of the hornblende crystals have irregular cores of pyroxene, while a few crystals grade into pale amphibole (Fig. 2.25). Common accessory minerals are magnetite and apatite. Sphene is rare.

The rock obtained from the low hill at Kampong Jelutong is more aptly a quartz diorite, and it is probably gradational to granodiorite. Under the microscope, anhedral oligoclase and quartz occurring as phenocrysts are seen. Clouded orthoclase, ragged green hornblende, and brown biotite are present in the

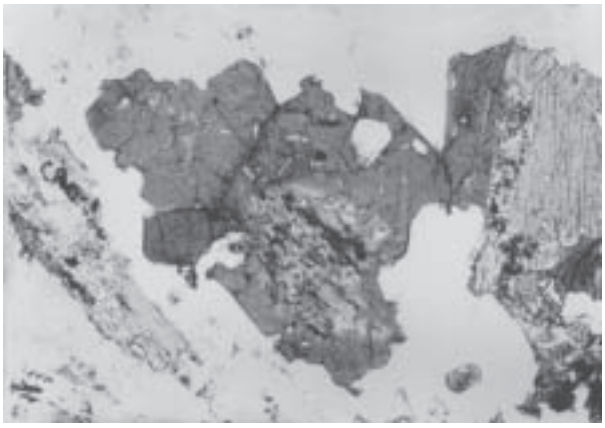


Fig. 2.25 Large plates of hornblende containing pyroxene in hybrid granodiorite from the Lian Moh Quarry, Pulau Ubin 45X Plain Polarised Light

groundmass. Biotite occurs as dense irregular clusters of smaller crystals, and they are accompanied by abundant tiny opaque grains. Magnetite, sphene, and apatite are common accessories. Precise comparison of this rock with the hybrid granodiorite obtained from the Lian Moh and Lip Seng quarries is uncertain. They appear to be similar in composition and show the same tendency for the ferromagnesian minerals to occur in clusters. However, pyroxene is not present in the quartz diorite.

The partially mixed hybrid rocks found in the HDB and Lee Hung Cheng quarries are similar to rocks from an area (GR 665530) about 1.3 km east-southeast from Fairy Point in Changi. Here, the granodiorite contains irregular aggregates of tiny flakes and prisms of biotite and hornblende associated with numerous opaque granules (Fig. 2.26). These dense clusters of tiny crystals

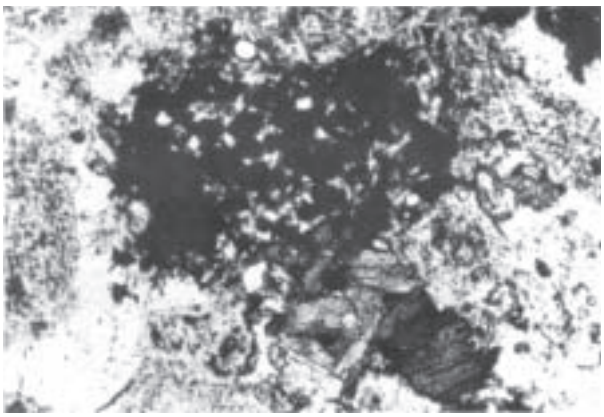


Fig. 2.26 Dense clusters of biotite, hornblende, and opaque grains of magnetite in granodiorite from location (GR 665530) in Changi 120X Plain Polarised Light

of biotite and hornblende are possibly remnants of original basic rocks now dispersed throughout the partly hybridised granodiorite. Other minerals present are albite, quartz, and orthoclase.

Field Relations

The Pulau Ubin Granite is not seen in contact with the Sajahat Formation. It is believed however that the granite belongs to the same intrusive suite as the Central Singapore Granite. A number of discrete bodies of older rocks were seen incorporated within the Pulau Ubin Granite, and their relationships are discussed in the next section.

Alexander (1950) noted that the granite from Pulau Sekudu is not metamorphosed whereas she considered Pulau Ubin to be. She suggested that the porphyritic granite of Pulau Sekudu is therefore younger and that it may be responsible for the metamorphism she recognised in the igneous rocks of Pulau Ubin.

INCLUSIONS IN THE PULAU UBIN GRANITE

Definition and Distribution

Inclusions have been found throughout the granitic and granodioritic host rocks on Pulau Ubin and at Changi. They vary greatly in size from less than a centimetre in diameter to tens of metres across. The large inclusions or rafted bodies (R) had rendered geological interpretation particularly difficult in the past. When rectangular-shaped inclusions of foreign rock were exposed without a break in their continuity on a quarry face, they had been mistaken for dykes or even as the host rock.

Some inclusions and dyke-like bodies exposed at different faces of the HDB Quarry show identical texture and mineralogy, and they undoubtedly originated from the same parent material. These dyke-like bodies are often very steeply dipping. Although this orientation is typical of dykes, they can be recognised as inclusions as the adjacent bodies of these masses can be seen to fit together if the intervening host granitic material between the adjacents bodies were to be removed (Fig. 2.27).



Fig. 2.27 Dyke-like inclusions at the HDB Quarry, Pulau Ubin

Composition

The variations in mineralogy and texture found in the inclusions have given rise to a diversity of rock names proposed by previous investigators. This is inevitable as the inclusions represent various stages of modification by an acid magma.

For the purpose of discussion, the inclusions have been classified here into two groups according to the presence or absence of pyroxene minerals in them. Within each group, two textural divisions are recognised: one showing a hornfelsic texture and the other without. Thus, inclusions with pyroxene minerals may possess a hornfelsic texture (Rphf) or a non-hornfelsic texture (Rp). Similarly, inclusions without pyroxene minerals may have a hornfelsic texture (Rnphf) or a non-hornfelsic texture (Rnp).

Most of the inclusions seen within the granite on Pulau Ubin and at Changi represent bodies of basic rock. Assimilation has, in some instances, continued until the granitic hybrid magma only carries inclusions of small clusters of ferromagnesian minerals (Fig. 2.28). Some inclusions may represent Palaeozoic sediments. The possible origin of these inclusions is inferred from the presence of garnet and the absence of pyroxene in the inclusions. Some inclusions might have been derived from andesites. These inclusions have biotite with quartz and an assemblage of plagioclase-hornblende-diopside that is often associated with minor magnetite, apatite, and sphene.

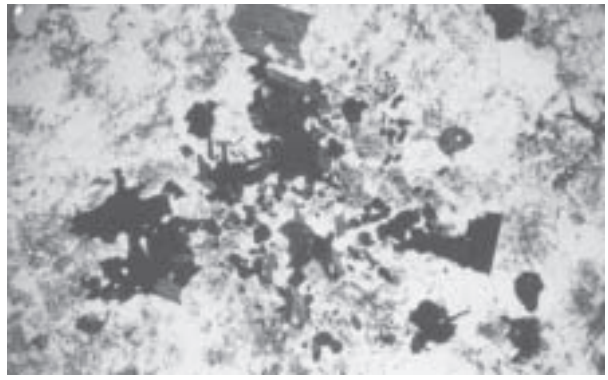


Fig. 2.28 Clusters of hornblende and biotite associated with accessory allanite in biotite-hornblende granite from Kampong Mamam, Pulau Ubin
45X Crossed Nicols

Inclusions of Rocks with Pyroxene

The inclusions that contain pyroxene minerals (Rp, Rphf) are characterised by the presence of augite with or without enstatite or hypersthene, plagioclase

of oligoclase to andesine composition, orthoclase, quartz, hornblende, and biotite in varying amounts. Phenocrysts of plagioclase are fairly common and augite phenocrysts relatively rare. Numerous green hornblende and brown biotite are present (generally as well-developed crystals), but in rocks that show a granoblastic texture, the hornblende and pyroxene frequently appear as small prismatic crystals or as needles and the biotite often occurs as small flakes or in clusters (Fig. 2.29). Quartz and orthoclase are abundant only in the hornfelsic varieties. The common accessory minerals are magnetite, apatite, and sphene.

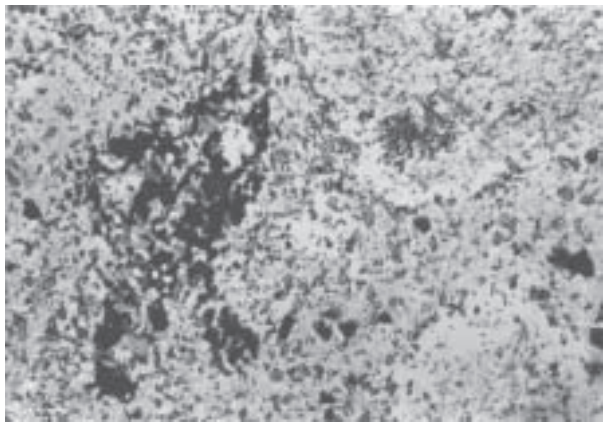


Fig. 2.29 Numerous stubby prisms of pyroxene grains and dense clusters of brown biotite in hypersthene hornfels from Kampong Jelutong, Pulau Ubin
45X Plain Polarised Light

Inclusions of Rocks without Pyroxene

Inclusions that do not contain pyroxene minerals (Rnp, Rnphf) are similar in hand specimen to those that do. They generally consist of a fine mosaic of quartz and feldspar as anhedral grains, varying amounts of small prisms of green hornblende, and many tattered, brown flakes of biotite. The ferromagnesian minerals occur as individual crystals or associated in clusters. The occasional phenocrysts are usually of quartz and feldspar with irregular crystal outlines. Common accessories are magnetite, apatite, and sphene. Garnet occurs in some inclusions of this group (Fig. 2.30).

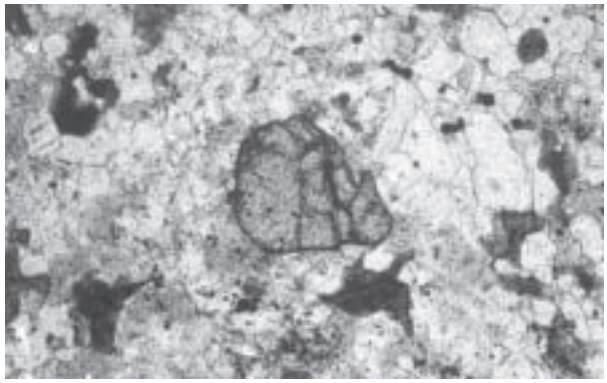


Fig. 2.30 Garnet, tattered biotite flakes, and hornblende in garnet-biotite hornfels from the Gim Huat Quarry, Pulau Ubin
120X Plain Polarised Ligh

Field Relations

The presence of numerous dark inclusions and large rafted bodies of rocks of different shapes, dimensions, colours, and mineralogy in the granitic host rocks has made the geology of Pulau Ubin complex.

The dyke-like masses are often very steeply dipping. Although this orientation is typical of dykes, their mode of occurrence on Pulau Ubin may be explained by the incorporation of rectangular-shaped masses of basic rock in a relatively immobile acid magma. This setting is suggested by the linear arrangement of disjointed, rectangular-shaped inclusions that are frequently seen on Pulau Ubin. In such an arrangement, the opposite edges of adjacent inclusions can be seen to fit if the intervening granite material between the disjointed inclusions were to be removed (Fig. 2.31). Such a disjointed rectangular-shaped inclusion could easily be misinterpreted as a dyke if it appeared without fractures in the confines of its outcrop.



Fig. 2.31 Rectangular-shaped inclusions in granite at the HDB Quarry, Pulau Ubin

DYKE ROCKS ASSOCIATED WITH THE PULAU UBIN GRANITE

Distribution

Dykes can be seen cutting the Sajahat Formation on Pulau Sajahat and Sajahat Kechil and along the northeastern and eastern coast of Pulau Tekong Kechil. Two parallel dykes are also exposed at location GR 733544 surrounded by present-day beach sand on the southern coast of Pulau Tekong. On Pulau Ubin itself, dykes have been seen cutting granite at location GR 595577 about 0.5 km to the northeast of Tanjong Tajam and in Aik Hwa Quarry No 1.

Composition

Dykes of Acid Affinity

Scrivenor (1931) described acid dykes of aplite, acid granite, granophyre, and hornblende-granite-aplite on Pulau Ubin. He mentioned that they represent the youngest phase of igneous activity. The dykes were observed by him to cut older dykes of enstatite-spessartite, hornblende porphyry, and hornblende granophyre in the area. Veins of pyroxene-bearing microgranite have been recorded at the Lian Moh Quarry and Tanjong Jelutong in Scrivenor (1931).

An acid dyke has been mapped by the PWD Geological Unit as intrusive into Palaeozoic sediment on Pulau Sajahat Kechil where the dyke has been cut by a basic dyke. No thin section of this acid dyke was prepared.

Dykes of Basic Affinity

The basic dykes are generally dolerites and lamprophyres. Dolerite dykes are observed on Pulau Sajahat and Sajahat Kechil.

Most of the dolerite is porphyritic with phenocrysts of severely altered plagioclase. The groundmass is made up of less altered small plagioclase laths, numerous small crystals of colourless augite, and occasional quartz found interstitially to the plagioclase. The composition of the plagioclase ranges from andesine to labradorite. In the more severely altered dolerite, most of the augite is replaced by pale green actinolitic amphibole. Such amphibole is abundant in the dolerite dyke exposed in the south of Pulau Sajahat. Ilmenite, sphene, magnetite, epidote, calcite and sericite are very common secondary minerals found in the dolerite.

The north-trending dyke on the west coast of Pulau Sajahat is a porphyritic microdiorite. Its fine-grained feldspathic groundmass contains phenocrysts of plagioclase and a few corroded quartz crystals. There are no phenocrysts of ferromagnesian minerals, but biotite and hornblende are widespread either as isolated small crystals or in irregular clusters. Apatite and magnetite are the common accessory minerals. Inclusions of granite up to 20 cm in diameter were seen in the dyke on Pulau Sajahat.

Scrivenor (1931) described a quartz norite in the HDB Quarry on Pulau Ubin. These rocks have not been sampled, but they are likely to represent rafted basic bodies.

The occurrence of lamprophyre dykes on Pulau Ubin and Changi have also been described in Scrivenor (1931). He described dykes of enstatite-spessartite cutting the granite at Tanjong Balai, at Changi, and at the HDB Quarry on Pulau Ubin. According to Scrivenor, vogesites intrusive into granite were also present in Tanjong Balai and Changi. However, these intrusives have not been located in the field.

Field Relations

Although the numerous acid dykes and veins described by Scrivenor (1931) on Pulau Ubin and Changi have not been observed, the intrusion of such dykes after the emplacement of the main granite can be expected. A later intrusive phase of basic dykes, already observed in the Central Singapore Granite, is also recognised here. The association of acid and basic dykes on Pulau Sajahat and Pulau Sajahat Kechil suggests a similar temporal relationship for these dykes on Pulau Ubin and Changi.

Age and Correlation

Hutchison (1973 A) placed the Bukit Timah Granite with the East Coast Granites of Malaya and discussed the granitic associations of the Malay Peninsula and their correlations in that paper and in the 'Geology of the Malay Peninsula' (Hutchison, 1973 C). No further discussion on correlation is given in this text.

The age of the Bukit Timah Granite has been obtained by isotope dating techniques. Biotite from granite sampled at the PWD Rural Depot and the Sin Seng

quarries has K-Ar ages averaging 200 ±9 million years. Granite samples from Rural Depot Quarry yield Rb-Sr ages that range from 210 to 221 million years (Bignell, 1972). An early to mid-Triassic age is hence suggested for the Bukit Timah Granite. Unfortunately, no isotopic age has been obtained from the Pulau Ubin Granite to confirm a parallel age. It has already been mentioned that the granite batholith postdates the gabbro of Gombak Norite. Consequently, the formation of hybrid rocks has been recognised on Singapore Island and Pulau Ubin.

The porphyritic granite from Pulau Sekudu is included in the Bukit Timah Granite although it has a similar appearance to the late Cretaceous to early Tertiary granite bodies situated at several points along the western margins of the Main Range Granite in the Malay Peninsula (Professor N.J. Snelling, pers. comm).

MINERALISATION

Evidence of mineralisation has been found in the granitic rocks at Bukit Timah, Bukit Mandai, and Bukit Panjang.

In the Hindhede Quarry at Bukit Timah, a mineralised contact zone between the host granite rock and a dyke of granite porphyry contains numerous flakes of silvery molybdenite crystals up to 4 cm long and patches of other sulphide minerals in quartz.

At the eastern face of the Sin Seng Quarry, there are conspicuous, thin veins containing quartz, pyrite, and some bornite. These minerals are restricted to the middle of the veins.

Alexander (1950) described the occurrence of numerous small veins containing quartz, calcite, tourmaline as radiating needles, pyrite, molybdenite, and some cassiterite in the PWD (Mandai) Quarry.

Scrivenor (1910) reported that tin ore was found in quantities sufficient for working at Bukit Mandai. He also recorded the presence of cassiterite, molybdenite, chlorite, and calcite in the granite at Bukit Panjang.

No mineralisation has been observed on Pulau Ubin.

JURONG FORMATION

Scrivenor (1924) described the presence of quartzite and shale interbedded with limestone in the southern and western parts of Singapore, under the heading of 'Shale and Sandstone'. Alexander (1950) remapped these rocks as the 'Older Sedimentary Rocks'. She recognised three rock types in the 'Older Sedimentary Rocks', and these rock types are an Older Schist, an Argillaceous Series, and an Arenaceous Series. However, their distributions were not mapped.

Wong (1960) mapped the southwest coastal area of Singapore and carried out petrological and sedimentological studies on the sediments. He recognised three subdivisions, namely A, B, and C (in ascending stratigraphic order). Series A consisted of quartz conglomerate, sandstone, and shale. Series B was described as a bluish conglomerate together with sandstone and clay, and Series C was a sandstone-quartzite sequence overlying a conglomerate. He mapped a continuous anticline-syncline couplet extending northwest from Tanjong Berlayar in Telok Blangah to Sungei Pandan.

Leow (1962) extended the structural mapping of Wong (1960) over the whole of the western side of the island. From analyses of heavy minerals, he could find no support for Alexander's division of her 'Older Sedimentary Rocks' into the Arenaceous and Argillaceous units.

Chin (1965) mapped the Pasir Panjang-Jurong area and proposed two formations: the Jurong Formation and the younger Pasir Panjang Formation. His Pasir Panjang Formation was described as a rhythmic conglomerate-sandstone-mudstone sequence with a dominant reddish colour. The Jurong Formation was described as an interbedded mudstone-sandstone sequence of grey to black colour.

Burton (1973 A), who worked in southern Johor, used the term Jurong Formation informally for the whole of the Triassic sedimentary sequence there but stated that further work on the unit was necessary before a formal nomenclature could be established. He chose 'Jurong' as he considered the area in west Singapore to furnish the best data for the establishment of a formal unit, possibly of group status. He recognised two members within his informal unit in southern Johor, and these he named the Gunong Pulai Member and the Bukit Resam Clastic Member. In the 'Geology of

the Malay Peninsula', Burton (1973 B) briefly discussed Chin's (1965) subdivision but considered that there was insufficient evidence then to separate the two formations and also pointed to both being of the same age. He recast them as members of the Jurong Formation and incorporated Chin's Jurong Formation in his Bukit Resam Member.

Lim (1974), however, adhered to Chin's subdivision status but postulated an upper Triassic age for the Jurong Formation and a lower to middle Jurassic age for the Pasir Panjang Formation.

Vilpponen (1988) noted that the Jurong Formation is made up of fining-upward sequences at a variety of scales that range from a few to 80 m.

Fontaine and Lee (1993) studied 185 m of limestone cores recovered by the PWD in 1987 from a borehole at Pandan Road (GR 396436) near the southeastern corner of Pandan Reservoir (Appendix 4). They have designated the limestone as the Pandan Limestone. Since 1987, limestone has increasingly been found underground in west and southwest Singapore. Guo (1998) has described a variety of carbonate rocks in Singapore. In this book, the carbonate rocks in Singapore have been grouped into the Pandan Facies.

It is proposed here that seven facies be established to illustrate the variations seen in the Jurong Formation. They are the Queenstown Facies (Jq), Jong Facies (Jj), Ayer Chawan Facies (Jac), Pandan Facies (Jp), Rimau Facies (Jr), St. John Facies (Jsp), and the Tengah Facies (Jt). Although a general spatial and temporal relationship can be recognised for each facies member, no one facies can be assumed to form a continuous unit. Similar sediments could have been deposited at different times and as discrete bodies, because closely similar environmental settings could have occurred periodically throughout the time of deposition of the Jurong Formation. It is not uncommon to find evidence for the superposition of one facies on another at one locality and the reverse relationship at another locality. Extensive weathering had hampered accurate mapping. Further difficulties were encountered in establishing satisfactory parameters for mapping each facies member. Much of the mapping was done before the facies member concept was adopted. Moreover, the mapping was done by geologists using different parameters; consequently, it has not always been possible to establish with certainty which facies member a particular observation should be assigned to.

Fossil collections have been made in the Jurong Formation by Scrivenor, Alexander, Chin, Lim, and others (Appendix 3). However, no systematic collecting was carried out during our survey. Most of the fossils are marine molluses and are found in the Ayer Chawan facies or interbeds within it. The age of the fossils is generally accepted as late Triassic. A collection has also been made from the Pandan Facies at Pandan Road (Fontaine and Lee, 1993). The probable age of the fossils from this facies is late Triassic (Appendix 4).

Extensive areas in the Jurong Formation have been affected by dynamic metamorphism resulting from tectonic activity. The grade of metamorphism is low, and it is still possible to determine the facies that has been affected. The metamorphosed areas have thus been mapped over the facies, and the rocks within these areas are referred to informally as the Murai Schist, a name first introduced by Alexander (1950). Moe Sein (pers. comm.) remarked that extensive areas of the Jurong Formation that are located far away from the Murai Schist and the major faults have also been regionally affected by dynamic metamorphism. Petrographic studies of fresh rocks sampled from many deep boreholes showed that many of the rocks are foliated, recrystallised and have low-grade metamorphic minerals such as sericite, chlorite, and other micas. Cleavages can also be readily observed in hand specimens of fresh, fine-grained, and coarse-grained rocks. The occurrences of slate, phyllite, and schistose sandstone with pronounced foliation have often been reported from bored piling and tunnel projects. Despite bearing the imprint of metamorphism, the primary rock type can be readily identified. Metamorphic features are absent in weathered rocks sampled from outcrops and shallow depths as the features would have been destroyed by weathering processes.

From the interpretation of seismic records secured during a geotechnical feasibility study on rock cavern construction in the Jurong Formation, Redding and Christensen (1999) presented an alternative view on the depositional history of the Jurong Formation. They remarked that the contents of the Jurong Formation had been laid down in a generally non-marine molasse basin that was developed consequently from the collision of the Sibumasu and the East Malaya continental fragments in the late Permian to early Triassic times. They remarked that the lowermost stratum in the Jurong Formation consists of calcitic or calcareous siltstone. It is equivalent to the Queenstown Facies that forms the northern fringe of the formation where it abuts the Bukit Timah Granite. The stratum

of calcareous siltstone and the red mudstone of the Queenstown Facies were overlain by limestone of the Pandan Facies. Stable basin conditions existed for the deposition of the calcareous siltstone, the red mudstone, and limestone. Their deposition was disrupted by the uplifting of the landmass to the south of the Jurong basin caused by the emplacement of the Malaya Main Range Granite. With active uplift, renewed erosion provided coarse clastic materials to the basin. They were deposited as localised submarine fans and deltas that later constituted the Rimau and Jong Facies. With later pulsatory uplift of the Jurong basin, coarse to fine clastic materials that make up the Ayer Chawan Facies were deposited away from the build-up of the submarine fans and deltas. The St. John Facies were late-stage delta-top clastic materials and fore-delta turbidites, and they could be remobilised materials from unstable delta fronts. Deposition probably continued till middle to late Jurassic.

QUEENSTOWN FACIES

Definition and Distribution

A 'Mudstone Unit' was established in the Singapore Public Works Department, Geological Report No. 3 (1974). This unit is adopted here as the Queenstown Facies (Jq). It is a distinctive purple-red clay, clayey sand, silt, or fine sand with minor tuff. The facies member was best exposed in excavations at GR 443448 near Holland Village for the Queenstown housing project extensions. The black shale and yellow sandstone included in the Mudstone Unit are excluded from the Queenstown Facies.

The member can be found running in a northwesterly direction from Peak Island towards Selat Johor, close to the granite or Old Alluvium boundary. It occurs as interbeds down the west coast and in the Jurong area. It is also found as thin beds in the southwestern and western islands. Its greatest development is in the Queenstown-Bukit Merah area (Fig. 2.32). Sand is more common in the northwest, and silt with clay are more common in the southeast.

Content

The member consists predominantly of thinly bedded red and purple mudstone with some red to purple shale. The mudstone often has a white coating on the joint surfaces. Vugs up to 5 mm in diameter, which possibly contain pyrite, are common. Greenish stains are often found on the fracture planes. Massive red to purple mudstone is also common, and being closely jointed or sheared, it readily disintegrates to a mass of lenticular fragments about a centimetre long on drying. In the more sandy portions, sharply angular quartz dominates. The quartz is coated with haematite and goethite to give the rock the red-purple colour (Fig. 2.33). A thin section of red fine sandstone shows the presence of quartz, subordinate clouded feldspar, a few flakes of muscovite, and pale brown biotite. Tuffaceous material could be identified in some of the coarser-grained rocks in this member, particularly in rocks from the west and northwest. This tuffaceous volcanic material may have also contributed to the red colouration of the Queenstown Facies.



Fig. 2.32 Exposure of the Queenstown Facies showing massive character of purplish red mudstone along Jalan Bukit Merah, Bukit Merah

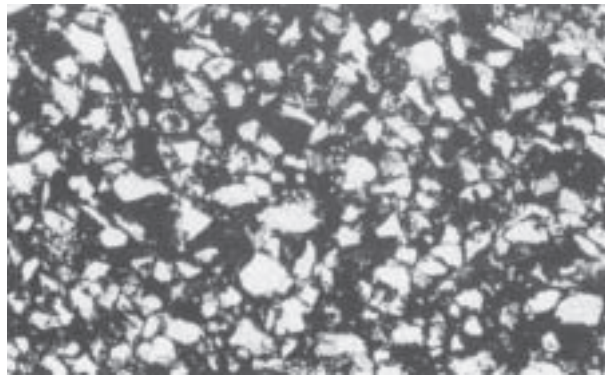


Fig. 2.33 Fine-grained red sandstone of the Queenstown Facies at GR 467434 near Kay Siang Road, Queenstown, showing very angular quartz grains in a limonitic clayey matrix 120X Plain Polarised Light

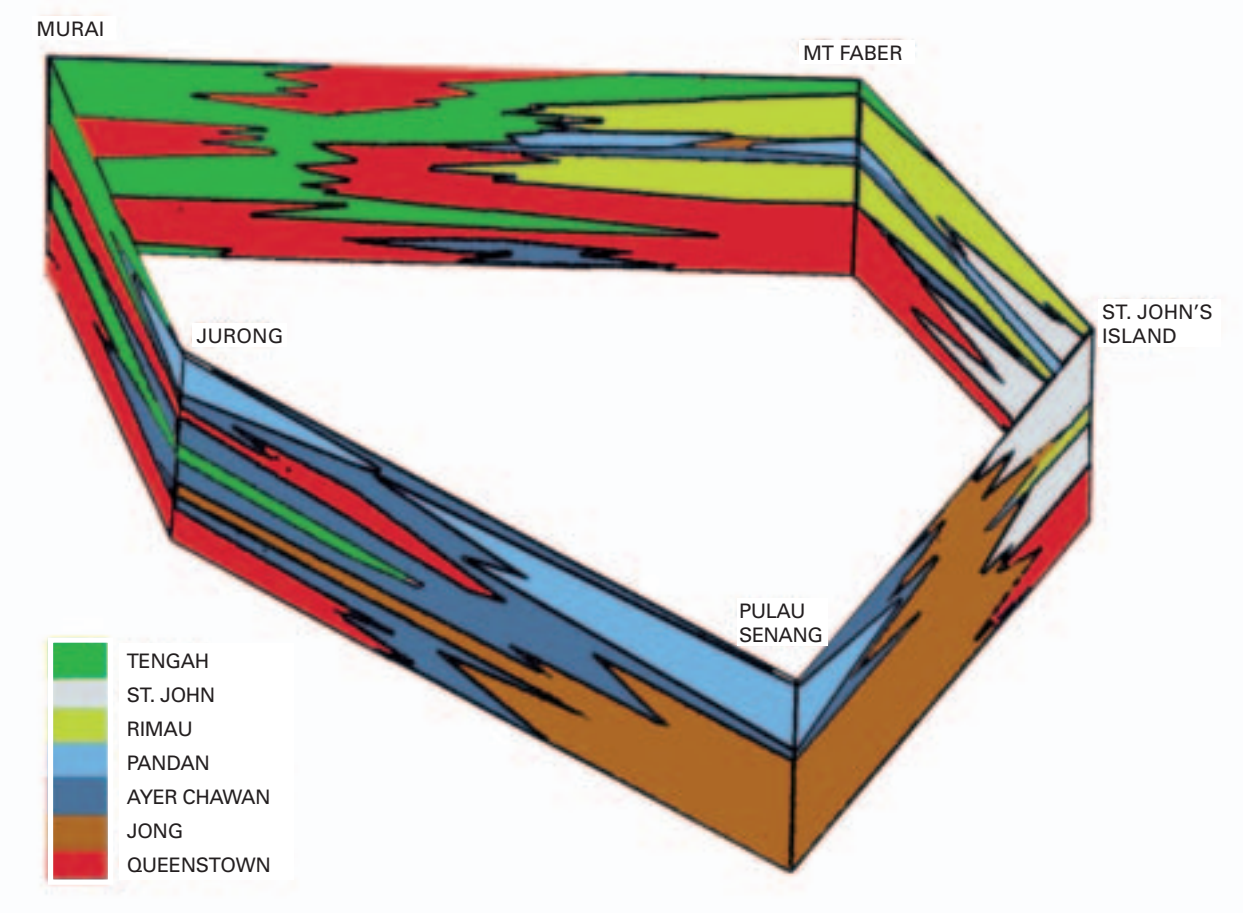


Fig 2.34 Diagrammatic representation (not drawn to scale) of facies relations in the Jurong Formation

Field Relations

The Queenstown Facies either lies directly on granite or is separated from it by yellow clayey sand or tuff beds of the Tengah or Ayer Chawan Facies. It appears to interdigitate with the Tengah Facies to the northwest and appears to pass up into the main outcrops of Ayer Chawan Facies in the Jurong Area where it is interbedded with the Ayer Chawan Facies there and elsewhere (Fig. 2.34). To the southeast, the facies passes into the St. John Facies. The Queenstown Facies appears to lie both above and below beds of the Rimau Facies and is normally separated from it by a tuffaceous bed. A sub-member of the Queenstown Facies can be recognised as lying within the Rimau Facies west of Henderson Road in Telok Blangah.

The member shows marked similarity with the Chilean red beds (Dr H.R. Katz, pers. comm), and it is suggested that the Queenstown Facies is a subaerial terrestrial deposit and a product of terrestrial weathering. The sand bodies are interpreted as representing ephemeral lacustrine or alluvial deposits. The existence of volcanic material within the facies, particularly in the coarser-grained horizons, has added further colouration to the rock.

The other facies associated with the Queenstown Facies are all interpreted as subaqueous and dominantly shallow marine. Thus, it can be presumed that the area of land available for the deposition of the Queenstown facies at that time was of limited extent.

Redding and Christensen (1999) remarked that the Queenstown Facies that abuts the Bukit Timah Granite represents the oldest sediments in the formation. It was deposited alongside the calcareous siltstones in a stable basin. The facies was subsequently overlain by the Pandan Facies. They further remarked that deltaic muddy sediments deposited later than the Pandan Facies have also been mapped as belonging to the Queenstown Facies by the PWD (1976).

JONG FACIES

Definition and Distribution

The Jong Facies contains alternating beds of roundstone conglomerate, sandstone, and (less frequently) beds of mudstone. The type locality is Pulau Jong, the place from which the facies takes its name. Conglomerate, sandstone, and minor mudstone are seen at the type locality.

Rocks of this facies are found mainly on the southwestern group of islands. The islands include Pulau Jong, Pulau Bukom, Pulau Sebarok, Pulau Semakau, Pulau Salu, Pulau Pawai, Pulau Senang, Pulau Biola and Pulau Satumu. It is also mapped on Pulau Subar Laut and Pulau Subar Darat, and at Tanjong Pangkong on Pulau Ayer Chawan (Map Sheets 4, 5 and 6). It has been mapped (although no outcrop was seen) on Pulau Bukom Kechil. A two-metre-thick bed of conglomerate similar to that of the Jong Facies was seen at the southeastern end of Mt Faber (GR 477397) in Telok Blangah, but the extent of the exposure was too small to be recorded on the map (Fig. 2.35).



Fig. 2.35 A two-metre-thick conglomerate bed containing clasts of sandstone, siltstone, quartz porphyry, and schist at Mt Faber, Telok Blangah

Content

Conglomerate with subrounded to rounded clasts, usually about 6 to 10 cm in diameter but frequently up to 30 cm, occurs in beds with thicknesses that range from 0.5 to 6 m thick and possibly thicker. These beds grade up into a muddy fine to coarse sandstone to make up the bulk of the facies. Beds of hard muddy sand grit, ranging from 0.2 to 2 m thick, form the remaining bulk of the facies member. Mudstone beds, usually dark grey to black and seldom more than 1 m thick, are less frequent. The sand in the unit is essentially quartz. Lithic-volcanic, tuffaceous, and pumiceous fragments are also present.

The dominanty siliceous clasts in the conglomerate are fine sandstone, siltstone, or quartz porphyry. Dark grey mudstone clasts are also common (Fig. 2.36). Fragments of vein quartz, as seen in the Rimau Facies, are conspicuously absent. Chert fragments are common within the facies and chert replacement of the original sediment can be seen in some beds. Occasional lensoid spilitic bodies have been observed within the member, and one spilitic body exposed on Pulau Salu measures some 25 cm thick and 2 m long. Heavy veining by quartz is also common within the facies (Fig. 2.37).



Fig. 2.36 Conglomerate of the Jong Facies containing clasts of sandstone, siltstone, grey mudstone, and quartz porphyry, Pulau Jong

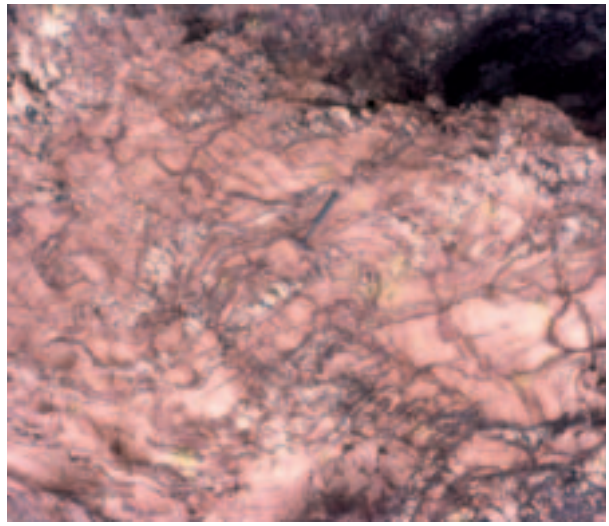


Fig. 2.37 Heavy quartz veining in fine sandstone of the Jong Facies, Pulau Senang

Field Relations

The Jong Facies is seen to lie below the Ayer Chawan Facies on both Pulau Salu and Pulau Senang but above it on Pulau Bukom. At Jurong, it is interbedded within the Ayer Chawan Facies. In the same exposure, beds of Tengah Facies also interdigitate with the Ayer Chawan Facies. On Pulau Salu, massive lithic tuff of the Ayer Chawan Facies ovelies one-to-two-metre-thick beds of well-cemented, slightly tuffaceous, muddy coarse sandstone of the Jong Facies.

Redding and Christensen (1999) remarked that the facies represents early coarse deltaic clastic materials that were associated with active uplift and renewed erosion of the area to the southwest of the Jurong basin due to the emplacement of the Malayan Main Range granite. The facies was deposited alongside sediments of the Rimau Facies. The Jong Facies overlies the Pandan Facies, and it is in turn overlain by the finer Ayer Chawan Facies.

AYER CHAWAN FACIES

Definition and Distribution

The Ayer Chawan Facies is characterised by tuffaceous debris, black sandstone, black mudstone, and minor black to red conglomerate. Spilitic lava is present in this facies. The type section is defined as the west coast of Pulau Sakra in the southwestern group of islands.

The member is found extensively in the Jurong area (Fig. 3.2) and particularly towards the southwest in the Ayer Chawan group of islands. It occurs as a single bed running from Sungei Ulu Pandan to the southern end of Sentosa and overlying the Rimau Facies. It occurs again as a conglomeratic bed underlying the Rimau Facies on Sentosa. Sub-members are also found towards the base of the Queenstown and Tengah Facies, close to the granite contact.

Content

The Ayer Chawan Facies is generally a well-bedded tuffaceous muddy sandstone facies. Bedding thicknesses vary from 1 to 1000 mm. Graded beds are common. Few other sedimentary structures are seen. Most common structures are minor scour channels filled with sandy foreset beds in the top of mud beds. Often, reddish-coloured beds of quartz grit, siltstone and mudstone, tuff, and tuffaceous conglomerate are common within the member. There is considerable evidence of reworking of the sediment by the biota at the time of deposition (Fig. 2.38).

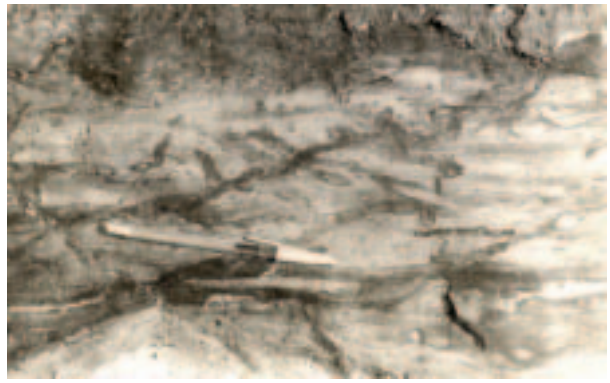


Fig. 2.38 A pale grey bed of mudstone exposed at location GR 327462 in Jurong Indusrial Estate showing features indicating reworking by biota

Chin (1965) recorded the presence of a volcanic breccia on Pulau Samulun, and Lim (1974) noted boulders of spilite about 400 m north of Selat Pulau Damar. Both of these rocks are included in the Ayer Chawan Facies. More such boulders were found in the Jurong area, and their locations are given in the section on the volcanic rocks within the Jurong Formation. Alexander (1950) recorded lavas associated with chert at Tanjong Kling, Tanjong Gul, and Pulau Sakra. Coarse tuff, again associated with chert, is also present on Pulau Pergam in northwest Singapore Island. Beds (1 to 2 m thick) of spilite flows occurs on Pulau Salu, and spilitic dyke cuts the Ayer Chawan Facies on Pulau Senang.

Lim (1974) produced detailed sections of sediments and discussed the petrology of the various lithologies he recognised. His sections A, B, C, D, and E come from areas mapped as Ayer Chawan Facies with interbeds of the Tengah Facies. The following descriptions are based on his work. The sandstone usually comprises a fine- to medium-grained sand made up of quartz with a significant amount of polycrystalline quartz grains, tuffaceous clasts, secondary chalcedony and chert. Biotite mica, zircon, tourmaline and opaque ore that are sometimes rimmed with haematite, are also present. Up to 20% clay can be found in the rock. The finer sediments are more characteristically black or grey and contain a few angular grains of sand-sized quartz in a clay silt matrix of silica minerals, sericite, opaque ore and heavy minerals.

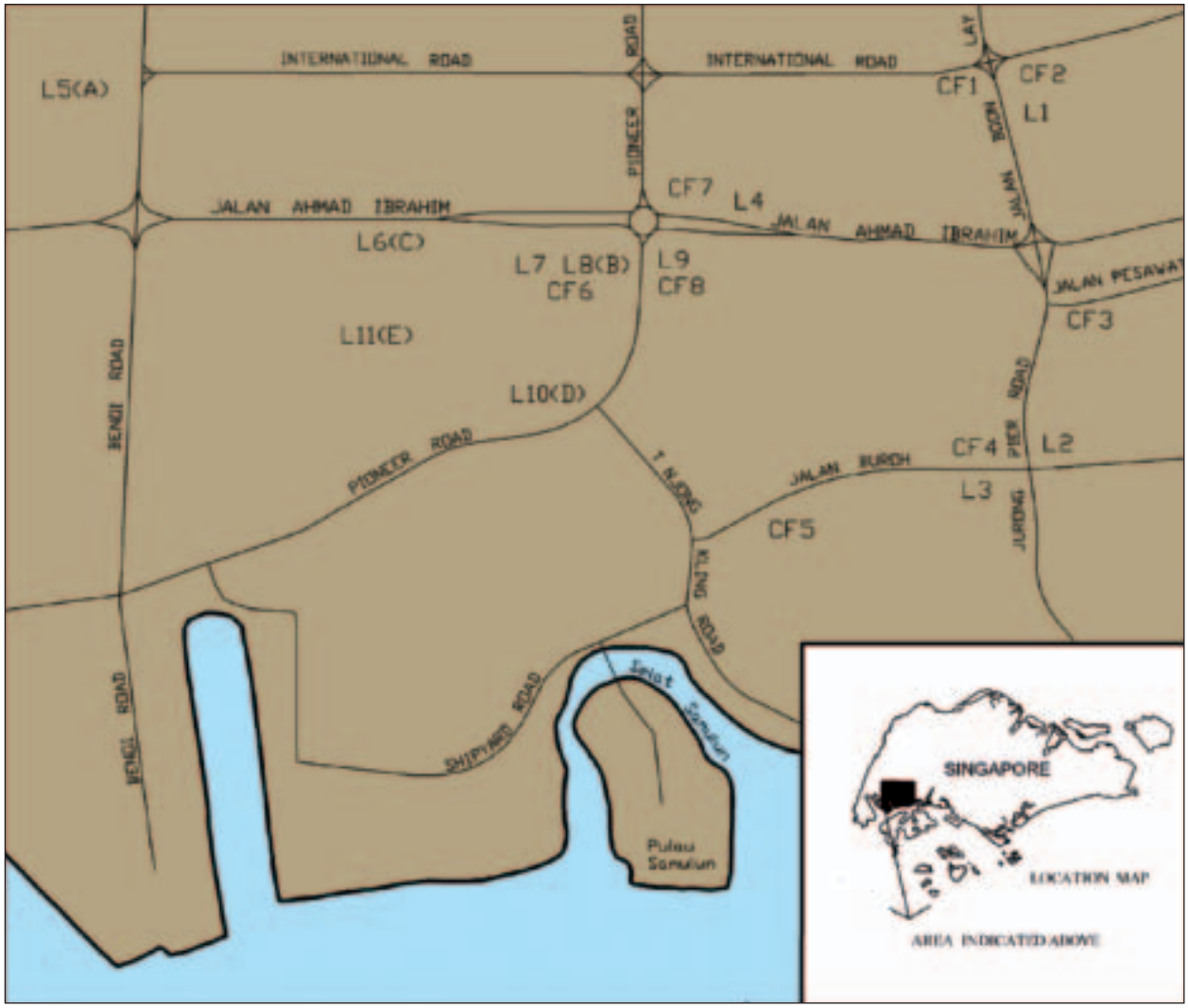


Fig. 2.39 Fossil collection localities of Lim (1975) (L1 — L11) and Chin (1965) (CF1 — CF8) from the Jurong Area. Locations (A) to (E) refer to Lim's detailed section.

No detailed work was done on the mudstone, but both Lim (1974) and Chin (1965) commented on the presence of carbonaceous matter. Chin (1965) reported 6.5% carbonaceous matter determined by loss of weight on ignition of a dry sample heated for a period of two to three days. It is suggested that part of the weight loss may have resulted from a driving out of water from the clay lattices under such severe treatment and that this figure may be overestimated. It is also thought unlikely that 6.5% carbonaceous matter would impart the degree of colouration observed in the Ayer Chawan Facies. Alexander (1950) suggested that the colour resulted from the presence of finely divided iron sulphide and this view is supported here. Efflorescence was observed to be common on the black sediments. Lim (1974) observed pyrite in one bed on Bukit Susop (GR 330460).

With the exception of the fossil collections from Mt Guthrie (now removed) at GR 507406 near Keppel Harbour and Alexandra Brickworks, it appears that all fossil collections from the Mesozoic rocks in the Republic of Singapore come from the black sediment of the Ayer Chawan Facies or from sandstone, presumably, of the Tengah Facies. This facies is intimately associated with the Ayer Chawan Facies. Most of the collections came from a restricted area in Jurong (Fig. 2.39), but collections have also been made from Pulau Ayer Chawan and from the black sediment near Mt Faber (Br Lawrence, pers. comm).

The first documented collection of fossils was made by Alexander (1950) from rock assumed to belong to the Ayer Chawan Facies. In her 1950 report, she described

them as lamellibranchs, gastropods, probably crustacea, and the brachiopod Lingula. At that stage, they still awaited identification, and Alexander did not document their location. Burton (1973 B) quoted a written communication from Cox who listed a number of species found near Jurong West Sports Complex (Appendix 3) and Estheria mangalensis from Pulau Ayer Chawan.

Chin (1965) collected fossils from the Jurong area. The specimens found were casts, as have been all subsequent finds. His localities CF3 to CF6 were in mudstone-sandstone interbeds while the other four localities were in black mudstone (Fig. 2.39).

Lim (1974) collected fossils from eleven localities, designated L1 to L11 (Fig. 2.39). His localities L1 and L9 are equivalent to Chin's localities CF1, CF2, and CF8 respectively. L5 is probably similar to the locality of Alexander's Huat Choe collection. Lim recognised two different faunal groups: one in the black mudstone and the other in the sandstone. His view supported Chin's inference that these groups indicate at least two sets of depositional environments. Both had recognised Gonodon and Posidonia from the sand beds and Myophoria from the black mudstone beds. Chin also recorded a fragment of an ammonite from one of the sand beds.

An effort was made to relocate the site of the Morse Road collection of Scrivenor (Newton, 1923). A single broken cast was found in place at the head of the valley above Morse Road. The cast was in a sequence (which exceeded 20 m in thickness) consisting of grey and pink, tuffaceous, carbonaceous, muddy fine sandstone sequence. It is assumed that this find is from the horizon from which Scrivenor made his collection. The sequence is mapped as belonging to the Ayer Chawan Facies. The sequence is capped by a two-metre-thick bed of roundstone conglomerate (similar to that of the Jong Facies) and beds of mudstone-conglomerate which lie between beds of angular-quartz conglomerate of the Rimau Facies. (A complete faunal list is in Appendix 3.)

The collection from black sediment near Mt Faber was a private collection made by Br Lawrence. It could not be traced and hence was not described.

Field Relations

The Ayer Chawan Facies appears to interdigitate with the Pandan Facies, the upper portion of the Tengah Facies, and the Queenstown Facies. A limb of the facies is also found to overlie the Rimau Facies. In the Jurong area, the Ayer Chawan and Tengah Facies interdigitate extensively, but much of the material identified as Tengah Facies may in fact be leached Ayer Chawan Facies.

Lim (1974) recorded six stratigraphic columns from this facies with his section A passing up into his Pasir Panjang Formation. In this area, the Tengah Facies was mapped as overlying the Ayer Chawan Facies but there is neither sufficient evidence to state categorically that this contact represents the top of the Ayer Chawan Facies nor is there sufficient evidence to place column 'A' relative to the other given sections. Re-mapping, however, confirms Lim's general sequence, but it places columns 'B' and 'C' at the same stratigraphic level.

The sediments of the Ayer Chawan Facies are dominantly fine-grained with laminae bedding and possess only minor current features. These features, together with the occurrence of finely disseminated carbonaceous matter, suggest a low energy environment. The characteristic black colouration (believed to be due to the presence of finely divided pyrite) points to a reducing environment. It is thought that these anaerobic conditions occurred after phases of volcanic activity and were responsible for the elimination of the biota present.

Dr I.G. Speden (pers. comm.) recognised six mode-of-life elements in the Myophoria assemblage (Table 1). Of these, Group '1' requires a firm to moderately firm substrate containing organic material for food and reasonably oxygenated sediments. The other five groups require firm to moderately firm, relatively stable substrates that can provide suitable holdfasts—shells, pebbles, wood or grain fragments, and marine plants—in a low to moderate energy environment with adequate oxygen and food. The species would not tolerate anaerobic or high energy scouring environmental conditions nor excessive turbidity. Most species of Group '5' can rebury themselves - if exhumed by currents - to moderate depths but would not tolerate frequent displacement. Dominance, both numerically and in diversity, of the assemblage of suspension feeders indicates low turbidity. All the groups frequently occur together in the stratigraphic record and are characteristic of shallow shelf to enclosed marine bay habitats of normal salinity, turbidity, oxygenation, and low to moderate energy conditions.

Redding and Christensen (1999) remarked that the facies represents finer deltaic clastic materials that were associated with pulsatory uplift of the Jurong basin. The finer clastic material was deposited on older coarser materials of the Jong and Rimau Facies and was in turn buried by even finer material of the St. John Facies. Some of the materials in this facies were also fine graded materials that had been transported further from the delta fans which were built up of coarser materials of the Rimau and Jong Facies.

The evidence of bioturbation and frequent incursion of sand often carrying the Gonodon-Posidonia fauna also argues against an anaerobic deep basin type setting. Rather, it points to a shallow water, sheltered, low energy setting (possibly similar to the brackish coastal areas found in tropical areas at the present time) but with a volcanic province nearby to supply the tuffaceous material. The Gonodon-bearing sand beds possibly came about through longshore drift during the longer periods of volcanic quiescence. Lim (1974) noted that the Gonodon and Posidonia shells were all orientated convex upwards, suggesting transport and reworking before burial.

Table 1

Mode of life elements for species listed by Lim (1974) as having been found in sediment mapped as Ayer Chawan Facies

1. Infaunal mobile deposit feeders: Palaeoneilo sp., Palaeonucula sp.
2. Endobyssate and epibyssate suspension feeders: Cassianella sp. Pteria pahangensis, Posidonia sp., Halobia sp., Buchia sp.*, Palaeolima sp., Aviculima sp., Lima spp. (2), Palaeopharus sp.
3. Epifaunal free-swimming suspension feeders: Amusium sp., Syncyclonema* (? = Entolium) sp., and possibly the Pecten spp.* (?4).
4. Epifaunal cemented suspension feeders: Plicatula sp., Spondylus dubiosus (Bittner).
5. Shallow burrowing siphonate and non-siphonate suspension feeders: Trigonodus sp., Anodontophora spp. (?2),Costatoria spp.* (?6), Gruenwaldia sp., Myophoria sp., Neoschizodus sp., Gonodon sp., Cardium scrivenori.
6. Other: Gastropod sp.

** Dr I.G. Speden (pers. comm.) stated that Pecten and Costatoria may be conspecific and that Syncyclonema may be Entolium sp. He also said that Buchia sp. is only known from late Jurassic to early Cretaceous, and there must therefore be a mis-identification in these collections.*

PANDAN FACIES

Definition and Distribution

Limestone was first discovered in 1987 during a site investigation project for a jetty at Pulau Merlimau. In the same year, limestone was obtained (from a borehole) at Pandan Road near Pandan Reservoir by the Public Works Department. Since 1987, limestone, dolomite, and marble have been frequently identified at locations near the Pandan Reservoir, the western end of Pasir Panjang Road, along West Coast Road, and the offshore near these roads. It is also present at Pulau Seraya, Boon Lay Way, North Buona Vista Road, and at the National University of Singapore.

Cores (totalling 185 m in length) of grey fossiliferous limestone (Appendix 4) obtained from a borehole at Pandan Road were examined by Fontaine and Lee (1993). They proposed that the limestone at the location be called Pandan Limestone. It is however proposed here that the carbonate rocks in the Jurong Formation be called the Pandan Facies (Jp).

This facies is found exclusively underground and often below 20 m from the ground surface (Jeyatharan, et al. 2003).

Content

Most of the carbonate rocks in Singapore are limestone with much lesser occurrences of dolomite and marble (Guo, 1998). Most have been recrystallised and a few have been totally recrystallised to marble.

At Pandan Road, the limestone is dark grey to black, occasionally laminated, fine-grained and the wackestone to packstone variety (Fontaine and Lee, 1993). Guo (1998) classified the limestone at this location as fossiliferous micritic limestone.

At Tuas, the limestones are fossiliferous limestone, arenaceous oolithic limestone and arenaceous micritic limestone (Guo, 1998).

At Pasir Panjang, the limestones are micritic limestone, fossiliferous limestone, sparitic micritic limestone, carboneous limestone, and some dolomite and dolomitic marble (Guo, 1998).

Limestone with inclusions of calcitic siltstone is found at Jurong Island (Cai, pers. comm).

The limestone at Pandan Road is richly fossiliferous (Fontaine and Lee, 1993). The fossils found at Pandan Road are listed in Appendix 4. Fossils are also present in limestone from Tuas and Pasir Panjang, but they have not been identified.

Solution cavities are common especially in the Pandan and West Coast areas (Jeyatharan, et al. 2003). They often occur to depths of 50 to 60 m below the ground surface. The deepest cavity recorded to date reaches a depth of 80 m. The size of the cavities ranges from less than 0.5 to 13.5 m, but they are generally less than 4 m. The average size is about 2 m.

Field Relations

Although the Pandan Facies is distinctive, its field relationship with the other facies of the Jurong Formation is unclear.

Fontaine and Lee (1993) suggested that the Pandan Limestone is part of the Ayer Chawan Facies. The facies at Pandan Road has been interpreted as a deposit laid down on a tidal flat and temporarily in slightly deeper water during probably late Triassic time (Fontaine and Lee, 1999). Its depositional environment was relatively sheltered and unaffected by strong currents.

Along the proposed East Jurong Fairway linking Jurong Island and West Coast Park (GR 408437) off West Coast Road, the limestone is interbedded and folded along with the grey mudstone, siltstone and sandstone of the Ayer Chawan Facies (Jeyatharan, et al. 2003). In Jurong Island, limestone containing abundant inclusions of calcitic siltstone is also interbedded and folded with the rocks of the Ayer Chawan Facies (Cai, pers. comm). The interbedding of the limestone in Jurong Island and its vicinity with the Ayer Chawan Facies suggests that at least part of the Pandan Facies has a close stratigraphic relationship with it. Yeap (1996), in his investigation for the Pasir Panjang Container Terminal off Pasir Panjang, however suggested that the limestone occur as interbeds or lenses either between the St. John Facies and the Tengah Facies or between the Rimau Facies and the Tengah Facies. Guo (1998) noted that contacts between limestone and other rock types of the Jurong Formation are often abrupt and suggested that the carbonate rocks could have been deposited in a series of grabens. He added that such a depositional environment could also explain the variety of carbonate rock types. Adding yet another view, in their account on the deposition of the Jurong Formation, Redding and Christensen (1999) remarked that the Pandan Facies is a distinct uniform unit that was deposited on the older red mudstones of the Queenstown Facies and calcareous siltstones under stable basin conditions. They were in turn overlain by the Jong and Rimau Facies. In their account, the limestone does not have a close stratigraphic relationship with the Ayer Chawan Facies.

RIMAU FACIES

Definition and Distribution

The Rimau Facies (Jr) is named after Sarang Rimau at the northwest tip of Sentosa where the facies is well exposed. The type section is defined as those beds lying above and to the southwest of the red mudstone exposed beneath the Siloso Jetty. A northwesterly trending fold axis lies 100 m southwest of the jetty, and the section is repeated to the southwest where 160 m of sediment are exposed.

The Rimau Facies is found on Mt Faber and Kent Ridge, in areas as far northwest as Sungei Ulu Pandan, and on the northern side of Pulau Brani. It forms the southwest coastline of Sentosa, the bulk of Pulau Tekukor and the St. John's Island group. It is also found on the southeast side of the main granite mass on Singapore Island.

Content

The Rimau Facies is characterised by quartzite and conglomerate. Both rock types are well lithified, and thus form the backbone of the prominent northwesterly trending ridges of Mt Faber, Kent Ridge, and the Southern Islands.

In these areas, the conglomerate contains subangular to rounded fragments of quartz, tuff, quartz sandstone, chert, rhyolite, basic igneous pebbles, pebbles of red sandstone (presumed to be derived from the Queenstown Facies), and schist (presumably derived from the Murai Schist). The fragments are usually less than 5 cm in diameter, but sometimes they are cobble-sized. Quartz, probably vein-quartz, is by far the dominant lithology of the clasts. The matrix is usually of subangular coarse quartz sand, but feldspar (usually weathered to clay) has been recognised. Tourmaline, fluorite, and zircon occur as heavy minerals. Haematite and magnetite were recorded in Chin (1965) as the

only observable opaque minerals. Wong (1960) also recorded limonite and leucoxene.

Thin sections of sandstones show a typical arrangement of closely packed, sub-angular and well-sorted grains (Fig. 2.40). Quartz may form up to 98% of the rock and occurs with biotite, tourmaline, and traces of iron oxide usually as a coating on the quartz grains. Quartz sandstone fragments can also be seen in the rock. In other thin sections, altered feldspar (to possibly kaolinite), chlorite and illite were recognised.

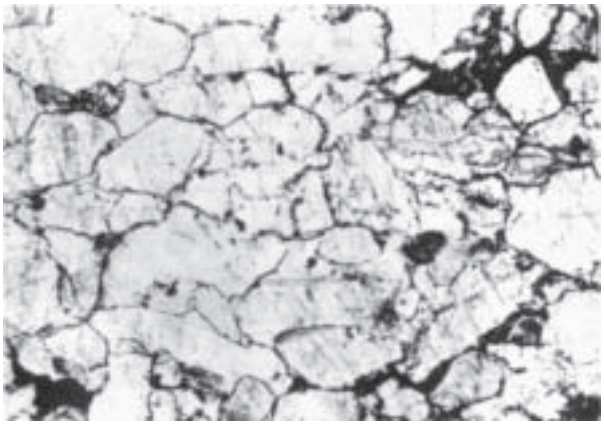


Fig. 2.40 Closely packed subangular quartz grains in a sandstone of the Rimau Facies from Kent Ridge 45X Plain Polarised Light

The quartz conglomerate beds along the Pasir Panjang coast and in the southern islands are usually grey-white, but the associated quartz sandstone is often stained pink. It thus shows similarities to the coarser horizons in the Queenstown Facies. On the southern islands, the pink colouration is also seen in the conglomerate.

Beds of the Rimau Facies are usually 500 to 1,500 mm thick. Coarse cross-beddings and scour features are common. Beds between 10 and 100 mm thick of silt and fine sand may be seen between the thicker beds, and these finer beds show a greater variety of sedimentary current features.

Field Relations

Although the Rimau Facies is a distinctive member in the field, its field relationships are not always clear. It can be seen to lie conformably on a volcanic conglomerate of the Ayer Chawan Facies on Sentosa, on the St. John Facies on Lazarus Island and is intimately associated with the Queenstown Facies and Ayer Chawan Facies along Kent Ridge. To the northwest, it appears to wedge out and pass laterally into Queenstown Facies and Ayer Chawan Facies. Identical rocks have been described in Priem (1975) from the Bintan Formation from the Riau Archipelago.

The general coarse nature of the facies and the presence of cross-bedding and current-bedding features suggest a shallow water, near-shore, probably deltaic environment close to a rising land mass that has been deeply leached. The clay residual has been removed and possibly has been incorporated into the Queenstown and Tengah Facies prior to the deposition of the Rimau Facies.

The juxtaposition of the Rimau Facies to the dominantly terrestrial Queenstown Facies and the marine St. John Facies supports the argument for a near-shore environment.

Redding and Christensen (1999) remarked that the facies represents early coarse deltaic clastic materials that are associated with active uplift of the area to the southwest of the Jurong basin from the emplacement of the Malayan Main Range granite. The facies was deposited alongside sediments of the Jong Facies. Like the Jong Facies, the Rimau Facies overlies the Pandan Facies, and they are in turn overlained by the finer Ayer Chawan Facies.

ST. JOHN FACIES

Definition and Distribution

The St. John Facies (Jsj) is a pale grey mudstone and muddy sandstone sequence found in the southern island group and possibly the site at GR 507406 once occupied by Mt Guthrie. The type area is defined as the southwest coast of Lazarus Island. The beds below the lowest quartz pebble bed that are exposed at the southwest tip of the island are included in this facies.

Content

The rock is a pale grey muddy sandstone with well-defined ripple-marked beds, current bedding, graded bedding and minor intraformational breccia (Fig. 2.41). Lenses of coal, less than 2 mm thick, are also characteristic of this facies. Coal was not recorded in any other facies. Because of the presence of coal and a marine fauna in them, the Mt Guthrie rocks, although now removed, are mapped as part of the St. John Facies. No petrographic or sedimentological studies were carried out on this facies.

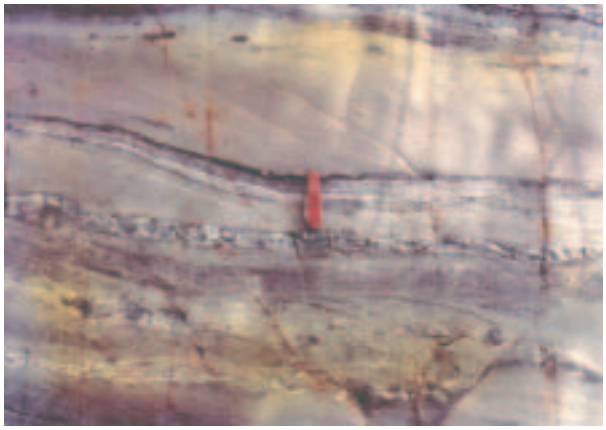


Fig. 2.41 Outcrop of the St. John Facies showing pale grey muddy sandstone and mudstone with intraformational breccia at St. John's Island

Newton (quoted by Burton, 1973 B) suggested that the Mt Guthrie collection had been recovered from either an estuarine or lagoonal deposit, in contrast to shallow but less stable basin enviroments in which the rest of the other facies (which often showing flysch-like characteristics) has been deposited.

Field Relations

The St. John Facies appears to pass laterally into the Queenstown Facies to the northwest. On Sentosa Island, it is interbedded with the Queenstown Facies and with other tuffaceous beds in the Ayer Chawan Facies. The St. John Facies is found mainly below the Rimau Facies, but at Mt Chermin in Telok Blangah it is found within the Rimau Facies.

Redding and Christensen (1999) remarked that the St. John Facies represents late-stage delta-top clastic materials and fore-delta turbidites and are remobilised deposited materials. They are probably the youngest sediments of the Jurong Formation.

TENGAH FACIES

Definition and Distribution

The Tengah Facies (Jt) includes all rocks of the Jurong Formation found in Singapore that have not been included within the other seven facies. No formal type area or definition is therefore proposed. They are typically recognised as being a muddy fine- to medium-grained poorly lithified sandstone. Since they show this specific unifying characteristic, they are not mapped as undifferentiated Jurong Formation. For the same reason, they are not mapped as a continuation of the Bukit Resam Member of Burton (1973 A).

The facies is found extensively in the west, from Tengah towards the coast and south from Tengah towards Jurong in the south. It is also found lying between the Rimau and Queenstown Facies between Sungei Ulu Pandan and Keppel Harbour. The facies is assumed to be present and in contact with the Rimau Facies to the east of the granite.

Content

No detailed work has been done on the sedimentology or petrology of this facies. The sediment is a muddy, quartz-rich, usually poorly indurated and fine- to medium-grained sandstone. As a result of weathering, there are few natural outcrops. The member is usually well-bedded with layers being 2 to 30 cm thick, but moderately well-lithified beds up to 1 m thick can be seen. The well-lithified beds are generally quartz-rich and appear to have been cemented by silica. Such beds are well exposed in an excavation made at a location (GR 387498) in Bukit Batok.

Occasional yellow-brown roundstone conglomerate and grit beds are found particularly to the northwest of Nanyang Technological University, along the Pasir Laba Ridge, and along the ridge at GR 310560 in northwest Singapore Island. Many of the pebbles, some of which are up to 10 cm in diameter, appear to be derived from the contemporaneous erosion of the Jurong Formation. The poor sorting of the sandstone and conglomerate and the rapid change in grain size from mudstone to grit suggest a flysch-like deposition.

Lim (1974) collected his Gonodon fauna from sediments assumed to be part of the Tengah Facies. These have been discussed together with his Myophoria collections under the Ayer Chawan Facies.

Field Relations

The Tengah Facies is interbedded with all the other facies with the exception of the St. John and Jong facies. It appears to be the lateral equivalent of the Queenstown and Rimau Facies and passes up into the main body of the Ayer Chawan Facies.

The presence of Gonodon and Posidonia in rocks that are supposed to belong to this facies suggests a marine environment. Both species are bottom surface dwelling or shallow borrowing marine types, preferring moderate to firm substrates. They could be exhumed and redeposited by low energy currents (Dr I.G. Speden, per comm). The orientation of Gonodon shells, as discussed by Chin (1965), points to the periodic influx of steady directional ocean currents. The presence of a marine environment is also suggested by the general bedding charactistics.

Redding and Christensen (1999) remarked that the facies is weathered material that has developed in-situ from the underlying bedrock. Structures from the underlying bedrock could be traced upwards into the weathered material.

MURAI SCHIST

Definition and Distribution

It is not proposed that the Murai Schist be recognised as a formal geological unit, but rather as a zone of well-developed cleavage in rocks otherwise recognised as sediments of the Queenstown, Jong, and Tengah Facies.

The Schist Zone forms a belt up to 0.5 km wide in Ama Keng, trending northeast from Tanjong Skopek to include the area originally described by Alexander (1950). A smaller schist zone was found on the north arm of the Pasir Laba Ridge (GR 295494) and another zone, not recorded on the map, was found in the Jong Facies in Jurong (GR 332452).

Content

The unifying characteristic of the Murai Schist is the presence of well-developed cleavage in the pelitic sediment and coarser fracture or foliation in the arenaceous sediment. The cleavages and the foliation in both sediment types are subparallel to the bedding (Fig. 2.42). While the cleavage planes are planar, the foliation planes in the coarser sediment are knotted and elongated with lenticular nodules lying along the foliation planes. Cleavage and foliation are absent in the harder tuffaceous material within the schist zone.

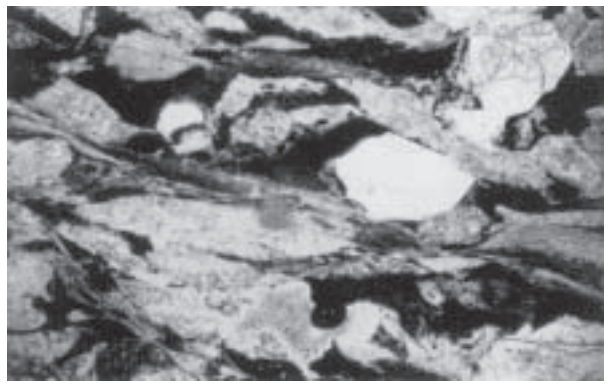


Fig. 2.42 Schistose sandstone of the Murai Schist , Ama Keng 45X Plain Polarised Light

From the thin sections, it is not clear if any metamorphic minerals have formed. White mica is the only likely metamorphic mineral. An apparent spotting, formed by the presence of a few small areas rich in fine-grained white mica, was recognised in one thin section of slate. Here, the quartz is the main recognisable mineral. Slately cleavage is well shown by the parallel orientation of small mica flakes and lines of iron oxide granules.

There are a few small irregular areas rich in fine-grained white mica. The coarse arenaceous material can be seen in thin sections forming a poorly schistosed rock that was originally quartz-rich sandstone. There appears to have been an overall reduction in average grain size by pronounced shearing. Many quartz grains show undulose extinction while others have been reduced to a mosaic of tiny interlocking grains. A few large grains have conspicuous lamella structures, probably resulting from crushing. In addition to the quartz, there are also occasional fragments of very fine-grained chert (usually grey or dark grey in colour) and finely divided opaque inclusions in them. There are scattered aggregates and streaks of mica, many of which are clouded by fine-grained alteration. The borders between many of the quartz and other grains are outlined by opaque material, probably iron oxide.

Field Relations

The rocks of the schist zone are in fault contact with the Tengah Facies to the northwest and there is a marked angular unconformity across this fault. To the southeast, however, the schist is seen to pass concordantly up into non-sheared Queenstown and Tengah Facies rocks.

VOLCANIC ROCKS WITHIN THE JURONG FORMATION

Introduction

Volcanic activity contemporaneous with the sedimentary deposition of the Jurong Formation has given rise to spilite, tuff, chert, and dolerite within the formation.

Spilite

Definition and Distribution

Spilite is defined as the soda rich basalts found as lavas and relict boulders in or associated with rocks of the Ayer Chawan and Jong Facies.

Alexander (1950) recorded lavas associated with chert at Tanjong Kling, Tanjong Gul, and on Pulau Sakra. Chin (1965) recorded the presence of a volcanic breccia on Pulau Samulun, and Lim (1974) noted boulders of spilite some 400 m north of Selat Pulau Damar (GR 352448). Lim's finding have been confirmed.

Spilite boulders have also been found during the survey in the Jurong area at GR 350452 along Jurong Pier Road and GR 363466 near Jurong Park (Fig. 2.43). On each occasion, the boulders were left after excavation activities within the Ayer Chawan Facies. Their relationship with the host rock cannot be seen. The sole spilite found interbedded with the Jurong Formation lies in sandstone of the Jong Facies on the northern shore of Pulau Salu. There, the rock forms an elongated pillow approximately 200 cm long and 25 cm thick (Fig. 2.44).



Fig. 2.43 A large boulder of spilite at Jurong Pier Road, Jurong Industrial Estate

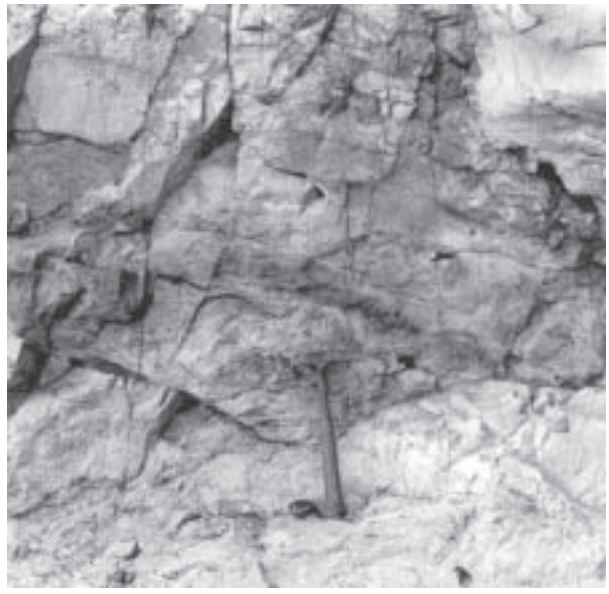


Fig. 2.44 Grey spilite body associated with coarse sandstone and chert, Pulau Salu

Composition

The spilite is described by Lim (1974) as being a dark green amygdaloidal rock containing phenocrysts of albite and epidote in a matrix of small albite laths

showing a trachytic texture in a matt of chlorite. He identified six groups of amygdales. They are filled with calcite, calcite and quartz, quartz, quartz and plagioclase, plagioclase, or plagioclase and chlorite. Large, angular inclusions in the boulders were observed near Selat Pulau Damar (Fig. 2.45). They were not mentioned in Lim (1974).

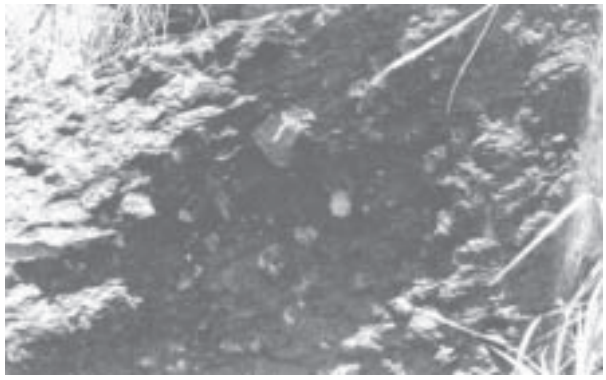


Fig 2.45 Angular inclusions in spilite boulder near Selat Pulau Damar, Jurong Industrial Estate

Tuff

Definition and Distribution

Lithified tuff found within the Jurong Formation is included under this heading. It appears as a crystal tuff made up mainly of crystals in a chloritised matrix. The poorly lithified ashy tuff cannot realistically be separated from the sediment and hence is not discussed under this heading.

Alexander (1950) recorded a coarse tuff associated with chert on Pulau Pergam. This occurrence was confirmed. One-to-two-metre-thick beds of tuff, interbedded with sandstone and associated with chert, were also found on Pulau Salu and are well exposed on the western shore of the island.

Composition

In hand specimen, the tuff appears as a hard, dense, green rock containing phenocrysts (up to 2 mm) of feldspar and quartz. The tuff on Pulau Pergam is coarse-grained and has crystals up to a centimetre in length (Fig. 2.46).

Dr W.A. Watters (pers. comm.) has described the rock from Pulau Pergam as a rhyolitic crystal tuff containing numerous crystals of acid plagioclase, orthoclase, and quartz set in a fine-grained matrix with abundant pale green chlorite. A few irregular elongate aggregates made up of chlorite and sericite may represent original fragments of pumica. There are also a few small areas of calcite and tiny aggregates of granular sphene (Fig. 2.46).

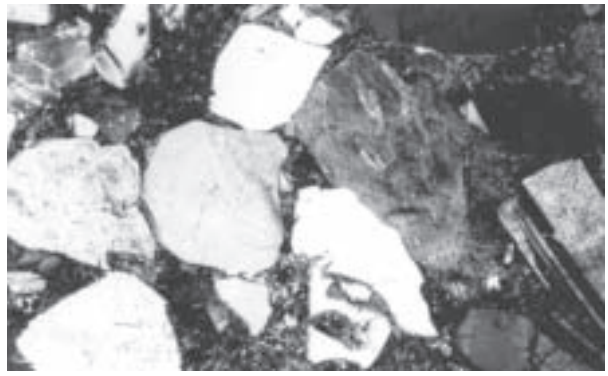


Fig. 2.46 Large crystals of quartz, orthoclase, and acid plagioclase in rhyolithic crystal tuff, Pulau Pergam 45X Crossed Nicols

The tuff from Pulau Salu is also a rhyolitic crystal tuff. It shows a fine-grained groundmass consisting mainly of plagioclase and quartz with subordinate orthoclase. All the minerals appear to be fresh.

Chert

The chert is a cryptocrystalline form of silica that occurs as bands or layers in the sedimentary rocks of the Jurong Formation. The chert is interstratified with spilitic lava, and it is therefore assumed that the silica owes its origin to the volcanism associated with the spilitic lavas.

Chert was found abundantly on Pulau Salu where it forms bands, layers, small lenses, and redeposited fragments within the Jong Facies immediately below the spilite.

No thin sections were made of the chert.

Dolerite

Definition and Distribution

The dolerite is a medium-grained basic hypabyssal igneous rock. Mineralogically and chemically, it is equivalent to gabbro or basalt.

Scrivenor (1924) reported the occurrence of dolerite boulders at the site of the Tan Tock Seng Hospital. At

the hospital site, sedimentary rock of the Rimau Facies is seen to outcrop, but no igneous boulders could be located despite detailed work and drilling in the area. The boulders reported by Scrivenor are probably not residual boulders weathered out from dykes cutting the Triassic sedimentary rock, but were instead derived from dykes in the granite which lie about 40 m to the west.

There is evidence of only one dolerite dyke cutting the Jurong Formation. On the western shore of Pulau Senang, an alignment of boulders, each measuring about 60 cm in diameter, indicates the presence of the dyke (Fig. 2.47).



Fig. 2.47 Relict doleritic boulders lying along the strike of a dolerite dyke cutting the Jong Facies, Pulau Senang

Composition

The dark grey, medium-grained rock from Pulau Senang is an olivine-bearing undersaturated dolerite. The rock is holocrystalline and is made up of long slender laths of calcic plagioclase whose intergranular spaces are filled with anhedral augite crystals. Occasional phenocrysts of strongly serpentinised olivine were found.

The abundance of augite and the relatively unaltered state of the feldspar and pyroxene of this basic dyke on Pulau Senang (Fig. 2.48) are in marked contrast with the dolerite dykes associated with the Bukit Timah Granite.



Fig. 2.48 Augite crystals, partly interstitial between calcic plagioclase and partly enclosing some of the plagioclase, in dolerite, Pulau Senang 45X Crossed Nicols

Field Relations

The field relations of each of the volcanic rocks listed above have been discussed in the sections under its host facies.

Age and Correlation

Scrivenor (1924) specified two fossil collections: one is from "a cutting near the top of the road that leads to the Mt Faber Ridge by way of Morse and Pender Roads" and the other is from Mt Guthrie. Newton (1923) reported on both collections and assigned them to the upper Triassic (Rhaetic) and possibly middle Jurassic respectively. Scrivenor considered that the sediment pre-dated the granite because the sedimentary beds dipped away from the granite and there was no evidence of the folding continuing into the granite.

Alexander (1950) accepted the same time sequence as Scrivenor and described a core boulder of granite, which she interpreted as a residual of an intrusive dyke penetrating 60 cm into the sediment. She extended the time range of the Jurong Formation to span from the possibly Carboniferous to the possibly Jurassic. Her fossil collections had not been studied when she wrote her report but were later reported by Cox (Burton 1973 B) as being late Triassic. Later collections by Chin (1965) and Lim (1974) confirmed this late Triassic age.

The age of the Mt Guthrie collection was re-assessed by Kobayashi and Tamura (1968). They suggested an early Jurassic age for this assemblage. With the stratigraphic detail now available, the Mt Guthrie site appears to lie some 200 m above the Morse Road site (Section E map 7) but still at least 400 m below the top of the Jurong Formation. The relationship to the collections in the Jurong area cannot be established. Redding and Christensen (1999) remarked that the Mount Guthrie

fossils are quite different from the fossils in Johor and South Malaya.

There was neither evidence of contact metamorphism of the Jurong Formation, nor was there intrusion seen. The example of intrusion cited by Alexander (1950) could not be located. She described a single residual boulder in deeply weathered material close to the granite pluton as representing a dyke. In view of the recent datings obtained for the two formations, it is thought that she misinterpreted this outcrop. Here, the Jurong Formation is interpreted as being younger than the granite. The palaeontological evidence suggests a late Triassic age for the Jurong Formation whereas the isotopic age for the Bukit Timah Granite is approximately 220 million years (early to mid-Triassic) or even older depending on the decay factor used in the isotopic age determinations. Poh, K. B.; Buttling, S.; Hwang, R. (1987) could not find evidence of metamorphism at the contact of the Bukit Timah Granite and the Jurong Formation during their observations of tunnelling works under Tan Tock Seng Hospital. Their observations lend support to the view that the granite pre-dates the sedimentary rocks of the Jurong Formation.

The age of the Jurong Formation is thus deduced to be late Triassic to early Jurassic.

The Jurong Formation is correlated with the Bukit Resam Member (Burton 1973 A) in Johor. No equivalent of the Gunong Pulai of Burton (op. cit.) is recognised in Singapore. The spilite, noted by Lim (1974) and the authors of PWD (1976), and the lavas, reported by Alexander (1950), are mapped in the Ayer Chawan Facies and are thus too high in the stratigraphic sequence to be correlated with the Gunong Pulai Member.

The Jurong Formation is also correlated with the Bintan Formation described by Priem, et al. (1975) from the Riau Archipelago, and Prof H.N.A. Priem (pers. comm.) stated that the sediment of the Rimau Facies shows a marked similarity to those sediments of the Bintan Formation. Burton (1973 B) has also correlated the Jurong Formation with the Kerdau and Jelai Formations of axial Malaya, and the top of the Jurong Formation with the Murau Conglomerate and the base of the Tembeling Formation.

The period of faulting of the Jurong sediment that gave rise to the Murai Schist is described in the chapter on structure. The views of Redding and Christensen (1999) on the deformation of the Jurong Formation are also described in the same chapter.

FORT CANNING BOULDER BED

Definition and Distribution

The Fort Canning Boulder Bed (FC) was first described by Nowson (1954) as consisting of generally lens-shaped or rounded fresh sandstone boulders ranging in size from 1 to 9 m in a hard, multi-coloured, mainly red and white silty clay which does not show any stratification. It was then referred to as boulder clay.

Later accounts are from Sehested (1960), Pitts (1984), Poh, et al. (1987), Shirlaw (1987), Han, et al. (1993), Broms and Lai (1995), Wong, et al. (1995), and Shirlaw, et al. (1990, 2003). In accounts published before 2003, the deposit was referred to as bouldery clay or boulder bed. It was called Fort Canning Boulder Bed in Shirlaw, et al. (2003). It is proposed that the latter name be used here.

The Fort Canning Boulder Bed is exclusively confined to the central business district in downtown Singapore. It underlies an area bounded by Fort Canning Park, The Cathay Building, Middle Road, and Raffles City (Map Sheet 8). It also underlies another area bounded by Telok Ayer Street, Boon Tat Street, Singapore River, and the old shoreline near Raffles Quay. In a much lesser extent, Shirlaw, et al. (2003) also documented its occurrence at Great Eastern Centre, Coleman Bridge, The Treasury Building, and the area between Mt Emily Park and Tekka Centre.

The Fort Canning Boulder Bed is covered by younger sediments of the Kallang Formation and fill material. Surface outcrops are rare. Shirlaw, et al. (2003) reported that it once outcropped at Raffles Place and the north and south flanks of Fort Canning Park. It also outcropped on the southern flank of Mt Sophia, a hill south of Mt Emily and next to Plaza Singapura.

Content

Shirlaw, et al. (2003) described the Fort Canning Boulder Bed as typically consisting of generally unfractured sandstone or quartzite boulders in a hard matrix of sandy clayey silt or sandy silty clay which is characteristically mottled in deep red, red and white, or red, yellow, and white. The weathered matrix can be coloured purple (Han, et al. 1993). Boulders of other sedimentary rock types and small angular rock fragments in the matrix are sometimes present. The boulders are generally distributed evenly and are not in contact with each other

(Shirlaw, et al. 1990). The hard silty or sandy clayey matrix consists mainly of quartz and kaolin (Wong, et al. 1995). Shirlaw, et al. (1990) noted that the characteristic distinguishing features of the Fort Canning Boulder Bed in core runs are the presence of relatively unfractured large pieces of sandstone and quartzite in a red or less often white clayey matrix and the presence of small angular fragments of rock material.

Nowson (1954) reported that the boulders at Great Eastern Centre ranges from 1 to 9 m. Far less impressive were the boulders excavated from a work shaft near Little India Station (GR 506445). At this site, Broms and Lai (1995) observed their maximum dimensions to be 1 to 2 m and their average dimension 0.4 m. Boulders over 200 m^3 have been documented in Pitts (1984), Poh, et al. (1987), and Shirlaw, et al. (1990).

The content of boulders in the Fort Canning Boulder Bed varies at different locations. At Republic Plaza (GR 507419), the content varies from 10% to 40% with an average of 20% (Broms and Lai, 1995). Wong, et al. (1995) estimated that the average content of boulders for several project sites ranges from 15% to 30%, and Shirlaw, et al. (1990) estimated an average boulder content of 25% based on observations from the first phase of MRT construction sites.

Field Relations

The Fort Canning Boulder Bed is bordered to the west by a ridge of Rimau Facies and to the east by the Old Alluvium. The Boulder Bed overlies the Jurong Formation, and at Beach Road, it is overlain by the Old Alluvium (Shirlaw, et al. 2003).

Age and Correlation

Sehested (1960) suggested the boulders are fragments from sandstone beds that had been rounded by vigorous wave action and subsequently, clay was deposited between them. Pitts (1984) suggested that the Fort Canning Boulder Bed originated from a landslide that occurred towards the end of the deposition of the Jurong Formation. Han, et al. (1993) suggested it was a Pleistocene landslide deposit. Redding and Christensen (1999) remarked that the Fort Canning Boulder Bed is probably displaced Rimau Facies material of the Jurong Formation partly enveloped in remoulded Queenstown Facies or possibly Cretaceous clay. The displacement occurred during a period of tectonic deformation in late Cretaceous. Shirlaw, et al. (2003) postulated that the Boulder Bed is colluvial material and represents collapsed material from a ridge of Rimau Facies located to the west of the Fort Canning Boulder Bed. The collapse occurred through a series of catastrophic mudflows that had been triggered by seismic activity associated with rapidly rising sea levels sometime between 0.5 to 1 million years ago during the Pleistocene.

As the Fort Canning Boulder Bed is older than the Old Alluvium, its minimum age is late Tertiary. Its maximum age is late Triassic when the contents of Jurong basin were deformed and uplifted and rock masses dislocated, slipped, and were transported down essentially by gravity during the late Triassic as suggested by Pitts (1984). Another probable age is late Cretaceous. This was when the contents of the Jurong Basin was lifted, faulted, tilted, and slipped when the Woyla continental fragment collided with the Sibumasu /East Malaya block (Redding and Christensen, 1999). It is taken here that the probable age of the Fort Canning Boulder Bed is late Cretaceous but it could be any time between late Triassic and late Tertiary.

OLD ALLUVIUM

Definition and Distribution

The term 'Older Alluvium' (OA) was introduced by Alexander (1950) to replace the term 'High Level Alluvium' first used by Scrivenor (1924) to describe the alluvial sand forming the hills traversed by Tampines Road, the PIE in Pasir Ris, Changi Road in Geylang, New upper Changi Road in Bedok, New Upper Changi Road East in Simei, and Upper East Coast Road in Siglap. Walker (in Stauffer, 1973) introduced the term 'Old Alluvium' together with 'Young Alluvium' for the deposits in the Kinta Valley in West Malaysia, and the term 'Old Alluvium' has been adopted by Stauffer (1973) for the Johor-Singapore area.

No formal definition of the Old Alluvium has been presented, and it is therefore proposed that those sediments exposed in the Bedok Sand Quarry and recorded in Public Utilities Board Test hole No. 1 at Bedok be taken as the type 'Old Alluvium' for the Singapore-Johor area. A summary log for Public Utilities Board Test Hole No. 1 is given in Fig. 2.49.

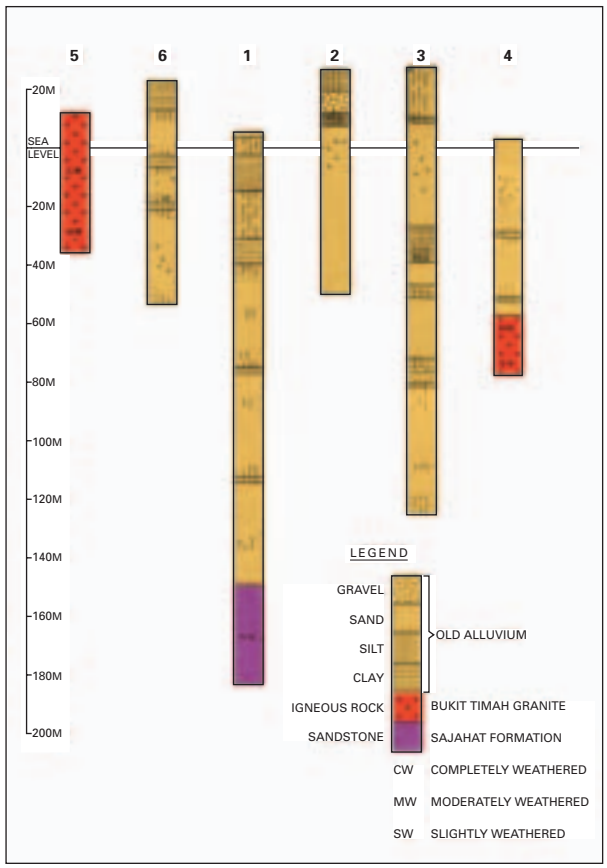


Fig. 2.49 Summary logs of PUB test holes

The Old Alluvium is found lying to the north and northeast of the Kallang River Basin between the central granite and the granite at Changi. Similar sediments, also assigned to the 'Old Alluvium', are found in Sungei Buloh Besar in northwest of Singapore island where they lie against the Jurong Formation.

In Public Utilities Board Test Hole No. 1, the Older Alluvium was found to lie at a depth of -149 m directly on quartz sandstone of the Sajahat Formation that has been contact metamorphosed. Thirty-five-metre-high hills (located close to the test hole) and forty-five-metre-high hills (located 3 km to the southwest) gives a possible aggregate thickness of 195 m. The granite basement probably occurs at -20 to -40 m in the area between Nee Soon and North Seletar. The granite basement was encountered at -53 m in Public Utilities Board Test Hole No. 4 (GR 633515) in Pasir Ris. Ong, et al. (2003), in their investigation on buried granite ridges in the Old Alluvium, documented the depths of the Old Alluvium-granitic bedrock interface at many locations. The two-to-sixteen-metre-thick Old Alluvium at Ang Mo Kio Avenue 1 to Ang Mo Kio Avenue 8 lies at -10.4 to -13.1 m on granitic bedrock. At the Kallang-Paya Lebar-Ubi-Eunos area, the fifteen-to-thirty-metre-thick formation lies at -25 to -50 m on granitic bedrock. The Old Alluvium at Loyang Avenue is 3 to 11 m thick and its interface with the granitic bedrock lies at -15 to -22 m. At Punggol, Lorong Halus, Kaki Bukit, Loyang Avenue, and Tampines, the Old Alluvium-granitic bedrock interfaces lie at -51 to -170 m, -30 to -50 m, -40 to -80 m, -25 to -100 m, and -50 to -200 m, respectively. Ong, et al. (2003) also noted that within the corridor between Punggol and Bedok, no site investigation has reported granitic materials at depths of less than 50 m from the ground surface. The Older Alluvium cuts out against the older sedimentary rock at Punggol and the Bukit Timah Granite at North Seletar. Pulau Ubin is also an igneous mass. There is thus little room to carry the Older Alluvium through to its correlatives in Johor at or below present sea level. However, on the western side of the island, the Older Alluvium can be mapped as continuing through in a northwest direction into Johor.

Content

Details of the content of the Old Alluvium have been documented by Alexander, (1950), Burton (1964, 1973 A), and Stauffer (1973).

At the time of mapping, the Old Alluvium was extensively exposed in excavations at Bedok, in the many sand pits to the north, in coastal exposures between Bedok and Changi, and on the northwest coast opposite Pulau Sarimbun. In all exposures, the formation is seen to be coarse, angular clayey sand with stringers of subrounded pebbles up to 4 cm in diameter (Fig. 2.50). It contains 15% to 30% silt and clay with these fines generally decreasing with depth (Chiam, et al. 2003). The beds are often cross-bedded with coset thicknesses of up to 1.5 m. Cut and fill structures are common and rare clastic dykes can also be seen (Fig. 2.51).

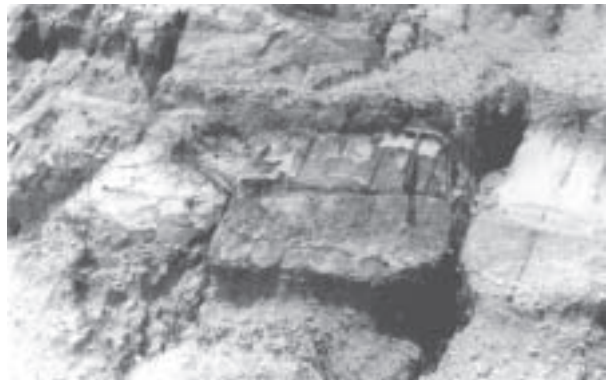


Fig. 2.50 Old Alluvium showing cross-bedded angular sand with layers of pebbles exposed near sand pit of the Nam Kee Sand Quarry (GR 617511) in Tampines

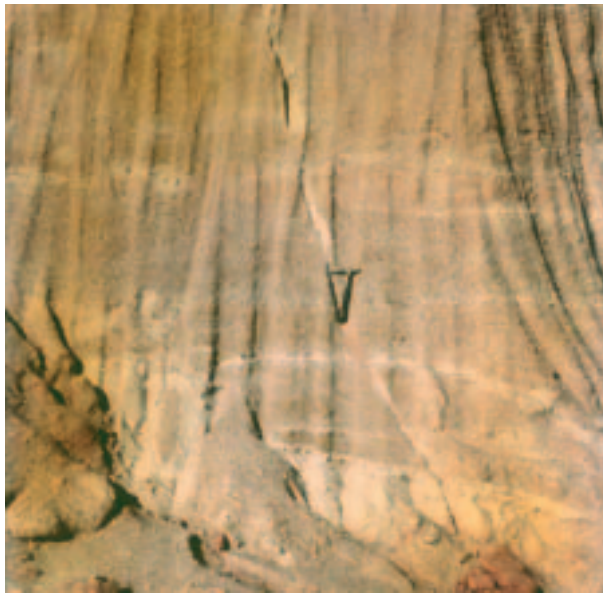


Fig. 2.51 Clastic dyke in Old Alluvium in sand pit at the Bedok Sand Quarry (GR 592475) in Bedok

Fine-grained beds are usually also present as small lenticular bodies. Alexander (1950) mentioned one such bed found near McPherson Road where it measures about 30 cm thick and 14 m in length. Several fine-grained horizons were recorded in the Public Utilities Board test holes usually as fine sand and silt beds. One clay layer at -36 to -45 m was recorded in Public Utilities Board Test Hole No. 1.

The pebbles within the Old Alluvium are dominantly quartz, but rhyolite, chert, and argillite pebbles are also found. The quartz pebbles are generally more angular, the others subrounded to rounded. No granite pebbles have yet to be found in Singapore but Burton (1973 A) recorded them in the Old Alluvium of Johor. The sand is quartzo-feldspathic with the feldspar weathering to clay to varying depths, usually in excess of 8 m. Fresh feldspar is present (Gupta, et al. 1987). Alexander (1950) found that 75% of the heavy mineral suite from a sample collected at the Changi Business Park near Simei was made up of magnetite and ilmenite with zircon, cassiterite and a few grains of monazite making up the rest. Scrivenor (1924) also found staurolite. In addition, Burton (1964), in discussing the Old Alluvium in both Singapore and Johor, also listed topaz, tourmaline, pyrite, anatase, and sphalerite. No plant matter has been found in Singapore, but again Burton (1973 A) recorded such finds in Johor. Tai (1972) collected a single marine fossil from the Old Alluvium in Singapore and Alexander (1950) recorded the presence of a water-worn gastropod preserved in black silica at the Kampong Eunos Earth Quarry, in the vicinity of GR 580490 and about 1 km from Paya Lebar Air Base. From her description, it is however assumed to be a derived specimen with a coating of the secondary silica mineral.

The formation is usually uncemented but quite dense with low permeability. Pfeiffer (1972) gives figures of 10^{-8} to 10^{-10} m/s for the permeability in the weathered zone of the Old Alluvium. Away from the weathered zone, permeability could be higher. Chiam, et al. (2003) found that the Old Alluvium has pockets of sand, confined aquifers with water under artesian pressures, and beds of permeable cemented sands. A one-kilometre-long bed of cemented sand, located along the Pan Island Expressway between Bedok and the Eunos Flyover (GR 568467), is reported in Chiam, et al. (2003). It has an in-situ permeability of 6×10^{-8} to 5×10^{-5} m/s. Zones of cementation are found but the cemented rock often disintegrates after a few days of exposure. Alexander (1958) suggested the rock was lightly cemented with silica or possibly alumina. She then proceeded to show that a loss in weight on exposure to the Singapore climate can be recorded for water-clear quartz crystals and suggested then that the silica cement dissolved on exposure disaggregating the rock. Her uncompleted experiment showed an average loss of 0.005% by weight over six weeks, but it is considered to be too slow to explain the breaking down of large blocks within days of exposure. Colloidal silica cement is thought to be more likely to respond to weathering within that interval of time.

Some areas of cemented Old Alluvium have been exposed for several years and yet have remained cemented. These areas around GR 575480 in Kaki Bukit and GR 653497 in Tanah Merah are plotted on the map. No study of the cement in these areas has been made.

Field Relations

The contact between the Old Alluvium and the older formations is presently buried or obscured by deep weathering. Alexander (1950) observed tongues of Old Alluvium lying on clay that she identified as weathered granite at Changi. The Old Alluvium is considered to lie on sandstone in Public Utilities Board Test Hole No. 1 and to lie on granite in Public Utilities Board Test Hole No. 4 at depths of -149 m and -53 m respectively. Basement was not reached when Public Utilities Board Test Hole No. 3 at GR 604501 in Tampines was terminated at -122 m.

To the north near Punggol, the Older Alluvium has been laid down against the Palaeozoic sediments and in Sungei Buloh Besar, the Old Alluvium was laid down against the Jurong Formation. In neither case is the contact seen. It is possible that part of the contact in the Sungei Buloh Besar area is faulted.

In the central business district in downtown Singapore, the Old Alluvium is in contact with the eastern edge of the Fort Canning Boulder Bed and postdates it (Shirlaw, et al. 2003).

The general texture of the Old Alluvium, as exposed in Singapore, is consistent with that of an alluvial fan or piedmont-plain type deposit. No evidence of marine incursion could be seen, but the presence of 9 m of clay and some fine-grained beds in Public Utilities Board Test Hole No. 1 and other deep boreholes may indicate marine deposition. No firm correlations could however be made from one hole to another.

Verstappen (1975) discussed the climatic changes in the Southeast Asia area during the Pleistocene and described conditions that would allow the development of such piedmont plains.

Age and Correlation

There is no direct evidence for the age of the Old Alluvium in Singapore. Scrivenor (1924) originally assumed it to be of Quaternary age and Alexander (1950) listed it as possibly Pleistocene. Burton (1964) stated that its deposition may at the most extend back to late Pliocene.

Stauffer (1973) listed finds of a mid-Pleistocene elephant tooth (Palaeoloxodon namadicus) from the Kinta Valley, Malaya and remains of a rhinoceros, suid deer, turtle shells, and catfish spines which are regarded as probably of Pleistocene age.

Three radiometric ages were listed in Stauffer (op. cit.) from Sungei Besi, Malaya. Two samples gave C14 ages of more than 41,200 and 41,500 respectively and are thus beyond the limit of the method. The third yielded an age of 36,420 (+1,255, -1,085) BP from wood apparently in the position of growth. Radiocarbon determinations on wood and peat from the Kinta Valley proved to be beyond the range of the method. The general character of the formation however suggests an age greater than 36,000 years and this date should be accepted with caution. It would require less than 1% contamination by present-day organic material to produce a figure of this order.

Apart from the characteristics of deep weathering and the common occurrence of slumping due to the solution of the underlying bedrock and the degree of dissection as listed in Stauffer (1973), the occurrence of Old Alluvium in deep troughs must also be considered in a discussion of its age. The Old Alluvium is found to a depth of -149 m in Singapore. Using an acoustic continuous profiler, Aleva (1973) plotted channels cut to a depth of -100 m and filled them with an 'Alluvial Complex' between the Singkep and Bangka Islands and again around the Karimata islands, an area about 3° of latitude to the south and southeast of Singapore. These channels lie beneath a 'Younger Sedimentary Cover' and below an extensive 'Marine Abrasion Surface' now at a depth of -20 to -30 m. It would appear reasonable to correlate Aleva's Alluvial Complex with the Old Alluvium and his 'Younger Sedimentary Series' with members of the Kallang Formation.

He described the Alluvial Complex as lying on an 'Older Sedimentary Cover' and filling deep troughs within this Older Sedimentary Cover. He recognised evidence for repeated vertical movements of the erosional base level and small-scale block faulting during the deposition of his Older Sedimentary Cover. Palynological determinations on the Older Sedimentary Cover indicate a Miocene-Pliocene age but gave no clear distinction between the Older Sedimentary Cover and the Alluvial Complex. The two units also showed similar lithologies in drill cores. Aleva suggested that an upper Tertiary to Pleistocene age would be reasonable for the deposition of both the Older Sedimentary Cover and the overlying Alluvial Complex.

From the evidence cited above, it appears that the Old Alluvium of Singapore must date from a period during which tectonic movements were still taking place. It cannot reasonably be argued that erosion has cut a channel in Singapore to a depth of 149 m below sea level, so far from the postulated edge of the Pleistocene Sundaland without it being a modification of a downfaulted or downwarped zone. It is possible, however, that the buried valley is partly filled with the equivalent of Aleva's Older Sedimentary Cover, but this has not been recognised in boreholes or geophysical exploration.

Tertiary warping has been recognised in the semi-craton of the Asian lithospheric plate. Folding of Tertiary sediment accompanying faulting has been recognised along the Kuala Lumpur-Endau fault zone in Malaysia (Gobbett and Tjia 1973). Renwick and Rishworth (1966)

reported late Tertiary sediments dipping at up to 25° to a minimum level of 144 m below sea level from kampong Bukit Kepong in Johor. The areas of block faulting cited above and those recognised by Aleva (1973) also lie within the Asian lithospheric plate.

As warping or faulting appears to continue through to at least to late Tertiary time and the deposition of the Old Alluvium appears to be tied to this faulting, deposition of the Old Alluvium must have commenced by the end of the Tertiary.

There is insufficient data to reconstruct the original aggradational surface of the Old Alluvium. The top of the unit in Singapore lies at 46 m. In Johor, it lies at 70 m. Marine beds occur at this height. It could even be at 138 m in Johor though this height is doubtful. The Old Alluvium is dominantly a terrestrial deposit; thus, its deposition must be tied to a base level that must at times be as high as 70 m. Aleva (1973) mapped his Alluvial Complex as predating his upper planation surface which he correlated with the Riss-Wurm Interglacial. Verstappen (1975) showed that a climatic change with lower precipitation, rainfall, and temperature would allow for the deposition of coarse, poorly weathered sediment similar to that of the Old Alluvium, and thus deposition can be associated with climatic changes accompanying the Pleistocene glaciations. It would follow that during the warm interglacials, a more deeply weathered, finer-grained sediment would be deposited. Such sedimentation has not been recognised within the Old Alluvium, but erosion of the Old Alluvium during the Wurm Glaciation is recognised and fine sediment is deposited in the eroded channels. Burton (1964) in his report on the Older Alluvium of Johor and Singapore concluded that it was related to a 75 m sea level stand during the pre-glacial of the First Interglacial (Gunz-Mindel or Aftonian) but added that the formation may be dated back to late Pliocene. The First Interglacial occurred at the end of early mid-Pleistocene.

The upper age limit of the Old Alluvium is thus taken to coincide with the climatic warming following, at the latest, the Riss Glaciation, but it is probably earlier at the end of early mid-Pleistocene because of the absence of 'normal' sedimentation within the Old Alluvium.

Variations in the term Old Alluvium and its predecessors have been used in Singapore and West Malaysia since Scrivenor first recognised the unit in Singapore. The correlatives in this region have been discussed in Stauffer (1973). The Alluvial Complex of Aleva, et al. (1973) is also correlated with the Old Alluvium, and deposits underlying the Palembang peneplain and described by Verstappen (1975) also appear to be correlatives.

It can also be argued that the bulk of the Old Alluvium is a late Tertiary deposit rather than an early Quaternary one. The formation occurs at heights of 46 to 70 m. Such heights suggest that sea levels must have been considerably higher than present sea level. Sea levels in late Tertiary were probably higher than sea levels during the Quaternary. The highest sea level during Quaternary times were a few metres higher than present sea level, and they were attained during the initial phase of the Riss-Wurm Integlacial from about 120,000 to 140,000 BP. It is suggested here that the age of the Old Alluvium spans from late Tertiary to early mid-Pleistocene.

HUAT CHOE FORMATION

Definition and Distribution

The Huat Choe Formation (HC) is defined as those mostly fine-grained, ponded, terrestrial sediments exposed in clay pits 1.5 km north of Huat Choe Village and 1 km northeast of Nanyang Technological University (Fig. 2.52). Huat Choe Village once stood on the ground across Pioneer Road, north from the Jurong West Sports Complex. Such deposits do not appear to have been recorded in Singapore before 1976.



Fig. 2.52 Shallow-dipping beds of clay and sandy clay of the Huat Choe Formation at GR 331494 near Pioneer Road

The formation has been recognised in only one area and is of limited extent. It covers an area of approximately 400 by 200 m, and from the lowest exposure to the upper eroded surface it is not more than 6 m thick. It has been separated out from the other Quaternary deposits because of its different depositional setting, its supposed younger age and its economic use as material for pottery making.

Content

The formation is made up of poorly bedded, white kaolin-rich clay with minor amounts of quartz gravel. The clay is similar to that in the low-lying areas of deeply leached Jurong Formation rocks nearby, and it is probable that the clay was derived from these rocks. Plant remains (mainly sedges) and the shells of an unidentified land snail were found within the deposits, but no systematic work has been done on these remains. Although the sedges were in-situ, it needs to be established that the land snails, whose shells show little sign of leaching, have not been introduced accidently into the clay pits during extraction of the clay for pottery works.

Field Relations

Exposures show the formation as lying unconformably on an eroded, moderately weathered surface of moderate relief cut in the Jurong Formation. It lies just to the east of a fault boundary, and thus lies within the down-thrown Jurong area. It is suggested that it had been laid down as a lacustrine sediment within this area during the period of faulting. It is probable that other (as yet unrecognised) similar deposits occurred close to the fault.

Age and Correlation

There is little evidence as to the age of the Huat Choe Formation. There is no evidence of lithification or obvious in-situ weathering (unless it was leached in place) suggesting any great age. Its position does not relate to the present-day drainage pattern, but more to the position of the fault mentioned above. There is no evidence to date the last fault movement. Burton (1973 A) observed tight folding and minor faulting in the Old Alluvium in Johor, and thus movements in the early Pleistocene or even younger can be postulated. It is suggested that the age of the Huat Choe Formation be taken as early mid-Pleistocene or younger.

TEKONG FORMATION

Definition and Distribution

Sediments forming the wide terraces (Fig 2.53) standing between 3.6 and 5.5 m, usually around 4 m, above sea level on Pulau Tekong Kechil and Pulau Tekong are assigned to the Tekong Formation (T). Sediments beneath surfaces within the same height range on Pulau Ubin and at Changi as well as remnants of terraces along the northeast and southwest coasts of Singapore Island are also correlated with the Tekong Formation. Also, correlatives of the formation are terrace deposits found at a similar height in the tidal reaches of Sungei Serangoon, Lower Seletar Reservoir, Kranji Reservoir, and Sungei Pandan.



Fig. 2.53 Terrace of the Tekong Formation on Pulau Tekong Kechil. Its average height is 4 m above mean sea level. Viewed from Pulau Sajahat

Boreholes in the type area on Pulau Tekong Kechil pass through some 5 m of sand before encountering yellow clay containing quartz pebbles. As drilling did not continue into this clay, it is not known if this represents the base of the formation. Quartz pebble beds were not recognised in the underlying Palaeozoic rocks although quartz veins were seen. Quartz pebbles were recognised within the Tekong Formation at the type locality.

Content

The borehole records describe the formation as being loose, fine to very fine, light brown sand with peat, wood fragments and occasional quartz pebbles. An exposure by the river near Changi Point (GR 659538) at Changi (GR 659538) reveals well-sorted, light brown, slightly iron-stained, fine to medium quartz sand. Drilling records also report the presence of fragmented shells within the cuttings.

Field Relations

The base of the unit has not been recognised with certainty. The formation is however seen to overlie each of the pre-Tertiary formations and it is assumed that it lies unconformably on them. Young terraces have been cut into the Tekong Formation, particularly along the Pasir Panjang coast.

Apart from being finer-grained, the Tekong Formation shows characteristics similar to present-day sand banks south of Pulau Tekong, and it is suggested that the unit represents beach and offshore sand bank deposits, which were tied to a previous high sea level stand of about 6 m. Shell fragments in the Tekong Formation supports the suggestion.

Bird, et al. (2003) states that it is best to consider the Tekong Formation as a transitional unit between the terrestial sediments of the Alluvial Member and the peaty sediments of the Transitional Member of the Kallang Formation.

Age and Correlation

The Tekong Formation was probably deposited when sea level stood at about 6 m. Its deposition could be associated with the 'Daly Levels'. They are variously quoted at 5 or 6 m and the 'Daly Levels' are dated as having occurred between 6,000 and 7,000 BP (Tjia, 1970, Tjia, et al. 1972, and Fairbridge, 1961). It could also have been deposited when sea levels were lower, at 2 to 2.5 m (Bird, et al. 2003). The highest astronomical tide associated with this sea level in mid-Holocene could have fashioned the 3.6 to 5.5 m high terraces of the Tekong Formation (Bird, et al. 2003).

It has been argued by Nossin (1964) that beach ridges are still being constructed to heights of 6 m along the east coast of Malaya. However, high terraces recognised in Singapore all lie in sheltered zones, on islands within the Johor River mouth area, and within the sheltered tidal reaches of rivers flowing north into the Selat Johor. Hence, such aggradation is not to be expected as wave energy, even at around the more exposed southern and southwestern islands, is low. Swan (1971) stated that waves of 1 m height were exceptional and these were observed to the south-west at Raffles Lighthouse on Pulau Satumu.

The subsequent cutting of terraces in the Tekong Formation on the southwest coast lends support to the hypothesis that these terraces are not present-day features.

Deposits of a similar elevation have been reported from Indonesia and Malaysia, and these deposits are correlated with the Tekong Formation. Tjia (1970) listed terraces at 6 to 8 m on Bangka and Billiton and referred to the presence of further terraces of similar heights in the Indonesian Archipelago.

It is suggested here that the Tekong Formation is a mid-Holocene deposit.

KALLANG FORMATION

This formation includes the sedimentary deposits discussed by Alexander (1950) under the heading of 'Recent Alluvium and related deposits' and the recent deposits referred to by Scrivenor (1924). The formation is named after the Kallang River Basin where it is most extensive, but no type area is proposed as the formation is poorly exposed. Most of the evidence for the existence and subdivision of this formation comes from boreholes and the physiographic setting of the deposit. The formation is found along the coastline and extends into the headwaters of the rivers draining Singapore. Its most extensive development is around Sungei Kallang, Sungei Jurong, Kranji Reservoir, Sungei Serangoon, and Lower Seletar Reservoir. Extensive deposits are also found on Pulau Ubin, the southwestern group of islands, and the southern group of islands. The formation also includes reef deposits exposed to the south and southwest of Singapore at low tides.

The deposits are generally low lying and are seldom recognised more than 4 m above sea level except in the more inland areas.

Five members are recognised within the formation, and these are referred to informally as the Marine Member, Alluvial Member, Littoral Member, Transitional Member, and Reef Member. Two divisions can be recognised in the Marine Member in certain boreholes.

MARINE MEMBER

Definition and Distribution

Two divisions can be recognised within the Marine Member (Km) in boreholes. The member is however not recognised anywhere at the surface. The two divisions, referred to as the Upper Marine Member and the Lower Marine Member, are best known in the Kallang River Basin area where they can be found within 1 m of the surface.

Undifferentiated Marine Member sediments are found in Sungei Jurong, Sungei Pandan, Kranji Reservoir, Lower Seletar Reservoir, Sungei Bedok, Sungei Punggol, and at Changi. The member has been recognised in boreholes as extending inland at least as far as Stevens Road in Bukit Timah Valley (GR 479461), Thomson Road in Whampoa Valley (GR 499459), and Braddell Road in Kallang Valley (GR 514485). The maximum thickness recorded to date on Singapore Island is some 35 m at GR 518553 close to the Rochor Canal Road (GR 518553). At 1 km further to the west at GR 511435 in the vicinity of Parco Bugis Junction and InterContinental Hotel, the upper surface of the Marine Member may lie between 1 and 1.8 m above sea level. Greater thicknesses in excess of 55 m have been reported in Tan, et al. (2002) for a reclamation site off Pulau Tekong.

Content

The base of the member is usually characterised by peaty clay sand that is seldom more than 3 m thick. This material passes up into a soft blue grey mud or sandy mud. Thin sand and peat layers may occur within the sequence. Tan (1972) stated that the mud contained an average of 50% clay fraction. However, Pitts (1983) noted the member has a clay fraction that is usually more than 50%, a constant silt/clay ratio, and a high but variable organic content. Tan, et al. (2002) remarked that its principal clay mineral is kaolinite with accessory amounts of illite and smectite and that the kaolinite in the Lower Marine Member has a compact structure, in contrast to the well-flocculated structure of the clay mineral in the Upper Marine Member.

If the Lower and Upper Marine Members occur, the boundary is recognised as the top of stiff reddish brown silty clay or sometimes as a bed of loose sand. The grey muds that are separated by the boundary are indistinguishable. This boundary usually lies at -10 to -20 m but can be found down to -28 m. It lies deeper at -30 m off Pulau Tekong (Tan, et al. 2002). Peat deposits

may also be associated with this horizon. It is assumed that the stiff silty clay represents a zone of desiccation, alluvial or marine reworking, and colonisation by plants during a period of low sea level.

Shell fragments are frequently found in the Marine Member. Dr Lim Chuan Fong (pers. comm.) identified the fragments to have come from a shallow muddy marine environment. Similar faunal assemblages are found in similar settings off the coast of Singapore in present times.

Field Relations

The Marine Member is the oldest unit in the Kallang Formation and it lies unconformably within valleys cut in the Bukit Timah Granite, Jurong Formation and Old Alluvium. The basal contact has been recognised only in boreholes, and the rocks beneath the basal contact are usually only moderately weathered.

The boreholes showing the inland extent of the Marine Member also indicate that the Marine Member interdigitates with the Alluvial Member.

Age and Correlation

The Marine Member fills valleys cut to at least 55 m below sea level at a time of low sea level. It has been deposited in two phases to attain a maximum height of 1.8 m. It is thus suggested that deposition started during a period of marine transgression following a low sea level stand. Deposition was interrupted by a second lowering of sea level to at least -30 m, after which, sea level rose again to 1.8 m.

Biswas (1973), working off the northeast coast of Malaya, recognised the last Quarternary fall in sea level as corresponding with the Pleistocene-Holocene boundary and dated plant matter from a buried mangrove swamp (now at 65 m below sea level) as having been deposited at 11,170 BP. There is neither evidence of any subsequent lowering of sea level of the order required to erode channels in Singapore to a depth of -30 m, nor is there evidence to suggest any major earth movements in Singapore during the Holocene. Thus, it is suggested that the Upper Marine Member had been deposited during the transgression following the 11,000 BP sea level. Deposition presumably continued until the fall in sea level from 1.8 m to present sea level some 3,000 BP (Tjia, 1970). It then follows that the Lower Marine Member was deposited during the Riss-Wurm Interglacial.

Bird, et al. (2003) describes the chronology of events that accounts for the deposition of the Marine Member. The Lower Marine Member was deposited when sea level rose following the end of the penultimate glacial (Riss) stage at about 140,000 BP. Deposition continued till the start of the last glacial (Wurm) stage at about 85,000 BP. The stiff reddish brown clay was developed at the top of the Lower Marine Member as it was subjected to soil-forming processes during the Last Glacial Maximum at about 20,000 BP, when sea level stood at -120 m. The Upper Marine Member was deposited when the sills to the east and west of Singapore Straits were breached as the sea level rose to about –25 m about 10,000 to 11,000 BP. As sea level began to drop after 6000 BP, the peat facies of the Transitional Member and sands of the Alluvial Member of the Kallang Formation were deposited over earlier sediments and the Upper Marine Member.

The age of the Marine Member of the Kallang Formation spans from the late Pleistocene to the present.

ALLUVIAL MEMBER

Definition and Distribution

The Alluvial Member (Ka) is found as valley fills throughout Singapore and as a thin veneer on the floor of the Kallang and Jurong River basins. It is of limited extent in the offshore islands. The member has been mapped from its physiographic setting and from borehole data.

Content

The deposits in the Alluvial Member vary from pebble beds through sand, muddy sand, and clay to peat (Fig. 2.54). No study has been made of these variations. Extensive peaty sand and mud are found in the Kallang and Jurong River basins.



Fig. 2.54 Horizontal layers of peat, muddy sand, and mud of the Alluvial Member exposed in excavation at GR 423480 near junction of Upper Bukit Timah Road and Jalan Anak Bukit, Bukit Timah

Field Relations

Like the Marine Member, the Alluvial Member overlies the older pre-late Pleistocene formations and was seen in boreholes to interdigitate with the Marine Member. It can be assumed to interdigitate with the other members of the Kallang Formation.

Age and Correlation

Interbedding of the Alluvial Member with the Marine Member can be recognised in boreholes at least to the base of the Upper Marine Member. Chang (1995) believed that the Alluvial Member is the oldest member of the Kallang Formation instead of the Marine Member. From samples of alluvial sand and clayey sand that underlie the Marine Member at Sungei Nipah in Pasir Panjang, he obtained dates that range from 60,000 BP to more than 137,000 BP. Similar studies by Taylor, et al. (2001) lend support to Chang's belief. Alluvial sediments continue to be deposited on the valley floors of our rivers in present day. The age of the Alluvial Member spans from mid-Pleistocene to the present.

LITTORAL MEMBER

Definition and Distribution

The Littoral Member (Kl) includes those sediments deposited in active coastal regions as beach deposits, immediate offshore deposits and tidal sandbanks. It also includes buried and raised-beach terraced deposits.

The member is found along the southern and eastern coast of Singapore and on the offshore islands. It also occurs as shoals south of Pulau Tekong, east of Pulau Ubin and Pulau Seletar, and as small isolated strips along the northern coast. The member is recognised in boreholes between Kallang and Bedok to -10 m, but it is normally less than 5 m deep. It is found building beach ridges and as terraces up to a height of 2.8 m on Singapore Island and the islands in the northeastern group of islands, and up to 3 m on the more exposed islands in the southwestern group of islands. Beach rock found on the southwest-facing coast of Pulau Tekulor and St. John's Island is also included within this member.

Content

Clean sand and pebbly sand is the most common lithology within the member. The sand is dominantly quartz, but lateritic, shell and lithic fragments also occur. Swan (1971) stated that the sand may contain up to 60% calcareous matter. He also reported feldspar as being present in the sand at Kampong Bahru on Pulau Ubin and on Pulau Sekudu. Heavy minerals derived from the igneous rocks may also be found on Pulau Ubin.

The beach rock at Pulau Tekukor is iron-cemented quartz sand or a lithic conglomerate. It dips seawards at the same slope as the present beach (Fig. 2.55). It lies at a height of a metre or less above the present beach and extends beneath present low tide level. Alexander (1950) reported the presence of primitive stone tools in the beach deposits.

Field Relations

The Littoral Member is seen to lie conformably on the Marine Member in boreholes and unconformably on the older formations. It is assumed to interdigitate with the Transitional, Alluvial and Reef Members, and can also be expected to interdigitate with the Marine Member.



Fig. 2.55 Hard iron-cemented beach rock of the Littoral Member exposed during low tide along the southwest coast of Pulau Tekukor

Age and Correlation

The Littoral Member is essentially a present-day deposit, but terraced deposits associated with it are found up to a height of 2.8 m. The terraced deposits are associated with the 'Daly' periods of higher sea level which have been dated at 5,000 BP and 3,500 BP (Tjia 1970) (Fig. 2.56). The extension of beach sand to -10 m may indicate deposition during the later stages of the post-Wurm transgression, but it is considered more likely to represent recent offshore deposits laid down below sea level.

The presence of beach rock below present low tide level suggests deposition when the sea stood at 1 to 2 m below the present level. Tjia (1970) recognised a -2 m sea level as being common on the Sunda Shelf and gave an age range of 1,500 BP to 3,000 BP for the occurrence of this -2 m sea level (Tjia op cit).

Though not investigated, the Littoral Member is expected to underlie the Upper Marine Member. It is therefore suggested that the age of the Littoral Member spans from the late Pleistocene to the present.



Fig. 2.56 A raised, dark brown pebble beach of the Littoral Member at 2 m above sea level lying on mudstone and siltstone at Pulau Jong

TRANSITIONAL MEMBER

Definition and Distribution

The Transitional Member (Kt) is found in the river mouths and tidal swamps surrounding Singapore, particularly in these environments at the western end of the islands. The enviroments are typified by the presence of mangrove swamps. Extensive areas of Transitional Member may be found on Pulau Ubin, Pulau Ketam, and Pulau Seletar. The member however is not recognised in the Kallang Basin, but mangrove swamps were plotted on the early maps of Singapore. These areas are presently buried under reclamation. The member also includes more deeply buried mangrove deposits.

Content

The Transitional Member is deposited in a low energy environment. The member is an unconsolidated black to blue-grey mud, muddy sand, or sand with a high organic content. The high organic content, general low energy conditions, and the slow rate of deposition give rise to a general reducing environment. The sand is mainly of quartz derived from the nearby rocks, and mica is found with the finer-grained sediments.

Field Relations

This member is found from borehole evidence to overlie the Marine Member and also the Littoral Member. It is also seen to lie on the cut surface of the older formations on Pulau Ubin and the southwestern group of islands. It was seen in excavations at Kranji Reservoir (GR 378573) to interdigitate with the Marine Member.

Age and Correlation

Much of the member has been mapped on its physiographic setting in the present-day environment, and hence by definition, it is a recent to present-day deposit. Buried older Transitional Members are however expected to be present between the Lower and Upper Marine Members. It is suggested that the age of the Transitional Member spans from late Pleistocene to the present.

REEF MEMBER

Definition and Distribution

The Reef Member (Kr) is recognised only around the southern, western, and southwestern group of islands. The reefs are exposed during periods of low tide as broad platforms surrounding the islands or as disconnected flat-topped shoals associated with former islands. It may also underlie parts of Tanjong Chek Jawa at the eastern end of Pulau Ubin (Teh, unpubl. data) and the Singapore Art Musuem (GR 505431) in downtown Singapore (Bird, unpubl. data). Hesp, et al. (1998) reported relict raised corals at mean to 0.5 m above mean sea level.

The coral sand making up this member is quite thin in the central portion of a number of the reefs, and it may wedge out exposing the underlying Jurong Formation. The thickness of the fore reef deposits however is unknown.

Content

The reefs are made up of coral and its detrital fragments. These fragments grade down to sand-size particles. Swan (1971) stated that 99.9% of the sand on the coral reefs is calcareous but that quartz and heavy secondary ferruginous grains can also be found. A higher percentage of quartz and secondary minerals can be found on the reefs attached to present-day islands.

Field Relations

The Reef Member lies unconformably on a marine cut surface in the Jurong Formation but may interdigitate with the Littoral and Alluvial Members.

Age and Correlation

The corals on the reefs are still actively growing and the top surface is thus related to present sea level. The underlying Jurong Formation rocks have been planned off to a level corresponding to present low tide level or to a level associated with a sea level 1 to 2 m below that of the present day. The base of the member is also related to these sea levels. Hesp, et al. (1998) suggested that the relict raised corals found at mean to 0.5 m above mean sea level grew during the mid-Holocene high sea level stand at about 6,300 to 6,500 BP. The Reef Member is thus a mid-Holocene deposit laid down from 6,500 BP to present.

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STRUCTURE 3

STRUCTURE

Singapore lies at the southern tip of the Malay Peninsula and is structurally an integral part of it. The rocks of the Republic show the same northwest trend as that running through from Kulau Lumpur to Johor Bahru in West Malaysia and on the south through the islands of the Riau Archipelago, Banka and Billiton. The structures that have affected the Triassic sediments are thought, in the Singapore area, to have resulted from tectonic activity which started prior to their deposition in early Triassic time and continued through as a relatively mild process until at least early Jurassic time. The tectonism appears to have been less violent than that recognised in Malaya and also to have lagged slightly behind it in time. Singapore was subjected to stronger tectonic activity in late Cretaceous.

Pre-Mesozoic rocks recognised in Singapore include the gabbro-norite described by Hutchison (1964) from Bukit Gombak, the Palaeozoic Volcanics (PV), and the sedimentary rocks of the Sajahat Formation (S).

A third period of faulting, dominantly block faulting, occurred probably in Pliocene time, and it is evident from the topographic expression in the Triassic sediment and granite, and by the depth to basement beneath the Old Alluvium (OA).

PRE-MESOZOIC STRUCTURE

The sediment of the Palaeozoic Sajahat Formation is folded, stretched and faulted, but the evidence for a pre-Mesozoic age is not positive. The folding style in the fine sandstone of the Sajahat Formation is different from that in the younger rocks and hence it is considered that the diastrophism affecting them is older. The minor folds are angular rather than rounded, the cleavage is better developed than in the Triassic rocks, and boudinage is more common. The rocks are not sufficiently well exposed to allow the collection of data for structural analysis, but those observations made show the two dominant strike directions to be clustered around 030°, dip 45W and 155°, dip 25E. No major folds were recognised and no data was collected on the minor folds, joints, or faults. Joints in the Palaeozoic Volcanics, quoted earlier as being vertical and trending due north or at 125°, were seen in only one locality. Grubb (1968) recognised a west-northwest (112.5°) and northeast (45°) trend in joints throughout the Pengerang area, Southeast Johor. This observation constitutes a clockwise rotation of 20° to 40° from those observed in the Singapore area.

MESOZOIC STRUCTURE

Hutchison (1973 B) postulated that the granite in Malaya was forced up during the uplift of the axial zone and spread out to form the Main Range and East Coast Batholith systems. The granite of Singapore lies in the East Coast Batholith system (Hutchison, 1973 A).

As uplift continued, a northwesterly trending downwarp paralleled the axis of uplift. The resulting trough was the back arc sedimentary basin mentioned in Hutchison (1973 A. Fig. 7). That this trough was mobile throughout the Triassic is evidenced by the presence of intraformational breccia (Fig. 3.1) and by the rapid variation in facies and alternations of biofacies within it.



Fig. 3.1 Intraformational breccia of black mudstone in sandstone at GR 378499, Bukit Batok

Sedimentation in this trough continued through to at least the early Jurassic when the St. John Facies (Jsj) containing the Mt Guthrie fossil fauna was deposited. The locus of uplift presumably migrated south so that the sedimentation was terminated by a later phase of the same event responsible for the uplift of the Malayan Peninsula in Triassic time.



Fig. 3.2 A syncline in the Jurong Formation along Benoi Sector (GR 327458) at Jurong showing interbeds of the fossiliferous and tuffaceous Ayer Chawan Facies and the red-brown sandstone of the Tengah Facies. The fold is asymmetrical about a sub-horizontal axis trending at 110°. The inter-limb angle is about 80° and the axial plane of the fold dips west-southwest at about 065°. Beds of both facies are deeply leached at the upper surfaces, and hence they can no longer be differentiated by colour.

Folding

It is suggested that the folding of the Triassic sedimentary pile started before the cessation of sedimentation in early Jurassic time by a sliding of the rock mass to the northeast against the buttress of granite. The folds vary in style, from open through vertical isoclinal to isoclinal over-folds but are normally open folds (Fig. 3.2 and Fig. 3.3). Parasitic folds can be found on the limbs of larger folds and these are responsible for the rapid variation in the strike and dip that may be observed. Most beds strike northwest but the dip may vary over a short distance from 10° to 90°, and field studies show that overturned beds dipping between 70° and 90° are common. Shallower dips in overturned beds were not found. Few fold axes can be followed for more than a kilometre, and although the fold axes are seen to be sub-horizontal in general, folds may plunge steeply at either end.



Fig. 3.3 An anticlinal fold at Tanjong Lokos, St. John's Island

The fine-grained sediment, particularly that of the Queenstown Facies (Jq), is sheared obliquely to the bedding (Fig. 3.4). It is suggested that these beds frequently failed incompetently during folding while the more rigid sandstone beds and conglomerate beds, particularly those of the Rimau Facies (Jr), were folded or tilted possibly as discontinuous bodies.

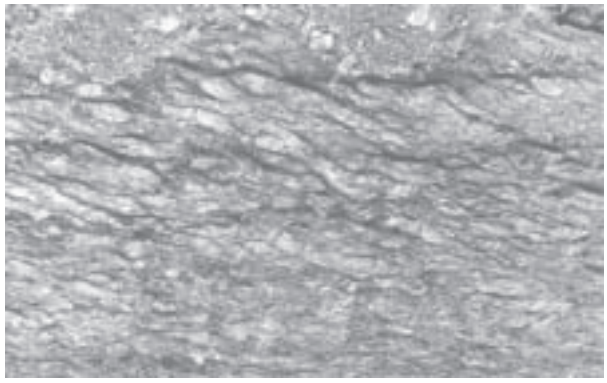


Fig. 3.4 Oblique shear developed in the Queenstown Facies exposed in an excavation off Pepys Road, Bukit Panjang

Isoclinal overfolds are recognised both in the city area and at Jurong, near Benoi Sector (Cross-Section B) and these show translation to the northeast. From these observations and with the existence of a low-angle thrust striking south-southeast from Nanyang Technological University, a northeasterly direction of slumping is postulated.

Redding and Christensen (1999) stated that much of the deformation of the Jurong Formation occurred not in the Triassic but in late Cretaceous times when the Woyla continental fragment collided with the Sibumasu/East Malaya block. The limestone, the red mudstone of the Queenstown facies, and the conglomerate and sandstone of the Rimau Facies and Jong Facies (Jj), being originally deeply confined, older, and more well lithified, have not been excessively deformed. The original sedimentary layering of the limestone at the Pandan Reservoir and the conglomerate and sandstone at Labrador Park has been retained. The shallower facies, especially the facies containing fine sediments such as the Ayer Chawan (Jac) and the St. John Facies, being less well confined and lithified, suffered much small-scale fold deformation. They also remarked that, on the whole, the Jurong Formation has not experienced major folding and in many cases, the original bedding planes have remained flat lying.

Faulting

Very few faults are exposed, and those that have been observed are usually small-scale features. Most of the faults plotted have been recognised on the alignment of erosional features, abrupt changes in topography, or detailed field evidence of lithological discontinuities. There is little evidence for the age of the faults. The only faults that can be assigned positively to the Mesozoic period are the three thrust faults and associated tear faults which are named as the Murai, Nanyang, and Pasir Laba faults (Map Sheets 1 and 4). The Murai Fault trends northeast for 3.2 km and then swings around to the southeast for a further 0.6 km. Movement on this fault has been sufficiently intense to be responsible for the dynamic metamorphism of the Triassic sediment on the southern overthrust side. It is suggested that the principal horizontal stress was from the southwest with the main thrust front now being represented by the comparatively small southeasterly trending arm of the fault and that the northeasterly trending arm is a tear fault with a reverse component. The whole mass has over-ridden the normally northwest-trending Triassic sediments.

Schist is found to be associated with the Pasir Laba Fault, but not with the Nanyang fault. Schist is also found within the Jong Facies in Jurong but no associated fault was recognised. The sediment associated with the Nanyang fault is a muddy and tuffaceous sand compared to the dense mudstone or sandstone found metamorphosed with the other two faults. Hence, a more plastic-type failure is proposed for the muddy and tuffaceous sand near the Nanyang Fault.

Schist pebbles have been found in the Triassic conglometrate in Singapore and these are identical to the Murai Schist. Thus, it must be argued that the thrusting described above occurred before the cessation of Mesozoic sedimentation in the Singapore area. The apparent association of an isoclinal overfold, assumed in the previous section to have been developed in Mesozoic time with the Nanyang Fault (Cross-Section B), also suggests that faulting took place in Mesozoic time.

Most of the faults affecting the Jurong Formation strike between north-northeast and northeast, with a smaller percentage striking between north and northeast. In the granite, the faults strike between northeast and east-northeast or around west-northwest.

The northeasterly trending faults affecting the Jurong

Formation are also considered to be tear faults between rafts of sediment sliding to the northeast. The Henderson Road Fault (Map Sheets 4 and 5) is well defined by the offset of conglomerate at Tanjong Berlayar and the discordance of bedding on either side of Henderson Road. The Tanjong Lokos Fault (Map Sheet 5) and Pepys Road Fault (Map Sheet 4) is also well defined by the juxtaposition of anticlinal and synclinal axes across the faults. In no case is the fault plane exposed, but the position of the nearby outcrops suggests they are high-angle faults.

The northwesterly trending faults are less common and are possibly thrust or bedding faults caused by the northeast translation of the Jurong Formation. These faults were seldom seen exposed. One, however, at Pearl Hill in the city area was seen to be near-vertical, and the two at Kent Ridge were also near-vertical. It has also been suggested that Selat Johor, on the western and northern side of Singapore and west of Sembawang, was formed by erosion along northeasterly and northwesterly trending faults. Recent studies on seismic records by Redding and Christensen (1999) suggested that northeasterly trending faults might be far more extensive than indicated by geologic mapping in the 1970s. Redding and Christensen (1999) states there is a major fault that lies slightly offshore and parallel to the Pasir Panjang/Sentosa coastline. Vertical displacement of this fault could explain the juxtaposition of the deeper Pandan Facies (Jp) under the offshore off Pasir Panjang and the wedge of sandstone and conglomerate under Labrador Park in Telok Blangah.

It has not been possible to determine whether the lineations seen in the granitic rocks are faults or major joints. The lineations have been mapped mainly from topography with the aid of aerial stereo-photographs. No offset features were recognised, but step-downs in topography were recognised to be associated with the Nee Soon Fault (Map Sheets 1 and 2) and the western end of the Seletar Fault (Map Sheet 2). As these lineations are confined to the granite, their age of formation is taken as being associated with the tectonic uplift of the granite in early Triassic or during the later periods of block faulting in early and late Tertiary, and possibly in early Pleistocene.

Redding and Christensen (1999) remarked that the Jurong Formation had been extensively faulted during the late Cretaceous, when the Woyla continental fragment collided with the Sibumasu/East Malaya block. The collision had caused uplift of the Jurong Fomation especially along pre-existing faults at the south. Uplift along pre-existing faults had caused major instability, resulting in northward slippage of materials. The slippage of materials aided by gravity along bedding planes has resulted in materials piled on top of one another. The Rimau Facies north of Singapore River are slipped material from the south. Thrust faulting is common in less well-confined materials of the Ayer Chawan, St. John, and Tengah Facies (Jt). These thrust faults have been especially concentrated towards the surface. The larger versions are the thrust faults associated with dynamic metamorphism in the northwestern area of Singapore and mapped by PWD (1976).

Metamorphism

Low-grade metamorphism has affected widespread areas of the Jurong Formation. Fresh rocks from deep boreholes are often observed to be foliated and recrystallised. They often possess cleavages and metamorphic minerals such as chlorite, sericite, and other micas. Slates, phyllites and schistose sandstones have been reported from many sites.

Redding and Christensen (1999) remarked that the slightly metamorphosed condition of the Jurong Formation was effected during general uplift and compression of the Jurong basin that accompanied the late Cretaceous tectonic event that was initiated from the collision of the Woyla and Sibumasu/East Malaya continental blocks.

CENOZOIC STRUCTURE

The collision of the Indian continental mass and the Eurasia block along the Himalayan suture in early Tertiary has affected Southeast Asia. This event had caused block faulting along pre-existing faults and is also responsible for the clockwise bending of the orogenic belt and the right-lateral shear faults in Peninsula Malaya.

Warping and block faulting aslo occurred in late Tertiary. A series of blocks standing at different elevations can be recognised in the Jurong Formation and Bukit Timah Granite (BT). The existence of a downfaulted block has been suggested earlier as occurring beneath the Old Alluvium (OA). This form of topographic relief suggests a period of block faulting which was possibly accompanied by warping at a period prior to or during the early phases of deposition of the Old Alluvium. The faulting was also sufficiently recent to control the

present-day topography. It has been suggested in the discussion of the age of the Old Alluvium that such faulting occurred in late Tertiary times.

Based on available evidence, it is not possible to separate those faults showing Mesozoic movement from those faults showing Cenozoic movement. It is probable that some of the older fault planes, particularly those to the west of Sungei Jurong and east of and including the Nee Soon Fault (Map Sheets 1 and 2) were reactivated during the Cenozoic. In particular, the Ama Keng Fault (Map Sheet 1), bounded by a down-thrown granitic block, allowed the Old Alluvium to be deposited in the northwest of Singapore, against the Jurong Formation.

It appears that tectonic activity was minimal after the deposition of the Old Alluvium. Burton (1973 A) recognised tight folds and a fault with an approximate throw of 6 m in the Old Alluvium in Johor. Small faults have been observed in the sand quarry at Bedok. A clastic dyke was seen associated with one of these faults, but the faulting and dyke were interpreted as compaction feature rather than of tectonic significance (Fig. 2.51).

Early writers have reported seismic activity in Singapore, but their papers were not available to the authors. It is suggested that the epicentres lay close to the subduction zone west of Sumatra and that these seismic waves do not necessarily reflect any activity in Singapore itself.

4 GEOLOGICAL HISTORY

GEOLOGICAL HISTORY

The earliest geological records in Singapore are associated with the Gombak Norite (GN) and the sediments of the Palaeozoic Volcanics (PV) and Sajahat Formation (S). Hutchison (1973 A) related the Gombak Norite to the Lower Palaeozoic Malayan Geosyncline. He drew the boundary of the subducting plate as passing through Singapore and suggested that the norite represented a portion of the ophiolite suite emplaced immediately to the west of the subduction zone. The Sajahat Formation sediments lie to the east along the site of the associated Malayan Geosyncline and could thus be interpreted as representing Lower Palaeozoic geosynclinal sediment.

As the geosynclinal axis migrated to the northeast, the sediment of the Sajahat Formation was folded against the stable shelf to the southwest and a general northwest structural trend was imposed on it.

It would follow that the Palaeozoic Volcanics represent rock from the volcanic arc, which followed behind the subduction zone as it migrated to the northeast, and Hutchison had plotted the volcanics in such a position in his palaeotectonic reconstruction (Hutchison, 1973 A).

An alternative interpretation is to consider the Gombak Norite as representing the primary stage of the Triassic granite association and the Sajahat Formation as being contact metamorphosed by a possibly younger Cretaceous granite and possibly represented by the pink porphyritic granitic rocks of Pulau Sekudu. If this interpretation is correct, the Sajahat Formation could be metamorphosed Jurong Formation and the dykes cutting it would be associated with the Cretaceous plutonic intrusion on Pulau Sekudu. The presence of the Palaeozoic Volcanics, which Grubb (1968) believed to have been metamorphosed by Triassic granite, and the occurrence of sedimentary inclusions (which were presumably derived from the Sajahat Formation) in the Bukit Timah Granite (BT) on Pulau Ubin, make this theory unattractive.

Granite emplacement in central and eastern Singapore may have commenced in Upper Palaeozoic (probably late Permian) time, but was most pronounced in early Triassic time. The granite was considered by Hutchison (1973 C) to be mesozonal, but Seet (1974) concluded from his study of the triclinicities of the feldspar that the rocks of Pulau Ubin were emplaced at a high level.

There is evidence of assimilation in the Bukit Timah Granite. The production of several phases of hybrid rocks and a mixing of the hybrid rock types can be observed in the granite. Large rafts of country rock and numerous smaller inclusions at various stages of digestion can be seen. As the larger inclusions, or rafts, bear little sign of metamorphism and usually lie in a near-vertical position, it is suggested that they have been incorporated at a relatively shallow depth, probably in a fairly viscous magma. The evidence of greater assimilation of the smaller inclusions and the production of completely fluid hybrid magmas suggest considerable assimilation at depth as well.

Evidence of rafting, assimilation, mixing and re-mixing suggests that there could have been considerable tectonic activity accompanying the emplacement and crystallisation of the acid magma.

The inclusions indicate a variety of host rocks such as norite, rhyolite, andesite, argillite and quartzite. Although most of these rock types can be found in the Gombak Norite, Sajahat Formation and Palaeozoic Volcanics, only the norite can be seen in contact with the granite. At the contact, the norite is metasomatised and the granite penetrates through the metasomatic zone into the relatively unaltered rock.

Late cooling phases of the acid intrusion have been injected into the granite pluton and older formations to form a set of dykes of varying composition. This has been followed quite closely by a second phase of dyke emplacement, this time of a basic nature. This event probably accompanied the further uplift of the batholith.

During cooling and uplift of the granite and associated norite, joints developed and faulting occurred. No detailed study was carried out during the survey on the joint patterns.

The pluton initially rose at a fairly slow pace to allow for the deep weathering of the covering strata. As the pluton continued to rise, deposition of the Jurong Formation took place in the back arc sedimentary basin. This basin was highly mobile throughout sedimentation as evidenced by the rapid variation in facies with changes from marine to terrestrial sedimentation and vice versa. Towards the middle of the sedimentary cycle, relatively coarse-grained facies become prominent, suggesting a more rapid uplift of the surrounding land. With the exception of the norite at Bukit Gombak, the quartzite from PUB Test Hole No. 1 at Bedok and the Palaeozoic rock at Punggol and Tanglin, the rock enclosing the original batholith has

either been removed from Singapore Island or is now buried beneath younger rock.

Uplift of the back arc basin continued with the uplift of the Main Range Granite* to the southwest, outpacing that of the Malayan East Coast Granite (to which the Bukit Timah Granite belongs). The basin tilted to the northeast and the Jurong Formation slid to the northeast. The failure within the Jurong Formation can be considered a semi-incompetent failure with slabs of coarse-grained sandstone tilted and folded in a more plastic fine-grained matrix. Tear faults in the direction of translation and thrust and bedding plane faults normal to the direction of translation occurred. In some places where the rocks were sufficiently rigid and the energy applied during failure sufficiently intense, local dynamic metamorphism occurred.

Both marine and terrestrial deposition (accompanied by volcanic activity giving spilite, dolerite and tuff) continued during this faulting. By middle Jurassic times, Singapore was above sea level and sedimentation ceased. However, faulting and folding may have continued for a short period.

An alternative view from Redding and Christensen (1999) is that the Jurong Formation was deformed in Late Cretaceous (rather than during the Triassic and Jurassic) when the Woyla continental mass collided with the Sibumasu/East Malaya Block. The contents in the southern and eastern parts of the Jurong basin were initially uplifted by about 750 m along a fault located offshore to Pasir Panjang, tilted and slipped. This event probably tilted the Rimau Facies (Jr) (that had originally been deposited south of the Jurong basin) and caused it to slip and pile up on the area north of the Singapore River. Some of the moved material could have been mixed with remoulded Queenstown Facies (Jq) or Cretaceous clay to form the Fort Canning Boulder Bed (FC). With subsequent shortening and compression that accompanied initial uplift, the formation experienced folding, faulting and regional metamorphism. The deeper and more lithified basin deposits of calcareous siltstone, red mudstone, limestone of the Pandan Facies (Jp), and conglomerate and coarse sandstone of the Rimau and Jong Facies (Jj) escaped excessive deformation. The shallower basin deposits of the Ayer Chawan (Jc) and St. John Facies (Jsj) suffered much small fold deformation and were extensively dislocated by shallow thrust faults. The larger versions of such thrust faults were the Murai and the Pasir Laba Faults. They also remarked that the Jurong Formation, on the whole, had not experienced major folding and tilting, and in most cases where the original bedding could be identified in the seismograms, the bedding has remained flat lying.

Faulting activity appears to have recommenced in late Tertiary time, giving rise to a trough that now plunges to the south beneath the Old Alluvium (OA) on the eastern side of Singapore Island. A second down-thrown block formed in the Sungei Buloh Besar in northwest Singapore, and Old Alluvium was deposited in this basin too. It is probable that a number of faults in the Jurong area were also reactivated (despite them being essentially surface tear faults) during this time. The main movement was apparently vertical as the evidence of rejuvenation is based largely on the existence of blocks of different height, but this topographic expression may also result from the juxtaposition of elements with different weathering characteristics by lateral movement. Some adjustment also took place on faults cutting the granite.

The Old Alluvium was deposited as a thick pile of alluvial to deltaic coarse sand, with occasional incursions of marine sediment. Sedimentation continued through into early mid-Pleistocene. The nature of the sediment suggests the existence of a well-exposed quartz-rich probably granitic terrain and a far lower rate of chemical weathering than is experienced today. Verstappen (1975) suggested that such conditions would be expected during glacial periods. The Old Alluvium was deposited as a delta-like body that extends back into Johor along the course of Sungei Skudai and was probably fed by the ancestral Sungei Johor as well. In Singapore, the deposit is over 195 m thick and extends down to at least -149 m. In Johor, it is found up to 70 m above sea level but its basement is unknown. There is a threshold of granite and Palaeozoic rock across the north of Singapore from Sembawang through Yishun, North Seletar, Punggol, and from Pulau Ubin to Changi. Unless the granite was uplifted after the deposition of the Old Alluvium, the sediment must have been carried across this threshold above present sea levels to be deposited in the trough plunging down to Bedok. This implies that the deposition of the Old Alluvium must have been tied to a base level above present sea level, or its deposition was graded to a more distant coastline.

** It follows that the Main Range Granite lay to the southwest of Singapore in late Triassic to early Jurassic time and not across Singapore as shown by Hutchison (1973 A, Fig. 7).*

The gradient of the depositional surface of the Old Alluvium measured from the maximum of 47 m above sea level recorded from the Old Alluvium immediately north of Johor Bahru and the maximum of 43 m above sea level measured just east of Paya Lebar Air Base is 1:750. Such a gradient would place the coast some 350 km distance, a very unlikely proposition.

A second depression must have formed in the northwest of Singapore and across into South Johor. There is little evidence to suggest that there was much faulting during the main period of deposition of the Old Alluvium, so it is assumed that trough formation terminated before the cessation of deposition of the Old Alluvium. Alluvial sedimentation must have kept pace with depression in the area as there is no evidence of any major marine incursion. Burton (1973 A) stated that marine incursions up to 70 m or possibly higher can be found in Johor and thus it can be assumed that the sea stood at that height or higher during the deposition of the Old Alluvium.

Because of the requirement for a cool dry climate to provide the necessary detritus and the observation of a maximum sea level of 70 m, it appears that the Old Alluvium must have been rapidly deposited in an initially subsiding trough during a transition from glacial to interglacial conditions.

Efforts to recognise distinct erosion levels subsequent to the deposition of the Old Alluvium, as a result of Pleistocene sea level fluctuations, were unrewarding. Alexander (1950) recorded beaches between 21 and 27 m near Punggol and Woodleigh Park and at 17 m near Seletar. Beach sand was found at 12 m adjacent to Changi Aerodrome, where it is mapped as Tekong Formation (T). Changi Aerodrome is now Changi Air Base (West) at GR 645492.

Ho, in Burton's paper (1964), drew attention to the common breaks in the slope of the Old Alluvium erosional surface of between 15 and 23 m, and Burton (op cit) related these surfaces to a temporary Pleistocene high sea level stand.

A hypsometric study was conducted over the area of Old Alluvium in Singapore by the Geological Unit using ten-feet contour maps prepared at a scale of 1:2,500. Three possible surfaces were indicated at 30 m, 24 m, and 17 m on either side of an east-west ridge east of Paya Lebar Air Base. Field studies, however, suggested that these surfaces, identified from the plotting of spot heights, are nothing more than the gently sloping spurs of the deeply dissected Old Alluvium. The existence of the 17 m surface on the right banks of Lower Seletar Reservoir may be more realistic. In general, the Old Alluvium is too sensitive to erosion and weathering for the preservation of ancient erosional surfaces.

A hypsometric study of the whole of Singapore (Pun 1975) showed a dominance of heights between 15 and 33 m, and hence revealed no additional information.

During the period of early Pleistocene faulting, ponded areas that were isolated from the zone of deposition of the Old Alluvium were also formed. A deposit in one such area in western Singapore is recognised as the Huat Choe Formation (HC).

Reworking and redeposition of the Old Alluvium is also recognised at two places, both at a height of 36 m, just east of Paya Lebar Air Base and at either end of the runway. The material is lithologically similar to the Old Alluvium but the bedding is better defined and more continuous. The environment of deposition cannot be determined.

The next event, of which there is a record, is that of low sea levels associated with the penultimate glacial (Riss) stage. Erosion was vigorous throughout Singapore then and rivers downcut their valleys to at least -55 m. As sea level rose following the end of the penultimate glacial stage at about 140,000 BP, the Lower Marine Member (Km) was deposited (Bird, et al. 2003). The Transitional member (Kt) and the Littoral Member (Kl) were deposited near the shoreline. Onshore, the Alluvial Member (Ka) was deposited in river valleys. Sea levels declined steadily during the Riss-Wurm Interglacial stage. When the sea level stood at -120 m during the Last Glacial Maximum at about 20,000 BP, stiff reddish brown clay was developed at the top of the Lower Marine Member as it was subjected to aerial conditions and soil-forming processes. At the end of the Last Glacial Maximum, sea levels rose rapidly. When it was about -25 m during the closing phase of the last glacial (Wurm) stage at about 10,000 BP, the sills to the east and west of the Singapore Straits were breached and Singapore was flooded (Bird, et al. 2003). Once again and in a similar manner, the Transitional and Littoral Members of the Kallang Formation were deposited near the advancing shoreline. A facies similar to the Tekong Formation was also built up (Bird, et al. 2003). Further offshore, the Upper Marine Member was deposited over the Transitional Member and Littoral Member, and onshore, the

Alluvial Member was deposited. The sea level stand of about 2 to 2.5 m above present sea level at 6,000 BP during the mid-Holocene allowed the deposition of the Tekong Formation (Bird, et al. 2003). Also, the relict raised corals found at mean to 0.5 m above mean sea level grew during the mid-Holocene from 6,300 BP to 6,500 BP (Hesp, et al. 1998). Since 6,000 BP, sea levels have declined steadily to the present. In the process, near-shore Upper Marine Member was overlain by the Alluvial and Transitional Members of the Kallang Formation. Tjia (1970) recognised two high levels, one at 1.5 to 2 m (5,000 BP) and one at 0.5 to 1 m (3,500 BP) in the general Sunda Shelf area, and beach ridges associated with these levels can be seen around Singapore. Beach rock on Pulau Tekukor was formed at a lower sea level than that of today. Also, rocks of the Jurong Formation must have been planed off when the sea was at its present level, or 1 to 2 m below present-day level to form a base for the corals of the Reef Member (Kr). All members of the Kallang Formation, with the exception of the Lower Marine Member, continue to be deposited in present day.

REFERENCES

ALEXANDER, F.E.S. (1950): The geology of Singapore and the surrounding islands. Appendix I in Report on the availability of granite on Singapore and the Surrounding Islands. Singapore Government Press. pp 24.

ALEVA, G.J.J.; BON, E.H.; BOSSIN, J.J.; SLUITER, W.J. (1973): A contribution to the geology of the part of Indonesian tin belt: the sea areas between Singkep and Banka Islands and around the Karimata Islands. Bulletin No.6 Geological Society of Malaysia. pp 257-272.

BIGNELL, J.D. (1972): The Geochronology of the Malayan Granite. Unpublished Doctor of Philosophy thesis. University of Oxford.

BIRD, M.I.; CHANG, C.H.; SHIRLAW, J.N.; TAN, T.S.; TEH, T.S. (2003): The Age and Origin of the Quaternary Sediments of Singapore with Emphasis on the Marine Clay. Proceedings of Underground Singapore 2003 and Workshop on Updating the Engineering Geology of Singapore. 27-29 Nov 2003. Singapore. pp 428-440.

BISWAS B. (1973): Quaternary changes in sea level in the South China Sea. Bulletin 6. Geological Society of Malaysia. pp 229-256.

BROMS, B.B.; Lai, P.H. (1995): The Republic Plaza in Singapore – Foundation Design. Bengt B. Broms Symposium on Geotechnical Engineering. Singapore. pp 3-24.

BURTON, C.K. (1964): The Older Alluvium of Johore and Singapore. Journal of Tropical Geography. 18. pp 30-42.

---------- (1973 A): Geology and mineral resources. Johore Bahru-Kulai area, South Johore. Map Bulletin No.2 Geological Survey of Malaysia. pp 72.

---------- (1973 B): Mesozoic. Chapter 5 in Geology of the Malay Peninsula.' West Malaysia and Singapore. Eds D.J. Gobbett and C.S. Hutchison. John Wiley-Interscience, New York. pp 97-142.

CHANG, C.H. (1995): The late Quaternary estuarine evolution of Sungei Nipah, Pasir Panjang, Singapore: a geomorphic and geographic information systems interpretation. Unpublished B.A. Honours Thesis. National University of Singapore.

CHIAM, S.L.; WONG, K.S.; TAN, T.S.; NI, Q.; KHOO, K.S.; CHU, J. (2003): The Old Alluvium. Proceedings of Underground Singapore 2003 and Workshop on Updating the Engineering Geology of Singapore. 27-29 Nov 2003. Singapore. pp 408-427.

CHIN, F. (1965): The Upper Triassic sediments of Pasir Panjang-Jurong area of Singapore. Unpublished Honours Thesis. Department of Geology. University of Malaya.

CHUNG, S.K. (1973): Geological Map of West Malaysia. Geological Survey of Malaysia.

Econ Geotech (PTE) LTD (1999): Site characterisation for a site at Mandai for Land Estate Organisation, Ministry of Defence, Singapore

FAIRBRIDGE, R.W. (1961): Eustatic changes in sea level. In Physics and Chemistry of the Earth. Ed L.H. Ahrens et al. Pergamon Press. London. pp 99-185.

FONTAINE, H.; LEE, K.W. (1993): A Triassic Limestone ("Pandan Limestone") discovered by drilling at Singapore. CCOP Newsletter. July-Sept 93. Vol 18 No.3. pp 9-19.

GOBBETT, Di. (1968): Bibliography and Index of the Geology of West Malaysia and Singapore Bulletin 2, Geological Society of Malaysia.

---------- (1972): Geological Map of the Malay Peninsula. Geological Society of Malaysia.

---------- and TJIA, H.D. (1973): Tectonic History. Chapter 10 in Geology of the Malay Peninsula. West Malaysia and Singapore: Eds D.J. Gobbett and C.S. Hutchison. John Wiley-Interscience, New York. pp 305-330.

GRUBB, R.L.C. (1968): Geology and Bauxite deposits of the Pengerang area, Southeast Johore. District Memoir 14. Geological Survey of West Malaysia.

GUO, C. J. (1998): Weathering and Engineering Properties of Carbonate Rocks in Singapore. Unpublished M.Eng. Thesis. School of Civil and Structural Engineering. Nanyang Technological University. Singapore.

GUPTA, A. (1986): Quaternary Geology in Singapore: a review. Proceedings of the CCOP Symposium on Progress in Quaternary Geology of East and Southeast Asia. pp 263-269. Bangkok Thailand.

GUPTA, A.; Rahman, A.; WONG, P.P.; PITTS, J. (1987): The Old Alluvium of Singapore and the Extinct Drainage System to the South China Sea. Earth Surface Processes and Landforms. 12. pp 259-275.

HAN, K.K.; WONG, K.S.; BROMS, B.B.; YAP, L.P. (1994): The origin and properties of bouldery clay in Singapore. Geotechnical Engineering Journal. South East Asia Geotechnical Society. 24(2). pp 151-166.

HESP, P.A.; CHANG, C.H.; HILTON, M.; MING, C.L.; TURNER, I.M. (1998): A first tentative Holocene sea level curve for Singapore. Jnl. Coastal Research. 14. pp 308-314.

HUTCHISON, CS. (1964): A gabbro-granodiorite association in Singapore Island. Quarterly Journal of the Geological Society. London. 120 pp 283-297.

---------- (1973 A): Tectonic evolution of Sundaland: a phanerozoic synthesis. Bulletin 6 Geology Society of Malaysia. pp 61-81.

---------- (1973 B): Volcanic Activity. Chapter 7 in Geology of the Malay Peninsula: West Malaysia and Singapore. Eds Di. Gobbett and C.S. Hutchison. John Wiley-Interscience, New York. pp 117-214.

---------- (1973 C): Plutonic Activity. Chapter 8 in Geology of the Malay Peninsula: West Malaysia and Singapore. Eds D.J. Gobbett and C.S. Hutchison. John Wiley-Interscience, New York. pp 215-252.

JEYATHARAN, K.; LEE, K.W.; PAKIANATHAN, L.J.; SEH, C.P. (2003): Limestones of the Jurong Formation. Proceedings of Underground Singapore 2003 and Workshop on Updating the Engineering Geology of Singapore. 27-29 Nov 2003. Singapore. pp 372-387.

LEOW, J.H. (1962): A glimpse of the sedimentary structure of Singapore. Malayan Nature Journal. 16. pp 54-60.

LIM, M.S.W. (1974): Biostratigraphy of the Jurong area, Singapore. Unpublished Honours thesis, Department of Geology, University of Malaya.

NEWTON, RB. (1923): On marine Triassic shells from Singapore. Annals and Magazine of Natural History (ser 9) 12. pp 300-21.

ONG, J.C.W.; MOE SEIN; FLANAGAN, R.F.; TANG, S.K. (2003): Buried Granite Ridges in Old Alluvium. Proceedings of Underground Singapore 2003 and Workshop on Updating the Engineering Geology of Singapore. 27-29 Nov 2003. Singapore. pp 353-370.

PFEIFFER, D. (1972): Investigations for groundwater, Singapore Islands, eastern end. No.2 (Final Report). U.N. E.C.A.F.E. Natural Resources Division, Mineral Resources Section. Sala Santitham. Bangkok 2. Thailand.

PITTS, J. (1983): The origin, nature, and extent of recent deposits in Singapore. Proceedings International Seminar on Construction Problems in Soft Soils. 1-3 December 1983, Singapore.

---------- (1984): A review of the geology and engineering geology in Singapore. Quarterly Journal of Engineering Geology. 17. pp 93-101.

---------- (1992): Landforms and geomorphic evolution of the islands during the Quaternary. In Gupta, A. and Pitts, J. (eds) The Singapore Story: Physical Changes in a Changing Landscape. pp 83-143 Singapore University Press, Singapore.

Poh, K.B.; Buttling, S.; Hwang, R. (1987): Some MRT experience of the soils and geology of Singapore. Proc. Singapore Mass Rapid Transport Conf. Singapore pp 177-191.

PUN, V.T. (1975): Report on the geology of Singapore. Unpublished Report. Singapore Public Works Department Geological Unit.

PRIEM, H.N.A.; BOELRIJK, N.A.I.M.; BON, E.H.; HEBEDA, E.H.; VERDURMEN, E.A.Th.; and VERSCHURE, R.H. (1975): Isotope geochronology in the Indonesian tin belt. Geologie en Mijnbouw. 54. pp 61-70.

REDDING, J.; CHRISTENSEN, J.B. (1999): Geotechnical Feasibility Study into Rock Cavern Construction in the Jurong Formation. Final project report by Ove Arup & Partners International Ltd and Norconsult International A/S.

RENWICK, A. and RISHWORTH, D.E.H. (1966): Fuel Resources (coal, lignite, and petroleum) in Malaya. Geological Survey Department of West Malaysia. Ipoh. pp 123.

SCRIVENOR, J.B. (1910): The rocks of Pulau Ubin and Pulau Nanas (Singapore). Quarterly Journal of the Geological Society, London. pp 420-434.

------- (1924): The geology of Singapore Island. Journal of the Malayan Branch of the Asiatic Society. 2. pp 1-8.

------- (1931): The geology of Malaya. McMillan and C.O. London. pp 217.

SEET C.P. (1974): The igneous complex of Pulau Ubin. Unpublished Honours Thesis, Department of Geology, University of Malaya.

SHIRLAW, J.N.; BROOME, P.B.; CHANDRASEGARAN, S.; DALEY, J.; ORIHARA, K.; RAJU, G.V.R.; TANG, S.K.; WONG, I.H.; WONG, K.S.; KYI YU (2003): The Fort Canning Boulder Bed. Proceedings of Underground Singapore 2003 and Workshop on Updating the Engineering Geology of Singapore. 27-29 Nov 2003. Singapore. pp 388-407.

SINGAPORE PUBLIC WORKS DEPARTMENT GEOLOGICAL UNIT (1973 A): Report No. 1. The geology of the Jurong-Tuas Area, Southwestern part of the main island, Republic of Singapore. Unpublished Report.

------- (1973 B): Report No.2 The Geology of the Jurong-Pengkang-Pandan Area, Southwestern part of the main island, Republic of Singapore. Unpublished Report.

------- (1974): Report No.3 The geology of the Pasir Panjang-Tanglin area, south-central part of the main island, Republic of Singapore. Unpublished Report.

SINGAPORE PUBLIC WORKS DEPARTMENT (1976): Geology of the Republic of Singapore. Public Works Department (Singapore).

STAUFFER, P.H. (1973): Cenozoic. Chapter 6 in Geology of the Malay Peninsula: West Malaysia and Singapore. Eds D.J. Gobbett and C.S. Hutchison. John Wiley-Interscience New York. pp 143-176.

SWAN, S.B.St.C. (1971): Coastal geomorphology in a humid tropical low energy environment: the islands of Singapore. Journal of Tropical Geography. 33. pp 43-61.

TAI, S.A. (1972): Abstract, Geology of Eastern Singapore Island with special emphasis on the Old Alluvium. Unpublished Honours Thesis, Department of Geology, University of Malaya.

TAN, S.B. (1972): Foundation Problems in Singapore marine clay. Asian Building and Construction. Special Report on The Third South East Asian Conference on Soil Engineering. pp 30-33.

TAYLOR, D.; YEN, O.H.; SANDERSON, P.D.; DODSON, J. (2001) : Late Quaternary peat formation and vegetation dynamics in a lowland tropical swamp at Nee Soon, Singapore. Palaeogeography, Palaeoclimatology, Palaeoecology. 171. pp 269-287.

TJIA, H.D. (1970): Rates of diastrophic movement during the Quaternary in Indonesia. Geologie en Mijnbouw. 49. pp 335-408.

------- and SHOJI FUJII, KUNIHIKO KIGOSHI, ARATA SUGIMURA, SAKARIA, T. (1972): Radiocarbon dates of elevated shorelines, Indonesia and Malaysia. Pt 1. Quaternary Research. 2. pp 487-495.

VERSTAPPEN, H.Th (1975): The effect of Quarternary tectonics and climates on erosion and sedimentation in Sumatra. Preprint from IVth I.P.A. Convention. Jakarta, June 1975.

VILPPONEN, A-M. B. (1988): The sedimentology and stratigraphy of the Jurong Formation, Singapore. Unpublished Masters Thesis. National University of Singapore.

WONG, I.F.T. (1960): The petrology and structure of the sedimentary rocks of southwest Singapore. Unpublished Honours Thesis. Department of Geology, University of Malaya.

WONG, I.H.; OOI, I.K.; BROMS, B.B. (1996) : Performance of raft foundations for high-rise buildings on the Bouldery Clay in Singapore. Canadian Geotech. Jnl. 33. pp 219-236.

WONG, PP. (1969): Surface configuration of Singapore Island: a quantitative description. Journal of Tropical Geography. 29. pp 64-74.

YEAP, E.B. (1996): Soil Investigation for Container Freight Station Complex and Blocks D & E of Stacking Yard (Phase 1) at New Container Terminal, Pasir Panjang. Report to the Port of Singapore Authority.

ZHAO, J.; CHEN, C.N.; CAI, J.G. (2002): A hydrogeological study of the Sembawang hot spring in Singapore. Bull Eng Geol Env (2002 61:59-71).

ZHOU, Y.; ZHAO, J.; CHOA. V. (1993A): Geological and Geotechnical Properties of the Bukit Timah Granite. Eleventh Southeast Asian Geotechnical Conf. 4-8 May 1993, Singapore.

ZHOU, Y.; ZHAO, J.; K.W. LEE (1993 B): Mechanical and Engineering Properties of the Bukit Timah Granite. Rock Caverns for Underground Space Utilization, 22 Oct 1993, Singapore.

ZHOU, Y. (2001): Engineering Geology and Rock Mass Properties of the Bukit Timah Granite. Underground Singapore 2001.

APPENDICES

APPENDIX 1

Analyses of Rocks

	1	2	3	4	5	6	7	8	9	10	11
SiO_2	76.55	75.60	69.98	68.98	59.83	51.52	49.95	69.92	69.39	69.69	72.80
Al_2O_3	12.78	13.03	14.58	14.80	16.21	13.29	21.03	14.30	14.33	14.82	13.90
Fe_2O_3	0.38	0.11	0.84	0.48	1.33	0.30	Nil	n.d.	n.d.	n.d.	n.d.
FeO	0.37	1.44	2.35	3.44	5.64	9.91	7.76	n.d.	n.d.	n.d.	n.d.
Total Iron as FeO								3.78	3.87	3.82	3.06
MgO	0.12	0.21	0.98	1.56	2.43	12.31	5.95	0.81	0.89	0.81	0.44
CaO	1.46	0.84	3.19	4.32	4.57	8.77	10.56	2.88	2.48	2.28	1.66
Na_2O	2.56	3.58	3.38	2.06	3.90	1.01	2.27	3.67	4.40	3.77	3.95
K_2O	5.04	4.17	3.60	3.26	2.63	0.39	0.19	3.32	3.36	3.29	3.29
H_2O_+	0.33	0.57	1.13	0.07	2.02	1.54	1.39	n.d.	n.d.	n.d.	n.d.
H_2O_-	0.22	0.13	0.28	Nil	0.42	Nil	n.d.	n.d.	n.d.	n.d.	n.d.
TiO_2	tr.	0.16	0.47	0.49	0.84	0.58	0.58	0.27	0.24	0.24	0.14
P_2O_5	tr.	0.06	tr.	0.08	tr.	0.06	0.06	0.05	0.07	0.11	0.08
MnO	tr.	Nil	0.04	0.02	0.10	0.29	0.24	0.08	0.05	0.05	0.01
CO_2	n.d.	Nil	n.d.	0.03	n.d.	Nil	Nil	n.d.	n.d.	n.d.	n.d.
Total	99.81	99.90	100.52	100.22	99.92	99.97	99.98	99.08	99.08	99.28	99.33

	12	13	14	15	16	17	18	19	20	21
SiO_2	70.00	71.69	68.33	71.61	64.58	66.75	52.41	48.12	64.50	72.19
Al_2O_3	18.13	17.26	19.11	17.71	14.51	14.79	15.57	22.51	15.96	13.25
Fe_2O_3	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
FeO	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Total Iron as FeO	2.52	2.47	2.80	1.30	6.99	6.73	9.52	10.01	5.08	3.23
MgO	0.23	0.28	0.30	0.29	0.69	0.69	6.67	7.40	1.58	0.29
CaO	2.02	2.63	2.13	1.11	3.78	3.48	8.16	7.51	4.52	2.38
Na_2O	2.93	3.16	2.98	3.69	5.09	3.20	2.25	1.60	2.81	3.37
K_2O	3.44	2.01	3.34	3.11	1.25	2.92	2.92	0.66	4.10	2.95
H_2O_+	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
H_2O_-	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
TiO_2	0.21	0.23	0.23	0.16	0.68	0.73	1.16	0.98	0.64	0.17
P_2O_5	0.10	0.13	0.09	0.07	0.25	0.21	0.05	0.80	0.19	0.65
MnO	0.07	0.06	0.06	0.01	0.08	0.01	0.16	0.24	0.13	0.05
CO_2	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Total	99.66	99.92	99.30	99.06	97.90	99.61	98.69	99.83	99.51	98.89

tr. = trace n.d. = not determined

Rock Types and Locations

1 Aplitic granite, Public Works Department Quarry (Mandai)

2 Adamallite, Singapore Granite Quarry (Bukit Timah)

3 Biotite-granite, Public Works Department Quarry (Mandai)

4 Biotite-hornblende-granodiorite, Swee Construction Quarry (Bukit Timah)

5 Quartz-monzonite xenolith, Public Works Department Quarry (Mandai)

6 Hornblende-gabbro, Yun Onn Quarry

7 Norite, Gim Huat Quarry

8 Hornblende microgranite, Gim Huat Quarry

9 Hornblende microgranite, Gim Huat Quarry

10 Hornblende microgranite, Aik Huat Quarry No. 1

11 Porphyritic biotite granophyre, Lee Hung Cheng Quarry

12 Hornblende granite, Public Works Department Quarry (Pulau Ubin)

13 Hornblende granite, Housing Developement Board (Pulau ubin)

14 Hornblende granite, Tanjong Chek Jawa (Pulau Ubin)

15 Hornblende granite, Aik Hwa Quarry No. 2

16 Actinolite hornfels, Aik Hwa Hung Cheng Quarry

17 Actinolite hornfels, Lee Hung Chung Quarry

18 Amphibolitised dyke rock, Lee Hung Cheng Quarry

19 Amphibolitised dyke rocks, Housing Developement Board Quarry

20 Granodiorite, South Coast of Pulau Ubin

21 Granodiorite, East Side of Pulau Ubin

Analyses Nos. 1 - 7 from Hutchison (1964)

Analyses Nos. 8 - 21 from Seet (1974)

APPENDIX 2

Analyses of Hot Spring Water

Tested Items	Sembawang Table 5 in Zhao, et al. (2002)	Sembawang PWD, 1976	Sembawang Table 1 in Zhao, et al. (2002) Values are from previous investigations in 1963, 1967, 1985	P. Tekong PWD, 1976
PH value	7.4	8.1	-	7.0
Colour	-	5	-	5
Loss on Ignition of Solids	-	191.0	-	295.0
Total Dissolved Solids	-	914.0	-	10170.0
Silica	-	-	80	-
Silica (as SiO_2)	-	16.0	-	22.0
Aluminium	0.04	-	0.11	-
Iron	0.02	-	0.01 0.55 1.10	-
Ferric Oxide	-	0	-	0.02
Calcium	48.6	-	24.1 34.7	-
Calcium Carbonate	-	-	69	-
Potassium	20.6	-	87.5	-
Magnesium	0.13	-	0.34 0.93	-
Magnesium Oxide	-	3.2	-	96.5
Sodium	253	-	260	-
Sodium Chloride	-	-	684	-
Chlorine	431	0	416 418	0
Fluorine	8.59	-	-	-
Sulphite (as SO_3)	0.65	-	-	-
Sulphate (as SO_4)	62.0	45.0	30.0 33.8	190.0
Carbonate (as CO_3)	<1.0	18.0	15.0	0
Carbonate (as HCO_3)	<1.0	-	75.0 91.5	-
Phosphate (as PO_4)	51.1	-	0	-

Tested Items	Sembawang Table 5 in Zhao, et al. (2002)	Sembawang PWD, 1976	Sembawang Table 1 in Zhao, et al. (2002) Values are from previous investigations in 1963, 1967, 1985	P. Tekong PWD, 1976
Arsenic	<0.05	-	0	-
Zinc	0.02	-	-	-
Cadmium	<0.02	-	-	-
Lead	<0.02	-	-	-
Chromium	<0.02	-	-	-

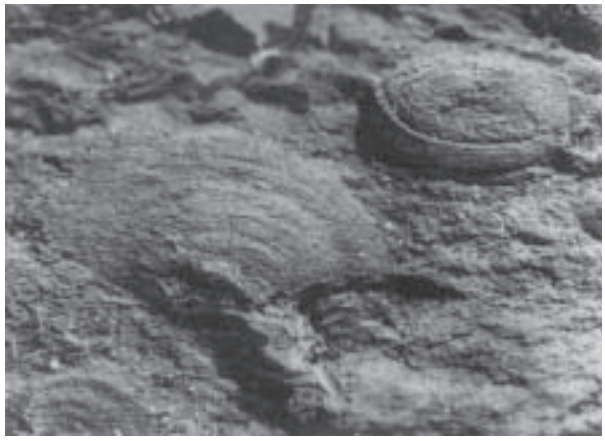
Results of analyses for chemical elements are in milligrams per litre, mg/l.

Sembawang hot spring is located at GR 475586 (about 400 m west-southwest from Gambas Avenue-Yishun Avenue 7 junction).

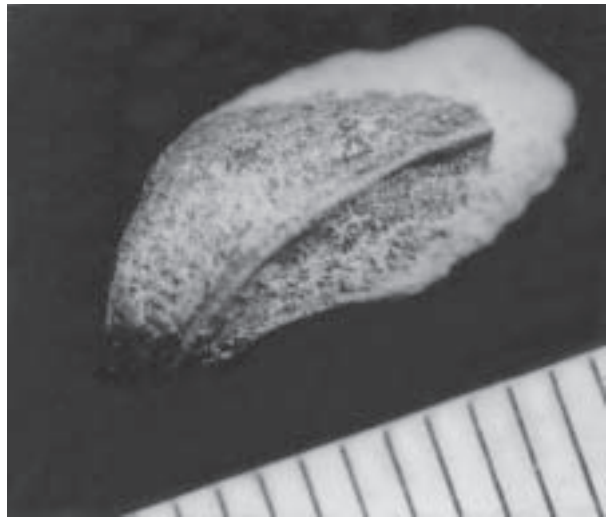
Pulau Tekong hot spring is located at GR 740581 (just east of Kampong Umum and 300 m west from Sungei Belang in north Pulau Tekong).

APPENDIX 3

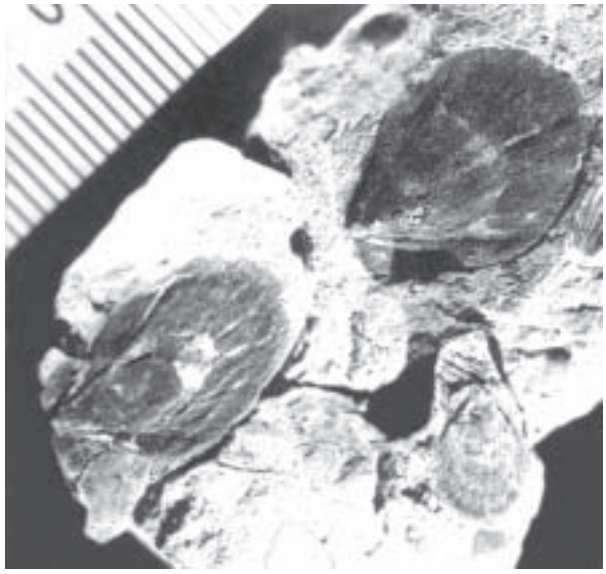
Plate of Fossils (PWD, 1976)



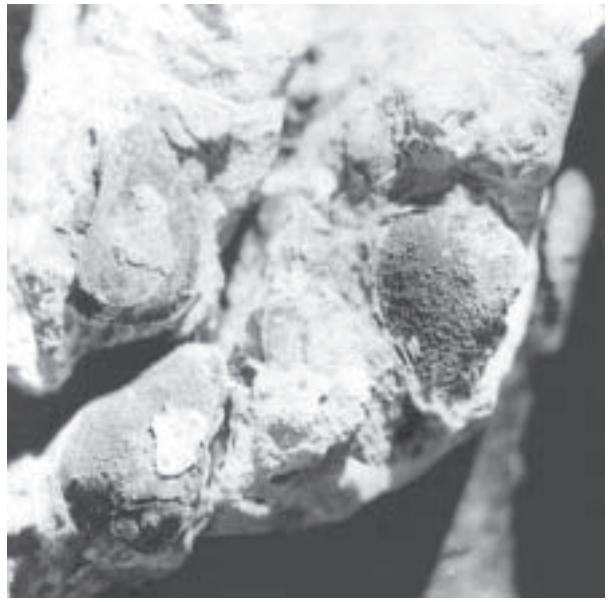
Gonodan sp. , Schafhaeutz
Triassic-Jurassic
GR. 333456
X4



Myophoria sp.,?
Low-Upper Triassic
GR. 316468
X4



Plicatula cf carinata (Healey)
Mid-Upper Triassic
GR. 333463
X2



Palaeonucula sp.,?
Triassic-Jurassic
GR. 333463
X2

List of Fossils: Jurong Formation

Fossil	Location																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Ammonite (Fragments)		N																	
Amusium sp.							L			L					L		L		
Anodontaphora A sp. (Kobayashi & Tamaru)					K		L			L							L		
Anodontaphora B sp. (Kobayashi & Tamaru)					K		L			L									
Arca sp	N		Co																
Astarte guthriensis (Newton)	N																		
Astarte scrivenori (Newton)	N																		
Aviculinia sp.										L									
Buchia sp.										L									
Cardium scrivenori (Kobayashi & Tamaru)													L						
Cardium scrivenori, Nov.					K					L									
Carpolithes sp.	N																		
Cassianella cf. tenuistriata (Munster)		N			K														
Cassianella cf. verbeeki (Kobayashi & Tamaru)					K														
Cassianella sp.			Co				L			L					L				
Chlamys cf. voloniensis (Deferance - Leymerie)		N																	
Costatoria aff. goldfussi					K														
Costatoria cf. myophoria (Boettiger, 1880)					K					L							L		
Costatoria chegarperahensis (Kobayashi)							L			L			L		L				
Costatoria malayensis (Newton, 1900)										L									
Costatoria pahangensis (Kobayashi)										L					L				
Costatoria singapurensis (Kobayashi)					K		L			L		L	L		L				
Costatoria (?) sp. ex. gr. goldfussi (Zieten)										L		L			L				
Cuculaea scrivenori (Newton)	N																		

Fossil	Location																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Cucullaea (?) sp.		N																	
Cuspidaria sp.			C																
Estheria mangalensis				Co															
Gastropod sp.							L			L				L	L				
Gen. et. sp. indet (Kobayashi & Tamura)					K														
Gervillia hanitschi (Newton)	N																		
Gervillia scrivenori (Newton) (Nov.)		N																	
Gervillia sp.			Co																
Goniomya scrivenori (Newton)	N																		
Goniomya singapurensis (Newton)	N																		
Gonodon sp. (Schafhautl, 1863)								C	C										
Gonodon sp. (Schafhautl, 1950)																		L	L
Gruenewaldia sp.										L									
Halobia n. sp. (Kobayashi & Tamura)					K														
Holabia sp.															L				
Holabia sp. ex. gr. H. verbeeki					K														
Holabia (?) sp. indet.					K														
Lima (Plagiostoma ?) sp.					K														
Lima sp.																			
Lima sp. A			Co												L				
Lima sp. B															L				
Lima sp. Bruguie, 1797						C	C												
Lopha of Montis-caprilis (Klipstein)		N																	
Lucina (?) sp.	N		Co																
Modiolopsis gonoides (Healey)		N																	
Modiolus cf. nachamensis			Co																
Myoconcha sp.			Co																

Fossil	Location																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Myophoria A & B sp. indet		N																	
Myophoria bittneri Newton (Nov.)		N																	
Myophoria cf. goldfussi			Co																
Myophoria cf. goldfussi Alberti		N																	
Myophoria cf. harpa?											C	C							
Myophoria inaequicostata			Co																
Myophoria ornata (Wissman)		N																	
Myophoria sp.											C					L			
Myophoria sp. A, (Brown 1834)						C													
Myophoria sp. B												C							
Myophoriopsis cf. carinata			Co																
Myophoriopsis sp.			Co																
Neoshizodus (Leviconcha cf. Ovatus)																L			
Nucloid gen. et. sp. indet.	N																		
Nuculana (?) sp.	N																		
Nucula sp. A						C					C	C							
Nucula sp. B						C					C	C							
Opis cf. heoninghausi (Klipstein)		N																	
Pachycardia ? sp.			Co																
Palaeocardita cf. crenata (Goldfuss)		N																	
Palaeoneilo A. sp. (Kobayashi & Tamura)					K					L					L				
Palaeoneilo B sp. (Kobayashi & Tamura)					K		L			L			L		L				
Palaeonucula sp.										L									
Palaeopharus sp.							L												
Paleolima sp.										L									
Paleopharus sp.										L					L		L		
Parallelodon Meek, 1866						C					C	C							

Fossil	Location																		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Pecten (Entolium ?) sp.					K														
Pecten (Entolium) sp. Meek, 1965						C	C												
Pecten sp.			Co									C							
Pecten sp. A.							L			L			L						
Pecten sp. B.										L			L		L				
Pecten sp. C.										L									
Pecten sp. D.										L					L				
Pecten (Syncyclonema) sp. Meek, 1864						C	C												
Pleuromya sp.			Co																
Plicatula cf. Healey							L			L		L	L		L				
Podozamites cf. lanceolatus (Lindley & Hutton)	N																		
Posidonia sp.			Co															L	L
Posidonia sp. (Brown)							C		C	C									
Posidonia sp. indet.					K														
Prolaria sp.			Co																
Promalthildia colon ? (Wissman)		N					C												
Prosospondylus comtus (Goldfuss)		N																	
Pteria pahangensis										L									
Pteria sp.			Co			C					C	C							
Schaufhaulia astartiformis (Wissman)		N																	
Spondylus dubiosus (Bittner)		N								L					L				
Spriferina cf. fragilis (Schlotheim)		N																	
Syncyclonema sp.		N					L			L				L	L				
Terebratuloid shell		N																	
Thracia (?) sp.	N																		
Trigonia cf. zlambachensis			Co																
Trigonodus sp.			Co													L	L		
Volsella cf. compressa (Goldfuss)	N																		

LOCATIONS

1 Mount Guthrie, Singapore (Newton, 1906)

2 Mount Faber, Singapore (Newton, 1923)

3 Near Huat Choe Village (Cox, 1952)

4 Pulau Ayer Chawan (Cox, 1952)

5 From 24 points on the Jurong Industrial Estate, about 3 square miles (Kobayashi & Tamura)

6 Behind Crown Cork Company; CF1, FIG. 3.32. (Chin, 1965)

7 Opposite Crown Cork Company; CF2 & L1, FIG. 3.32. (Chin, 1965 and Lim, 1975)

8 Drain cutting beside Jalan Buroh; CF4, FIG. 3.32. (Chin, 1965)

9 Floor of levelled hill beside Jalan Buroh; CF5, FIG 3.32. (Chin, 1965)

10 North face of Bukit Susap behind General Electric Factory; CF6, L7 & L8, FIG. 3.32. (Chin, 1965 and Lim, 1975)

11 Hill cutting beside Pioneer Road; CF7, FIG. 3.32. (Chin, 1965)

12 Drain cutting about 8 m south of Pioneer Circus; CF8 & L9, FIG. 3.32. (Chin, 1965 and Lim, 1975)

13 Levelled area 27 m northeast of Jurong Pier Circus; L2, FIG. 3.32. (Lim, 1975)

14 Drain cutting 13 m west of Jurong Pier Circus; L3, FIG. 3.32. (Lim, 1975)

15 Drain cutting beside S.T.I Factory; L4, FIG. 3.32. (Lim, 1975)

16 East face of Bukit Sembawang; L5, FIG. 3.32. (Lim, 1975)

17 Hill reserve for Singapore Port Authority; L6, FIG. 3.32. (Lim, 1975)

18 Drain cutting opposite Singapore Clay Products; L10, FIG. 3.32. (Lim, 1975)

19 Benoi Sector syncline; L11, FIG. 3.32. (Lim, 1975)

Abbreviations:

N = Newton, Co = Cox, K = Kobayashi & Tamura, C = Chin and L = Lim Meng Sze Wu

Notes:

Locality 13 - all fossils collected from this locality have been doubtfully identified.

APPENDIX 4

Log of PWD Borehole in Pandan Facies describing Lithology, Palaeontology, and Sample Locations (Fontaine and Lee, 1993)

Depth (m)	Lithology	Palaeontology	Samples	Studied Corals	Organic matter
0	Cover and weathered rock. Recovery: 30% to 60%				
23	Dolomite, occasionally with black angular inclusions. Recovery: 70% to 100%				
44	Dolomite to dolomitic limestone. Recovery: 100% Rare stylolites	Very rare fragments of corals and shells, poorly preserved	Sp 1:57.4m		
62	Dolomitic limestone. Thin calcite veins. Recovery: 100%				
77	Laminated limestone (tidalite), gray laminae interfingering with black laminae. Recovery: 100%	Very poor in fossils. Occasional burrows. Small bioclasts with rare fragments of urchin spines	Sp 2+3:77.2 m Sp 4:82 m Sp 5:90.2 m Sp:6:97 m Sp 7:98 m Sp 8:102.5 m Sp 9:104 m		Sp 7
110	Dark gray limestone. A few stylolites. Recovery:100%	Very rare brachiopod shells	Sp 10:110 m Sp 11:112 m Sp 12:119.5 m Sp 13:121 m Sp 14:123 m		Sp 14
125	Laminated limestone. Recovery: 100%	Fragments of small shells, not in abundance. Rare urchin spines and calcareous sponges. Tubiphytes	Sp 15:131 m Sp 16+17:136 m Sp 18:141 m		
145	Black to dark gray limestone (Wackestone to packstone), coaly between 151 to 153 m. Recovery: 100%	A few fragments shells and crinoids. Rare corals (fasciculate and massive) Sponges. Foraminifers (Endothyra, Duostominidae), Ladinellas, Tubiphytes, a few ostracods, stromatolites at a depth of 157.3 m	Sp 19:145 m Sp 20:148 m Sp 21:151 m Sp 23:154 m Sp 22:156 m Sp 24:157.3 m Sp 25:158.5 m	Sp 21	Sp 19
162	Laminated limestone. Recovery: 100%	Thaumatoporella (alga)	Sp 26:163 m Sp 27:165 m		Sp 26

Depth (m)	Lithology	Palaeontology	Samples	Studied Corals	Organic matter
173	Dark gray limestone (packstone). Calcite veins at depths of 173 to 177.5 m, 182 to 186 m and 201 to 207.5 m. Occasional stylolites. Recovery: 100%	Fasciculate corals are common; massive corals are less abundant. A few fragments of shells, rare gastropoes, crinoids, urchin spines, Tubiphytes, bryozoans at depths of 188 and 199 m. Stromatolites at depth of 199 m. Very rare ostracods. Occasional peloids	Sp 28:172 m Sp 29:173 m Sp 30:175 m Sp 31:176 m Sp 32:179 m Sp 33:183 m Sp 34:188 m Sp 35:190 m Sp 36:192 m Sp 37:193 m Sp 38:195 m	 Sp 35 Sp 37 Sp 39	 Sp 33
203	Sheared limestone. Fossils are crushed. Calcite veins. Probably a fault zone. See pl.1, Fig. 4 in original article		Sp 39:199 m Sp 40:203 m Sp 41:206 m		
207.5			Sp 42:207.5 m		

Plate 1 of Fossils in Pandan Facies (Fontaine and Lee, 1993)

Scale bar = 1 mm

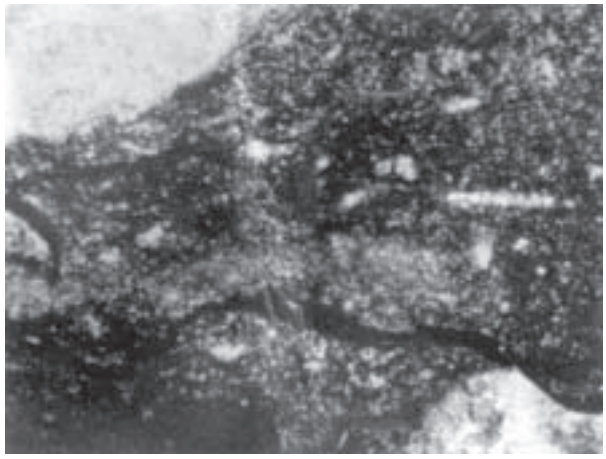


Figure 1: Sample Sp 37 displaying thin black streaks in a similar way as sample Sp 33 which has been analysed for organic matter. Recrystallised corals at two corners of the photograph.

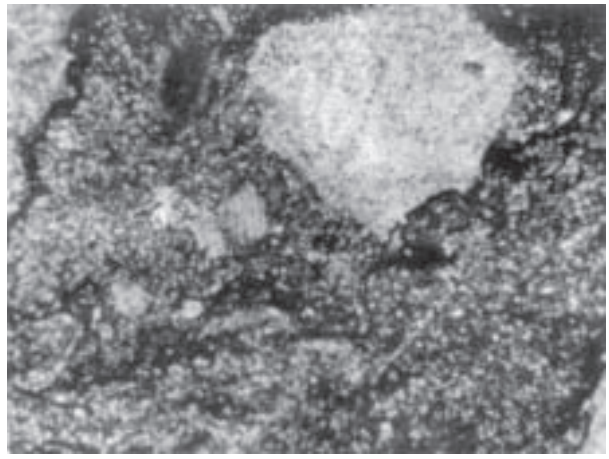


Figure 2: Sample Sp 24, dolomitised limestone containing a fragment of echinoderm.

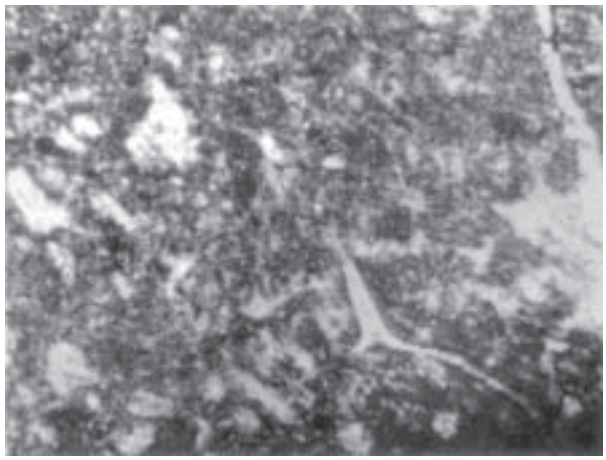


Figure 3: Sample Sp15 showing a fragment of brachiopod shell with a spine.

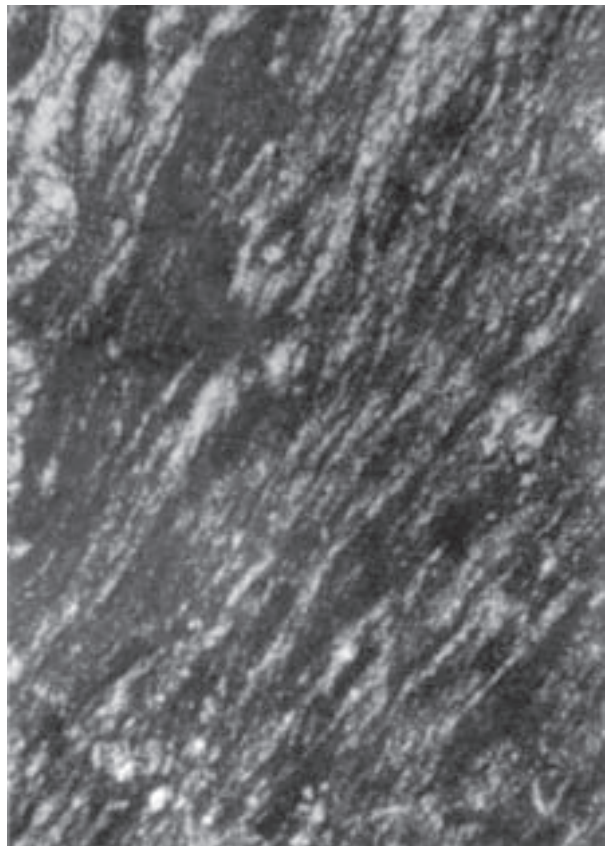


Figure 4: Sample Sp 40, sheared limestone at bottom of the borehole.

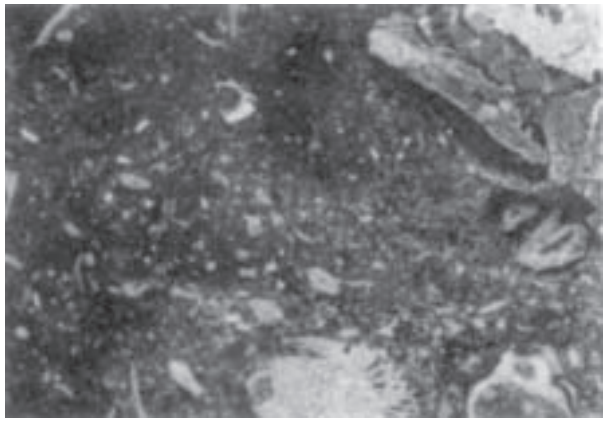


Figure 5: Sample Sp 39, fragment of bryozoan in a limestone rich in bioclasts.

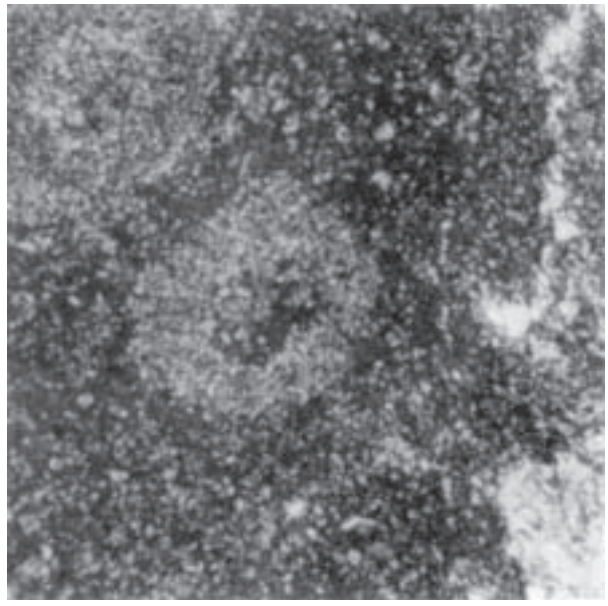


Figure 6: Sample Sp30, dolomitic limestone.

Plate 2 of Fossils in Pandan Facies (Fontaine and Lee, 1993)

Scale bar = 1 mm

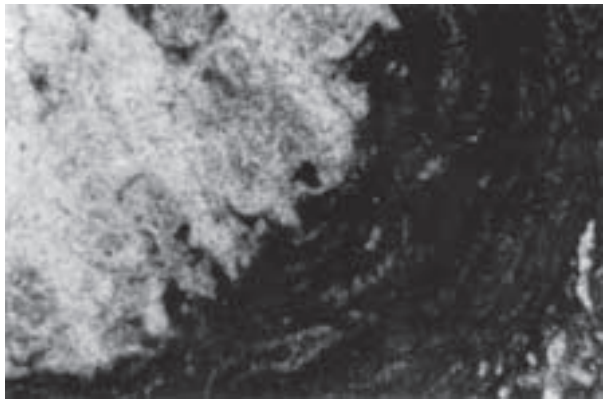


Figure 1: A coral encrusted by stromatolite (thin section from Sp 39)

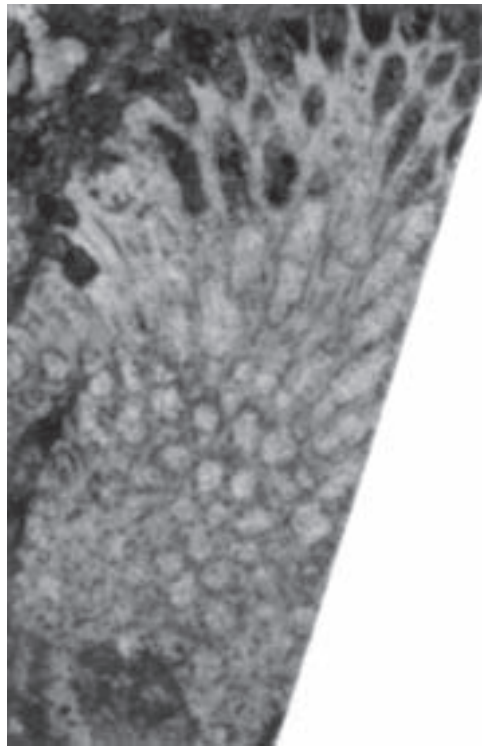


Figure 2: Paralioclema sp., a bryozoan (thin section from Sp 39)

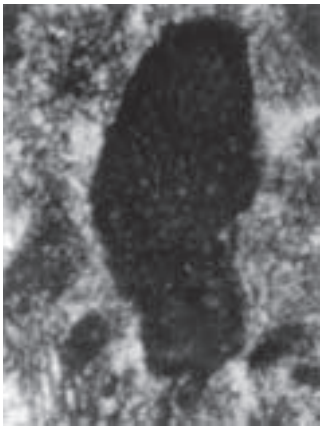


Fig. 3



Fig. 4



Fig. 5

Figures 3 to 5: *Ladinella sp., an alga (thin section from Sp24)*

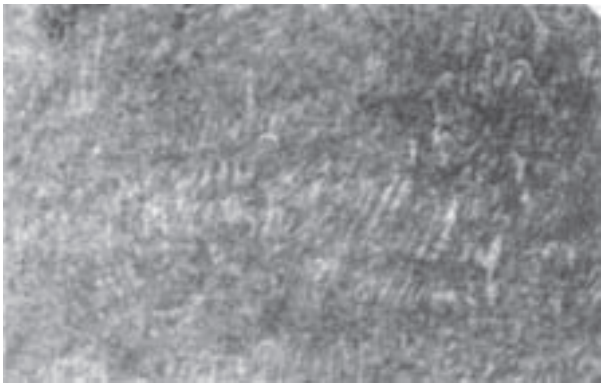


Fig. 6

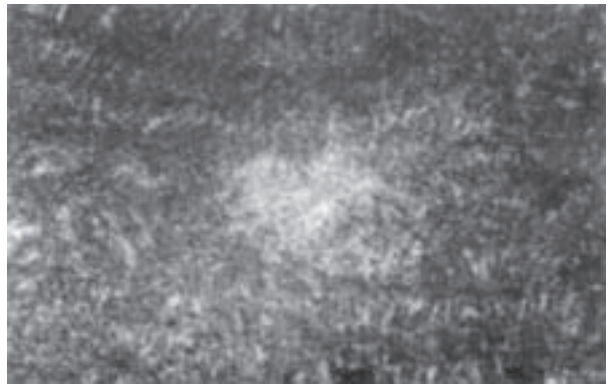


Fig. 7

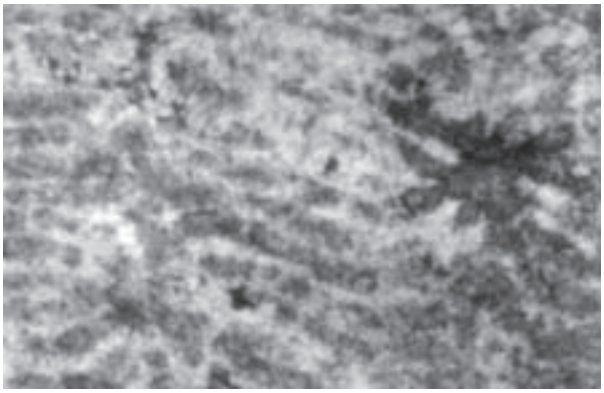


Fig. 8

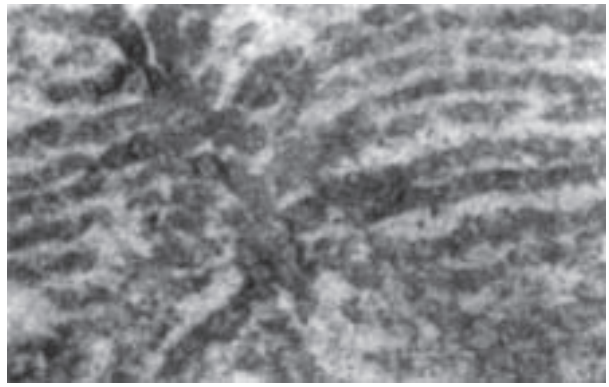


Fig. 9

Figures 6 to 9: *Pamiroseris minima nov. sp. Holotype. (thin section from Sp21) 6 and 7: transverse sections showing the long and parallel series. 8 and 9: transverse sections in the axial part of the corallite, showing the compact biseptal plates, lack of septa of valley and the small trabecular columella.*

Plate 3 of Fossils in Pandan Facies (Fontaine and Lee, 1993)

Scale bar = 1 mm

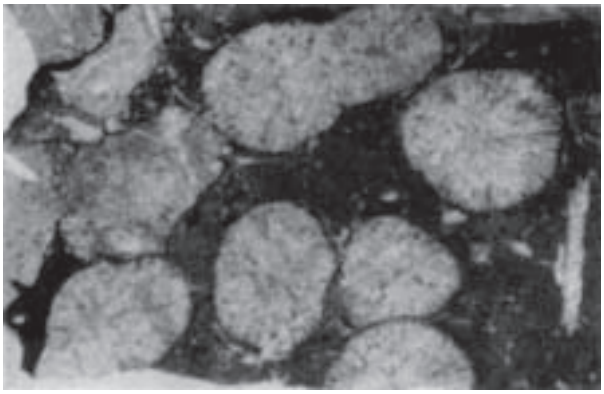


Fig. 1a

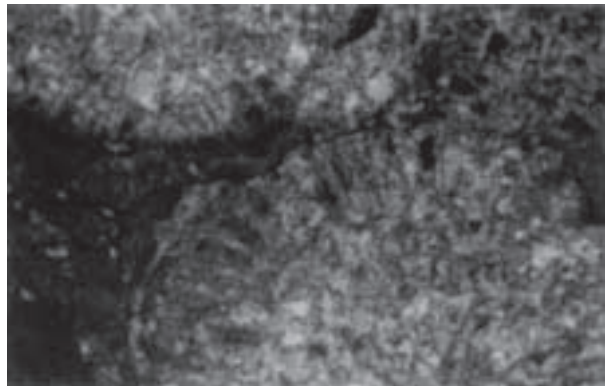


Fig. 1b

Figure 1: Retiophyllia clathrata (Emmrich) (thin sections from Sp 37)

a = transverse section of the corallites showing the intracalicinal budding

b = transverse section showing the internal septotheca overlain by the epicostal wall

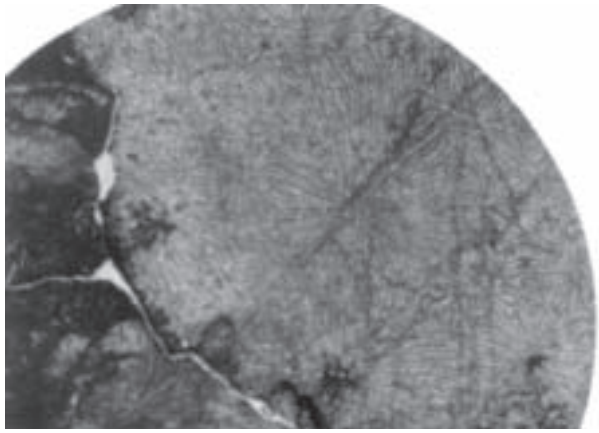


Fig. 2a

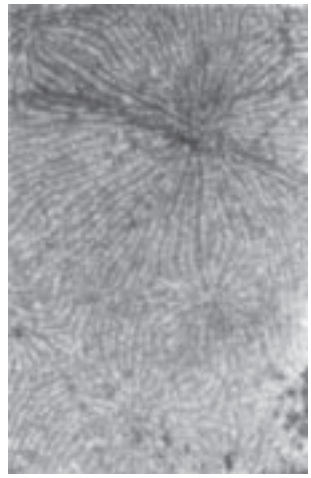


Fig. 2b

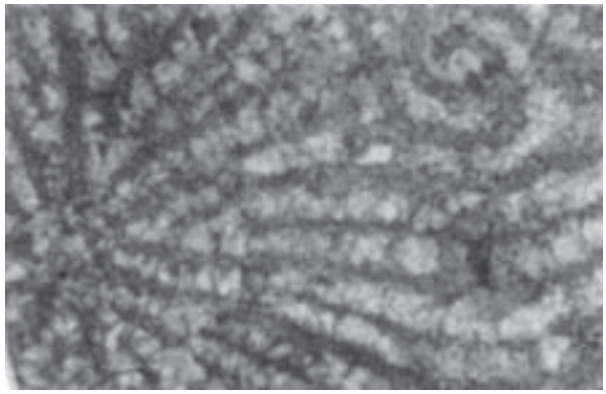


Fig. 2c

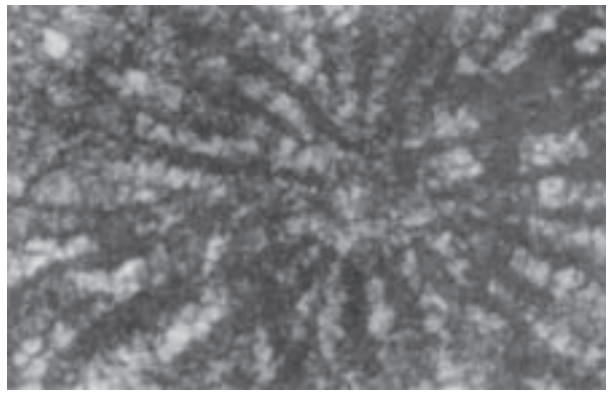


Fig. 2d

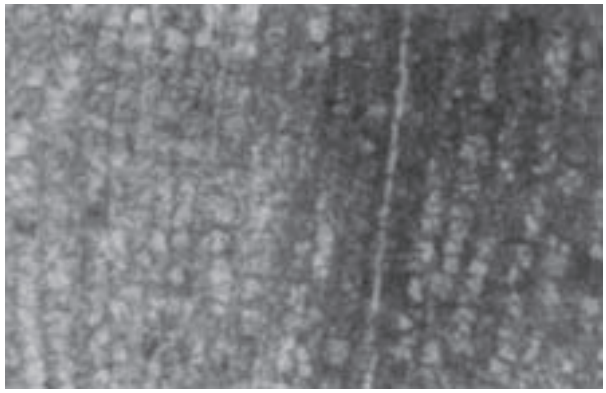


Fig. 2e

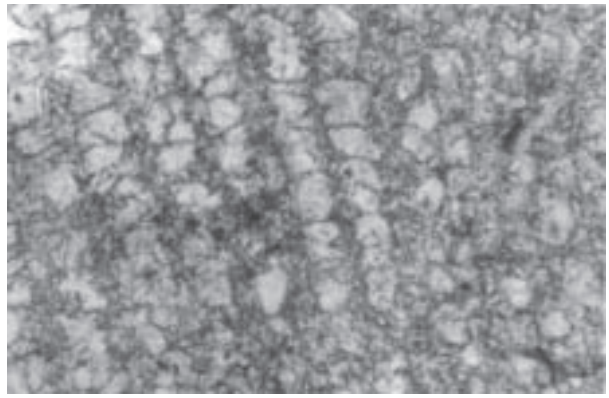


Fig. 2f

Figure 2: Thamnasteriamorpha multisepta nov. sp. Holotype. (thin section from Sp39)

a = transverse section showing the protocorallite surrounded by smaller corallites

b = the same showing the compact biseptal plates

c & d = the same showing the styliform columella and the synapticules

e & f = vertical section showing the endotheca and the menianes

APPENDIX 5

Engineering Properties of Singapore Rocks

This appendix aims to provide the reader with a basic appreciation of the engineering and mechanical properties of the major rocks found in Singapore. The engineering properties presented are those of fresh intact rock and have been compiled and derived from published literature and site investigation reports that have been made available to the project team. They are presented here as typical ranges and typical average values from previous tests and may not necessarily represent the actual values for rocks found at a specific site. Accordingly, project-specific investigations and tests should be conducted.

Table of General Engineering Properties of Singapore Rocks

STRATI-GRAPHY	LOCATION	γ (g/cm^3)	σ_c (MPa)	ν	E (GPa)	V_p (km/s)	V_s (km/s)	Reference
Jurong Formation								
Sandstone	Kent Ridge Park	2.30 - 2.68	5 - 117	-	1.5 - 25.2	-	-	1, 2, 3
	Jurong Island	2.16 - 2.77	46.4 - 61	0.11 - 0.27	14.5 - 98.0	5.0 - 5.7	2.8 - 3.3	4, 5
	Others	-	35 - 137 (68)	-	57.0	-	-	6
Siltstone	Kent Ridge Park	2.31 - 2.73	0.1 - 36	-	0.01 - 16.0	-	-	1, 2
	Jurong Island	2.32 - 2.77	30 - 67	0.10 - 0.25	20 - 35	4.9 - 5.2	2.8 - 3.1	4, 5
	Others	-	21 – 53	-	38.0	-	-	6
Limestone	Sungei Pandan River	2.28 - 2.68	16.8 - 182	-	10.0 - 56.0	-	-	7, 8, 9
	West Coast Park and Harbour Drive Road	2.4 - 2.70	13.4 - 56	-	-	-	-	7, 8, 9
	West Coast Highway	2.35 - 2.70	18.5 - 89	-	-	-	-	7, 8, 9
	Jurong Island	2.4 - 2.70	31 - 49	0.11 - 0.38	36.3 - 91.5	5.4 - 6.0	2.6 - 3.4	4
	Others	-	45 – 162 (98)	-	98.0	-	-	6
Conglomerate	Kent Ridge Park	-	23 - 42	-	4.0 - 60.0	-	-	2
	Jurong Island	2.69 - 2.70	31 - 63	0.19 - 0.28	23.5 - 46.1	5.6 - 5.7	3.2 - 3.5	4, 5
	Others	-	32 – 102 (52)	-	33.0	-	-	-
Bukit Timah Formation								
Granite	Central Singapore	2.62 - 2.83 (2.65)	38 - 260 (157)	0.03 - 0.40 (0.2)	20.8 - 150.3 (70.4)	5.2 - 6.5 (5.9)	2.7 - 4.5 (3.6)	10, 11, 12, 13
Gombak Norite								
Norite	Central Singapore	2.62 - 2.92 (2.75)	125 - 245 (174)	0.21 - 0.30 (0.27)	86.5 - 101.4 (95.8)	5.6 - 7.8 (6.6)	-	14

Numbers appearing in the ( ) represent the typical value found for the rock type

Bulk Density	γ	Modulus of Elasticity	E
Compressive Strength	σ_c	Compression Wave Velocity	V_p
Poisson Ratio	ν	Shear Wave Velocity	V_s

References to Appendix 5

1 Soil and Foundation (PTE) LTD (1998)
Site Investigation for The Underground Science City at Kent Ridge Park, Singapore.
(Volume II, Part I – Engineering Geological Assessment)

2 J. ZHAO, J.G. CAI, A.M. HEFNY
Creation of the Underground Science City in rock caverns below the Kent Ridge Park in Singapore
NTU Geotechnical Research Centre Geotechnical Engineering Monograph 4, pp 31

3 Soil and Foundation (PTE) LTD (1998)
Site Investigation for The Underground Science City at Kent Ridge Park, Singapore.
(Volume II, Part I – Drilling Exploration)

4 Tritech Engineering and Testing (Singapore) PTE, LTD (2003)
Geological and Rock Investigation at Jurong Island (Volume IV – Laboratory Test)

5 J. ZHAO, W.L. NG, J.G CAI, X.H. ZHANG
Feasibility of Underground Hydrocarbon Storage Caverns in Jurong Island (2003)
Proceedings of Underground Singapore 2003 and Workshop 'Updating the Engineering Geology of Singapore', pp 62-71

6 J. ZHAO, K.W. LEE, V. CHOA, Q. LIU, J.G. CAI
Underground Cavern Development in the Jurong Formation of Sedimentary rocks (1999)
NTU-PWD Geotechnical Research Centre, Geotechnical Engineering Monograph 2, pp 40

7 L. J. PAKIANATHAN, K. JEYATHARAN, C. F. LEUNG AND V. CHEPURTHY
Recent Experiences in Singapore Limestone Rocks (2006)
Proceedings of ISRM International Symposium 2006 & 4th Asian Rock Mechanics Symposium, pp 330

8 L. J. PAKIANATHAN, K. JEYATHARAN
Limestone Rocks of the Jurong Formation Engineering Experience (2005)
Proceedings of Underground Singapore 2005, pp 68-76

9 K. JEYATHARAN, L. J. PAKIANATHAN, K.W. LEE, C.P. SEH
Limestone of the Jurong Formation (2003)
Proceedings of Underground Singapore 2003 and Workshop 'Updating the Engineering Geology of Singapore', pp 372-387

10 Y.X. ZHOU, J. ZHAO, K. LEE
Mechanical and Engineering Properties of the Bukit Timah Granite (1993), Rock Caverns for Underground Space Utilisation, Singapore

11 Soil and Foundation (PTE) LTD (1996)
Seismic Survey & Site Investigation Works at Mandai for Lands & Estates Organisation, Ministry of Defence, Singapore

12 Econ Geotech (PTE) LTD (1999)
Site Characterisation for a site at Mandai for Lands & Estates Organisation, Ministry of Defence, Singapore

13 K. ORIHARA, T. NONAKA. G. TABE, G. PAGE, N. KUROSAKI, J. YAMASHITA, L.W. WENG
Rock Exploration for Tunnelling in Bukit Timah Granite (2005)
Proceedings of Underground Singapore 2005, pp 94-101

14 Structural Lab, School of Civil and Environmental Engineering, Nanyang Technological University (2007)
Rock materials come from SI WORK Project at Bukit Gombak. Samples were taken from a shallow depth.

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